



**GREEN
CLIMATE
FUND**

Meeting of the Board

4 – 7 October 2021

Virtual meeting

Provisional agenda item 15

GCF/B.30/02/Add.04

14 September 2021

Consideration of funding proposals - Addendum IV

Funding proposal package for FP172

Summary

This addendum contains the following six parts:

- a) A funding proposal titled "Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Secretariat's assessment;
- d) Independent Technical Advisory Panel's assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- f) Gender documentation.

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Funding Proposal

Project/Programme title:	Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)
Country(ies):	Nepal
Accredited Entity:	Alternative Energy Promotion Centre, Ministry of Energy, Water Resources and Irrigation, Government of Nepal.
Date of first submission:	[2020/12/30]
Date of current submission	[2021/09/08]
Version number	[V.10]



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. Choose an item		
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of <u>GCF budget</u> devoted to it. The total of the percentages when summed should be 100%.		
	Mitigation: Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☒ Buildings, cities, industries and appliances: ☐ Forestry and land use: Adaptation: Increased resilience of: ☐ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:		GCF contribution: <u>Enter number</u>% <u>Enter number</u>% <u>100</u>% <u>Enter number</u>% <u>Enter number</u>% <u>Enter number</u>% <u>Enter number</u>%
A.5. Expected mitigation impact	6,513,629 tCO ₂ eq	A.6. Expected adaptation impact	

A.7. Total financing (GCF + co-finance)	49,151,817 USD	A.9. Project size	Small (Up to USD 50 million)
A.8. Total GCF funding requested	21,128,224USD <i>For multi-country proposals, please fill out annex 17.</i>		
A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i>		
	☒ Grant 21,128,224USD ☐ Loan <u>Enter number</u> ☐ Guarantee <u>Enter number</u> ☐ Equity <u>Enter number</u> ☐ Results-based payment <u>Enter number</u>		
A.11. Implementation period	<i>Indicate the number of years and months the project/ programme is expected to be implemented. 60 Months</i>	A.12. Total lifespan	<i>Indicate the maximum number of years over which the impacts of the investment are expected to be effective. 24 Years</i>
A.13. Expected date of AE internal approval	November 2021	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> C
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes No ☒

A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☐ No ☒
A.20. Executing Entity information	<i>If not the Accredited Entity, please indicate the full legal name of the Executing Entity (ies) and provide its country of registration and ownership type. Note that there can be more than one Executing Entity. Also indicate if an Executing Entity is the National Designated Authority. Refer to the definition of Executing Entity in the Accreditation Master Agreement.</i> Alternative Energy Promotion Center (AEPC)
A.21. Executive summary (max. 750 words, approximately 1.5 pages)	
<i>Provide an executive summary of the project/programme including:</i> 1. <i>Climate change problem</i> Nepal is highly vulnerable to climate change and is ranked as the 9th most affected country to climate change, as per Global Climate Risk Index 2020. <i>Terai</i> region of Nepal is the most affected region in the country compared to <i>Hills and Mountains</i> in terms of death toll due to disasters. <i>Terai</i> is bordered by Indian Gangetic plain in the south and the <i>Churia</i> Physiographic Region in the north. Penetration of clean cooking solution technologies is very low in <i>Terai</i>. According to the 2011 census¹, around 75% of entire country's populations were using traditional biomass- 63% used fuel-wood, while 10.38% relied on cow dung (dung cake) and 21% used Liquefied Petroleum Gas (LPG), whereas, during the same period, 56% household in <i>Terai</i> used fuel-wood, 22% cow dung and 15% used LPG. Burning of solid fuels using traditional cook stoves (TCS)², cause health problems and emit a massive amount of black carbon and GHGs (such as carbon dioxide, methane). Even though, fuel-wood consumption in <i>Terai</i> has remained fairly constant, LPG consumption is in increasing trend, as around 40% of the households are using it now. As per the Nepal Oil Corporation (NOC) import and sales report for FY 2019/20, Nepal imported and consumed 449063 MT of LPG. LPG has to be imported, it has negative macroeconomic impact in the country, due to widening trade deficit with India, and has made the country very susceptible to supply risk as well. Nepal relies heavily on hydropower for its electricity generation, which is a clean and renewable energy source with no CO₂ emissions. Currently it has installed capacity of around of 1278 MW electricity of which 95.7% is from hydroelectricity³. Nepal has enormous potential for hydro-electricity, touted around 83,000 MW and 80% of its population is connected to national grid⁴. <i>Terai</i> region has relatively better electricity access rate, compared to <i>Hills and Mountains</i>⁵. However, use of electricity for cooking has yet to pick up, as merely 0.08% of the target population relies on it⁶.	

¹ CBS, 2011. Nepal Population and Housing Census 2011. <https://unstats.un.org/unsd/demographic-social/census/documents/Nepal/Nepal-Census-2011-Vol1.pdf>

² TCS has efficiency of <10%. These stoves emits >18.3 g/MJ carbon monoxide and >1031 mg/MJ fine particulate matter. Example: three-stone fire

³ Department of Electricity Development, 2020. Licence. Operating projects <https://doed.gov.np/>

⁴ MoF, 2020. Economic survey 2076/2077. https://mof.gov.np/uploads/document/file/Economic_Survey_2076-77.pdf

⁵ NEA, 2019. *Nepal Electrification Statistics*. Kathmandu: Nepal Electricity Authority

⁶ Baseline survey of *Terai* clean cooking program, 2019

2. Proposed interventions

To mitigate the impact of climate change and strengthen resilience of the most vulnerable communities to adapt to climate change, Alternative Energy Promotion Centre (AEPC) has proposed to implement the project “**Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)**”. The main objective of the project is to surge use of Clean Cooking Solutions by instigating innovative concepts in Nepal of (i) bulk tendering via reverse auctioning for cost effectiveness (ii) output based financing for de-risking of investments and (iii) mainstreaming and capacitating local governments in renewable energy (RE) sector. These concepts have never been introduced in the country up to now and could be game changers in the clean cooking sector. The proposed project will collaborate with Nepal Electricity Authority (NEA) to explore and develop cost effective tariff structure incentivizing electric cooking for the households underpinned by the immense hydropower potential and support NEA in the upgradation of electricity power supply in the project locations ensuring reliability, connectivity and system improvement. Additionally, this project actively aims to ensure effective women participation in each stage of value chain, and especially create jobs and volunteering opportunities for them. For the first time in Nepalese cookstove sector, it aims to generate women technical workforce.

The project aims to promote clean cooking solutions, such as Tier 3+ Improved Cooking Stoves (ICS)⁷, Biogas and Electric Cook Stoves (ECS) in 1,000,000 Households (HHs) of *Terai* region. It will be implemented in 22 districts of the *Terai* region, which also includes certain parts of the *Churia*⁸. The project targets to switch 500,000 households from LPG and fuel-wood stoves to electric stoves; 490,000 households from loose biomass, dung cake and fuel wood to Tier 3+ ICS; and to introduce biogas system for 10,000 households that have sufficient livestock. Electric cooking is one of the potential clean cooking solutions for Nepal as 95.7% of electricity is from Hydro plants.

3. Climate impacts/benefits

The project will address specific barriers from transitioning traditional inefficient cooking practice to an efficient and climate friendly cooking solutions. Scaling up the government initiative on CCS, the proposed project aims at reducing an estimated 6.51 million tons of CO₂eq by bringing transformative change in cooking pattern with wider usage of modern clean cooking solutions via ECS, Tier 3+ ICS and biogas. Moreover, the project will have additional benefits such as improved health due to reduced indoor air pollution, time savings due to reduced fuel usage and more employment opportunities in clean cooking sector. There will be no CO₂ emissions as electricity from hydro will be used for cooking.

The project will also help to conserve fragile ecosystem of *Churia*, a source for fuel wood, by reducing deforestation and forest degradation due to reduced demand for fuel wood consumption. It is estimated that, Tier 3+ ICS can reduce fuel wood consumption by 30% on average and biogas and ECS will significantly reduce the use of fuel wood. In addition, more than 30% of reduction in indoor air pollution is expected by replacing TCS by Tier 3+ ICS, which will primarily benefit women and children as women

⁷Tire 3+ ICS has efficiency of ≥30%. These stoves emits ≤7.2 g/MJ carbon monoxide and ≤218 mg/MJ fine particulate matter.

⁸The project is not linked with FAO project (Building a Resilient Churia Region in Nepal) as FAO targets the Central and Eastern Churia-Terai Region, few geographic area is overlapped. This projects targets to reach 1 Million HH (potential is more than 1.5 Million Households). So, the targeted households by the FAO project and AEPC will be different. Furthermore, AEPC will coordinate with FAO project for avoiding the duplication as the subsidy is mobilized through AEPC, the duplication will not happen.

will be a direct user of the stoves. In average, clean cookstoves can save more than 50% time used for fuel wood collection and preparing meal will be saved⁹, which can be utilized for other work and care for their families¹⁰. Besides, in cases where fertilizer needs to be purchased, the cost for fertilizers can be saved by using biogas slurry as fertilizer.

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<https://genderandenvironment.org/wp-content/uploads/dropbox/Publication/Guatemala%20Stakeholder%20Consultation%20Alliance%20Strategic%20Business%20Plan.pdf>

¹⁰<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6480161/>

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate change problem: Describe the climate change problem the proposal is expected to address. Describe the mitigation needs (GHG emissions profile) and/or adaptation needs (climate hazards and associated risks based on impacts, exposure, and vulnerabilities) that the proposed interventions are expected to address. Also describe the most likely scenario (prevailing conditions or other alternative) that would remain or continue in the absence of the proposed interventions. Include baseline information. The methodologies used to derive such information, including the mitigation and adaptation needs, should be included in the feasibility study.

Context: In describing the mitigation and/or adaptation needs, briefly describe the target region/area of the proposed interventions including information on the demographics, economy, topography, etc.

Nepal is a landlocked country bordered by India and China. Topographically, it is divided into three ecological regions, namely *Mountains*, *Hills* and *Terai* (southern plains). *Terai* is bordered by the Indian Gangetic plain in the south and the *Churia* Physiographic Region in the north. The project will be implemented in 22 districts of the *Terai* region, including some parts of the *Churia* (Figure 1)¹¹. *Terai* represents around 17% of the total land but its inhabitant represent about 50% of the country's population¹². It has very fertile soil and thus is known as the country's rice basket. It consists of dense forests- about 20.41% of total *Terai* area¹³.

Access to clean energy cooking solutions is a massive problem for Nepal, including *Terai*. Nepal depends highly on traditional emissive energy sources, such as firewood, animal waste and crop residues for cooking purposes. According to the 2011 census¹⁴, around 75% of the populations were using traditional biomass- 63% used fuel wood, while 10.38% relied on cow dung (dung cake). This has not changed much over the last decade, as the recent report published by the World Bank in 2019, showed that still 73.5 % of the households depend on firewood for cooking¹⁵. Furthermore, majority of them are using highly in-efficient traditional cookstoves (TCS)¹⁶, which not only consume more fuel, thus putting more pressure on the forests, but also cause indoor air pollution. Burning of solid fuels using TCS (efficiency of 10%), cause health problems and emit a massive amount of black carbon and GHGs (such as carbon dioxide, methane).

¹¹The project is not linked with FAO project (Building a Resilient Churia Region in Nepal) as FAO targets the Central and Eastern Churia-Terai Region, few geographic area is overlapped. This projects targets to reach 1 Million HH (potential is more than 1.5 Million Households). So, the targeted households by the FAO project and AEPC will be different. Furthermore, AEPC will coordinate with FAO project for avoiding the duplication as the subsidy is mobilized through AEPC, the duplication will not happen

¹² CBS, 2011. Nepal Population and Housing Census 2011. <https://unstats.un.org/unsd/demographic-social/census/documents/Nepal/Nepal-Census-2011-Vol1.pdf>

¹³ FRA, 2014. Terai Forests of Nepal

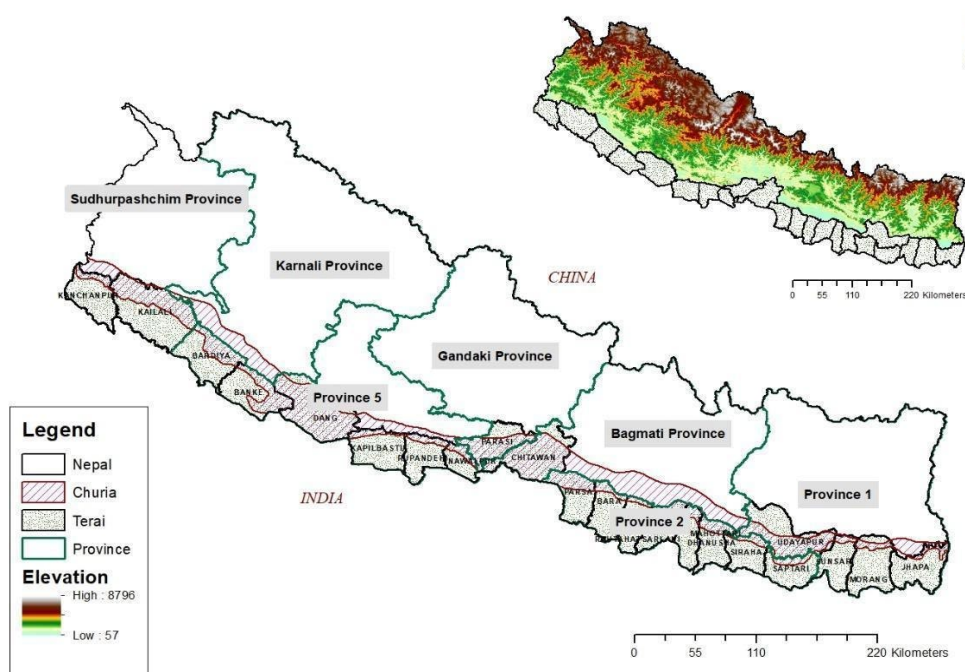
¹⁴ CBS, 2011. Nepal Population and Housing Census 2011. <https://unstats.un.org/unsd/demographic-social/census/documents/Nepal/Nepal-Census-2011-Vol1.pdf>

¹⁵ World Bank, 2019. NEPAL | Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework

¹⁶ibid

Household combustion is estimated to produce 25% of global emissions of black carbon, which is the second largest contributor to climate change after carbon dioxide (CO₂)¹⁷. World Health Organization (WHO) estimates 8,700 deaths a year from indoor air pollution in Nepal.

Figure 1: Project Location

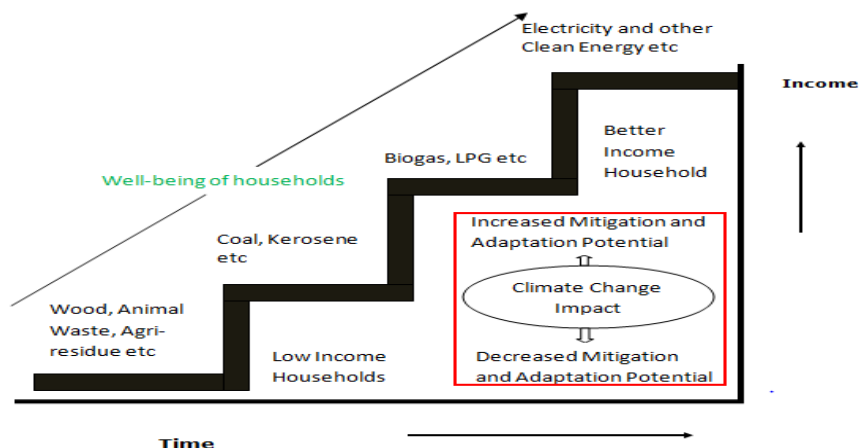


In *Terai*, as cow dung are more easily available and free to collect, households rely heavily on dung cake as cooking fuel; it is used by 25% of the population. This cooking pattern is typical to the *Terai* region compared to other topography of the country as the dependence on dung cake in hills and mountains are at meager 0.1% and 0.4% respectively¹⁸. Dung cake is the most polluting fuel for cooking, and therefore lies at the bottom of the energy ladder (Figure 2). Dung cakes are produced at the household level and burnt in traditional open fires or mud stoves, resulting in the emission of a dangerous cocktail of hundreds of pollutants to which women and young children are exposed daily.

¹⁷ WHO, 2016. Burning opportunity: Clean Household Energy for Health, Sustainable Development, and Wellbeing of Women and Children https://apps.who.int/iris/bitstream/handle/10665/204717/9789241565233_eng.pdf?sequence=1

¹⁸ CBS, 2011. Nepal Population and Housing Census 2011. <https://unstats.un.org/unsd/demographic-social/census/documents/Nepal/Nepal-Census-2011-Vol1.pdf>

Figure 2: Energy ladder representing household cooking in Nepal



While TCS is predominantly used for cooking in the rural areas, Liquefied Petroleum Gas (LPG) is the most commonly used cooking technology in the urban areas. Due to smoke-free operation, flexibility in use and traditional kitchen practices dependent on this technology, there is a growing trend of LPG consumption. In 2011 census, around 21% of the entire populations were using it, while the 2019 World Bank survey showed its share had increased to 34.9%. LPG consumption is even higher in *Terai* region; it has increased from 15% at 2011¹⁹ to 40.25% at 2019²⁰.

Even though, LPG is a cleaner fuel compared to fuel wood and dung, as it does not produce smoke and contribute to indoor air quality, it is derived from fossil fuel, hence contributes to global GHG emissions. Moreover, LPG has to be imported from India. In 2018-19, Nepal imported 429,609 MT of LPG, which is a 270% increase since 2008/09²¹. This over dependence on LPG import has put Nepal in a perilous situation, as it has not only widened the trade deficit²², but has also made Nepal very susceptible to supply risk as experienced during the 2015 fuel crisis, when the supply of fossil fuels to Nepal from India was severely disrupted²³. Nepal must curtail its dependency on LPG in order to strengthen its economy and for its sovereignty.

Electricity is one big option that Nepal could rely on for its transition to clean cooking solution. Though Nepal has enormous potential for hydro electricity generation, which is touted to be around 83,000 MW, until couple of years back, Nepal was reeling under an acute power shortage that resulted in up to 16 hours of scheduled load shedding. As such, the environment was not conducive to promote electric cooking, despite the benefit it offers. However, now the time is ideal to intervene with electric cooking, as the country is officially free from load shedding²⁴ and by 2028, Government of Nepal has envisioned to increase its electricity generation to 15,000MW; with the long awaited 456MW Upper Tamakoshi hydropower project set to be connected to the grid within 2020. Currently, nearly 95.7% of the total electricity generation is supplied by hydro power projects. With new addition of hydro projects the share

¹⁹ibid

²⁰ Baseline survey of *Terai* clean cooking program, 2019

²¹<http://noc.org.np/import>

²²https://npc.gov.np/images/category/Nepals_Trade_Deficit1.pdf

²³<https://kathmandupost.com/miscellaneous/2015/09/27/fuel-crisis-worsens>

²⁴<https://thehimalayantimes.com/business/nepal-is-totally-load-shedding-free/>

of electricity would be cleaner. A simulation conducted by Nepal Electricity Authority, the state owned utility, shows that Nepal will be in the state of seasonal surplus by 2027 from which Nepal will be a pure exporter of power on RoY (Round-the-Year) basis²⁵. With an annual electricity consumption growth rate of 15%, the country cannot use surplus power because it lacks energy-intensive industries and electric mass-transit systems.

At present, 80% of populations of Nepal are connected to national grid. Terai has relatively high electricity access rate compared to *Hills* and *Mountains*²⁶. However, use of electricity for cooking has yet to pick up, as merely 0.08% of the population relies on it²⁷. Electric stove will be provided to those households who have grid access. For those who are not connected to grid would be offered the facility of Tier 3+ ICS and biogas. As a consequence, there will be no CO₂ emissions as electricity from hydropower would be used for cooking. Therefore, GoN has been prioritizing programmes that increase national electricity consumption. Electricity Regulation Commission of Nepal has made reforms to the household electricity tariff rates by reducing the cost structure by 10% so as to incentivize increase in use of electrical appliances²⁸. Nepal Electricity Authority has publicly requested to use the electric appliances in the kitchen as much as possible.

AEPC has conducted a user survey along 284 local governments among the 22 districts of *Terai* Region, spread across six Provinces, regarding the usage of cooking solutions²⁹. The survey provides baseline information on cooking solutions in the region. It shows that out of 2.7 million households in *Terai* region, 58.13% use TCS, 3.26% use biogas plant, 40.24% use LPG, and 0.42% use electric induction stoves. Through this project, AEPC has envisioned replacing 500,000 TCS and Tier 1 ICS by 10,000 domestic biogas plants and 490,000 Tier 3+ ICS. Similarly, it seeks to replace another 500,000 LPG users to electric cookstoves (ECS). Transitioning from TCS and LPG to more cleaner cooking solutions would reduce unsustainable consumption of fuel wood, cow dung and LPG, and also create demand to use electricity generated from its hydropower projects. However, these changes are less likely to occur on its own. TCS users are among the poorest and socially excluded groups.

Overall, HDI (Human Development Index) ranking for the Province, which encompasses most of the *Terai* region is very low³⁰. Furthermore, the target beneficiaries that the project aims to intervene on, the women, are disadvantaged on socio-economic fronts at BAU. For instance, estimated income (at PPP) reflects severe gender disparity (0.63 female to male ratio) in this region. Hence, affordability to switch to energy efficient technologies without external interventions remains a challenge.

In the absence of the project intervention, there would be no reduction in solid biomass and LPG consumption. This will further aggravate indoor air pollution and GHG emissions.

²⁵ <https://myrepublica.nagariknetwork.com/news/anxiety-of-surplus-electricity/>

²⁶ NEA, 2019. *Nepal Electrification Statistics*. Kathmandu: Nepal Electricity Authority

²⁷ Baseline survey of *Terai* clean cooking program, 2019

²⁸ <https://myrepublica.nagariknetwork.com/news/nea-proposes-tariff-reduction-for-household-users>

²⁹ AEPC conducted a survey in 2019/20 to identify the baseline information on cooking solutions (fuel type and Stoves) from 284 local levels of the 22 districts of the Terai Region.

³⁰ Dhungel, 2019. Provincial Comparison of Development Status in Nepal: An Analysis of Human Development Trend for 1996 to 2026. <https://www.nepjol.info/index.php/jmds/article/download/24958/21005/>

Related projects/interventions: Also describe any recent or ongoing projects/interventions that are related to the proposal from other domestic or international sources of funding, such as the Global Environment Facility, Adaptation Fund, Climate Investment Funds, etc., and how they will be complemented by this project/programme (e.g. scaling up, replication, etc.). Please identify current gaps and barriers regarding recent or ongoing projects and elaborate further how this project/programme complements or addresses these.

With the backdrop of ambitious campaign of Clean Cooking Solution for all (CCS4ALL) initiated in 2013, the Government of Nepal had endorsed Investment Prospectus for clean cooking solutions³¹ and Biomass Energy Strategy (BEST), 2017³². BEST has the aim of attaining CCS4ALL in two stages i.e. by improved solutions of cooking by 2022 (no traditional way of cooking with open fire burning) and modern solutions of cooking with at least Tier 3+ level by 2030.

The Government of Nepal has announced the implementation of a multi-year programme: “*Terai Clean Cooking Programme*”, starting from the fiscal year 2019/20. Under this programme, traditional fuel such as animal dung cake will be replaced in 22 Districts of *Terai*, by Tier-3+ ICS, biogas and electric cooking. Despite many efforts by the Government of Nepal, a number of challenges and barriers continue to inhibit the transition towards a clean, modern and efficient cooking solution in the Terai region. Government of Nepal and External Development Partners (EDPs) have invested around 4.5 million USD for CCS in fiscal year 2018/2019³³. Predominantly, AEPC distributed mud ICS which had immediate positive impacts but in the long run its regards to its durability the mud ICS could not provide a long lasting impact. Promotion of electric cooking is a shift from existing renewable energy technologies that are under the current subsidy regime. In terms of the implementation arrangement the technology had been highly promoted but the vendors (private sector) had to create a market place which required massive investment and resources. Government and non-government agencies are implementing climate change initiatives in different parts of Nepal. GEF Small Grants Programme works with local communities in Nepal, to achieve global environmental benefits by addressing their local needs. The Nepal Small Grants Programme primarily contributes to Goal 7: Ensure Access to Affordable, Reliable, Sustainable and Modern Energy to All; Goal 13: Take Urgent Action to Combat Climate Change and its Impacts; and Goal 15: Protect, Restore and Promote Sustainable Use of Terrestrial Ecosystems, Sustainably Manage Forests, Combat Desertification and Halt and Reverse Land Degradation and Biodiversity Loss while also contributing to other goals. In relation to the project, the GEF SGP has supported in the Installation of bio-gas, solar home and micro hydro systems, as well as solar tukis, rice husk stoves, Puxing biogas and ICSs—saving altogether 16,422 tonnes of carbon dioxide per year. Furthermore, the Ministry of Forests and Environment (MoFE) is implementing the Nepal Climate Change Support Program (NCCSP), with funding support from the UK Department for International Development (DFID) and the European Union (EU) in 14 districts of mid- and far-west Nepal. This project is under implementation with particular focus on the preparation and implementation of LAPAs. There are other projects such as the Community-based Flood Risk and GLOF Risk

³¹World Bank, 2017. *Investment prospectus for clean cooking solutions in Nepal : a roadmap to national goal of providing clean cooking solutions for all* (English). Group <http://documents.worldbank.org/curated/en/916571494854063344/pdf/Investment-prospectus-for-clean-cooking-solutions-in-Nepal-a-roadmap-to-national-goal-of-providing-clean-cooking-solutions-for-all.pdf>

³² Biomass Energy Strategy, 2017. [https://www.aepc.gov.np/uploads/docs/2018-07-29_Biomass%20Energy%20Strategy%202073%20BS%20\(2017\)%20English.pdf](https://www.aepc.gov.np/uploads/docs/2018-07-29_Biomass%20Energy%20Strategy%202073%20BS%20(2017)%20English.pdf)

³³ AEPC, 2018. Budget of FY 2075-76. <https://www.aepc.gov.np/uploads/docs/2018-07-09Budget%20of%20FY%202075-76.pdf>; AEPC, 2019. Progress at Glance: A year in Review FY 2075/76 (2018/19). <https://www.aepc.gov.np/uploads/docs/progress-at-glance-a-year-in-review-fy-207576-201819-1572879503.pdf>

Reduction Program, which has received funding from the Least Developed Countries (LDCs) Fund through the United Nations Development Program (UNDP), as the Global Environment Facility (GEF) implementing agency.

The proposed project is designed incorporating the lesson learnt from the past. The proposed project will create a market in it itself, thus providing the initial big boost to the CCS market. In terms of the project interventions, the project brings in the paradigm shift much required in the sector by introducing the innovative approaches of bulk tendering via reverse auctioning, output based financing and capacitating local governments while providing a boost to the CCS market which has seen the dearth of Tier 3+ ICS and Electric Cooking. Thus, the project will catalyze the sector and underpin the scale up in future post the project interventions.

The investment in CCS by the GoN focuses exclusively on distribution of technology. During the FY 2018/19 less than 2% of the entire budget spent on programs under CCS was spent on activities in line with capacity development and strengthening of supply chain. The consequences of that gap can be seen in lack of after sales capacity at the user level which ultimately affects in up-scaling and dissemination of CCS.

In a business-as-usual (BAU) approach for deployment of CCS technologies, AEPC sets annual target for technology deployment, fixes certain amount of subsidy for each technology for households and institutions located in different districts that are grouped in terms of remoteness. Then companies are shortlisted based on a set of qualification criteria to implement those targets- such shortlisted companies are called prequalified companies (PQ companies). PQ companies collect demand from households, and once verified by AEPC they install the system. After successful commissioning, PQ companies collect payment from households and submit the installation completion form to AEPC at which point they qualify to claim the first installment of subsidy amount from AEPC. Finally, AEPC conducts sample based monitoring to release the final installment of the subsidy amount to the PQ companies. The bureaucratic and lengthy process of this BAU approach has made it difficult for AEPC to achieve its annual targets. Furthermore, since the specification and price of technology, and the subsidy amount are predetermined, there is no incentive for PQ companies to reduce the cost of technology or innovate better technology. To shift towards clean cooking, Nepal requires a multi-pronged approach. First, “increase the cost effectiveness by instigating the concept of bulk tendering via reverse auctioning” then “de risk the investment by introducing the output based financing”. Second, ensure quality assurance and adequate after sales service by strengthening the existing testing centers and expanding its services at the regional and local level. Third, local governments can play a crucial role in improving access to clean cooking solutions. Hence, their capacity needs to be developed.

Below mentioned point summarizes these barriers, as well as how the project will address them.

1. Socio-economic barriers

The main socio-economic barriers for transition to CCS include **consumer’s limited awareness about benefits of CCS and subsidy policy on renewable energy and their low willingness to pay higher prices due to low economic levels of the household**. Most of the LPG users are unaware of cost savings from using ECS. Likewise, TCS users are unaware about the health benefits of Tier 3+ ICS and

biogas. There is an existing subsidy provision for building domestic biogas, however many households are unaware of the subsidy scheme. In this context, users need more awareness raising and access to affordable financing to adopt CCS. Furthermore, household cooking is challenged from gender stereotyping that women are responsible for household chores.

The project will address the issue by bringing behavioral change among users by raising awareness about the cost and health benefits of ICS through campaigns and promotional activities. Also, the project will stress on gender parity and women empowerment. Within the project period of five years it will reach the number of 1 million households in the Terai region of Nepal, the five year period that focuses on pre and post purchase behavior enforcement will instigate and maintain the behavior change among the consumers of the Terai Region. The will ensure the acceptability of the proposed technologies. Post project interventions, beneficiaries are expected to switch to the proposed technologies on their accord given the change in behavior coupled with the intervention on value chain through awareness and promotional campaigns, promotion of emerging CCS technologies, training and support with enhanced repair and maintenance services and technology availability. This will create a virtuous cycle in the sector through which further interventions will be no longer required and the uptake of the technologies will follow a natural course given the benefits and the change in behavior.

2. Technological barriers

Low level of technical knowledge and unavailability of technicians is another major constraint to shift towards CCS. Very few stove designs are available in the country and consumers face a lack of choice in models. Dearth of Research and Development in developing models and products that meet user's requirements is also a challenge. Currently there are not many manufacturers and vendors of Tier 3+ ICS and ECS in Nepal. They are being imported, mainly from India and China. There is no technical capacity in the community to manufacture advanced ICS. There is already limited technical workforce availability; the scene is further alarming on women's involvement. At community level, capacity building, repair and maintenance training would eliminate such barrier.

The project will support capacity enhancement at community level, human resources, technical expertise and technology in the form of Tier 3+ ICS, biogas and ECS. In addition, the project will strengthen the existing testing center to ensure the quality of the stoves. As stated in GAP, the project will address gender issues on technological aspects.

Access to grid electricity, quality of electricity power supply and voltage fluctuations are other challenges to shift towards electric cooking. The quality of electricity power supply in different parts of the country including the Terai is still poor and should be upgraded. Electric cooking would be promoted in the grid-connected households exploring and introducing a cost effective tariff structure for household consumption. Nepal Electricity Authority, the state owned sole utility, is also a member of the AEPC board. In the short term, beneficiary selection will be a key process while disseminating the different types of technologies provisioned by the project. Electric Cooking will be mainly targeted based on the demand received from the Local level where in it is required to assure that the beneficiaries of ECS will have reliable electric supply. This will include selected clusters in rural settings, urban and peri-urban areas which will have the required infrastructure. As a corrective measure for short term, AEPC will provide beneficiary mapping to Nepal Electricity Authority (NEA) and NEA will be upgrading capacitors and conductors in the overloaded areas. Thus beneficiary selection process is the specific measures to ensure the electric supply is reliable at the downstream household level. In terms of the medium and

long term measures, AEPC will work with Nepal Electricity Authority on increasing distribution transformer size and improving distribution and transmission lines to control voltage fluctuations in the project area. Through the proposed project, the government will play a major role in setting up reliable infrastructure which will assure the wider usage of electric cooking among the household population. With vigorous technological reliability, electric cook stoves will diminish the usage of LPG in the country.

3. Financial Barriers

Most of the TCS users are among the poorest households, and they lack capital to switch to CCS technology. Furthermore, they are usually deprived of formal banking channel, hence do not have access to bank financing. Although few cooperatives provide loans for CCS at community level, the interest rate is too high for the poorest households. For extensive promotion of electric cooking, cost effective tariff structure needs to be introduced, which is also an ongoing policy of the Ministry of Energy, Water Supplies and Irrigation. As such, Electricity Regulatory Commission has revised the tariff structure with completely subsidized electricity to the low electricity consuming households consuming up to 10 units per month, 25 percent discount for those consuming up to 150 units per month and 15 percent discount to those consuming 250 units per month in electricity charges applicable from Fiscal Year 2020/21. The revision has been made to promote Electric Cooking in the households. A further cost effective tariff structure will be introduced in coordination with Nepal Electricity Authority analyzing the increment in the revenue due to addition of electric cooking loads to the distribution system during project implementation.

In addition, there is a high **investment risk due to existing subsidy delivery mechanisms**. Current subsidy delivery mechanism is a very bureaucratic and lengthy process that makes it difficult for the government to achieve big target in a short period. Bulk tendering via reverse auctioning would help to reduce the price of CCS through competition. Furthermore, Capital resources provided by Green Climate Fund (GCF), GoN and other possible development partners would help in addressing the financial barriers. Also, the procurement will follow bulk tendering via reverse auctioning (least cost based selection) approach for cost effectiveness and output based payment to de-risk the investment (detail has been included in section B3).

From the beneficiaries' perspective, the targeted beneficiaries who shall be the poor households of the urban and peri urban areas for electric cookstoves and poor households of the rural area for the Tier 3+ ICS lack the financing to the purchase these clean technologies. Based on our analysis in terms of their portion of annual income to the amount that needs to be invested on the installation (CAPEX) and operation (OPEX), considering a conservative approach of with bare minimum cost allocation households with annual income level between \$250 to \$2000, the electric cookstoves without the subsidy support from the project will require between 5% to 32% of their annual income on the CAPEX and between 8% to 49% of their annual income on the OPEX over the five year period. The percentage of income level analysis has been presented as a separate annex and also in section D.6 of the funding proposal.

4. Institutional Barriers

After the promulgation of new constitution in 2015, Nepal has become a federal republic, with country being divided into seven Provinces. The old village development committees have been merged to

create rural municipalities, while new municipalities and metropolitan cities have also been created. In a nutshell, Nepal has now different governance modality, with newly formed Local Governments (LG) and Province Government (PG). The new constitution has given mandate to the Local Governments for development of Renewable Energy. However, they have weak institutional setup vis-à-vis renewable energy technologies. They also lack

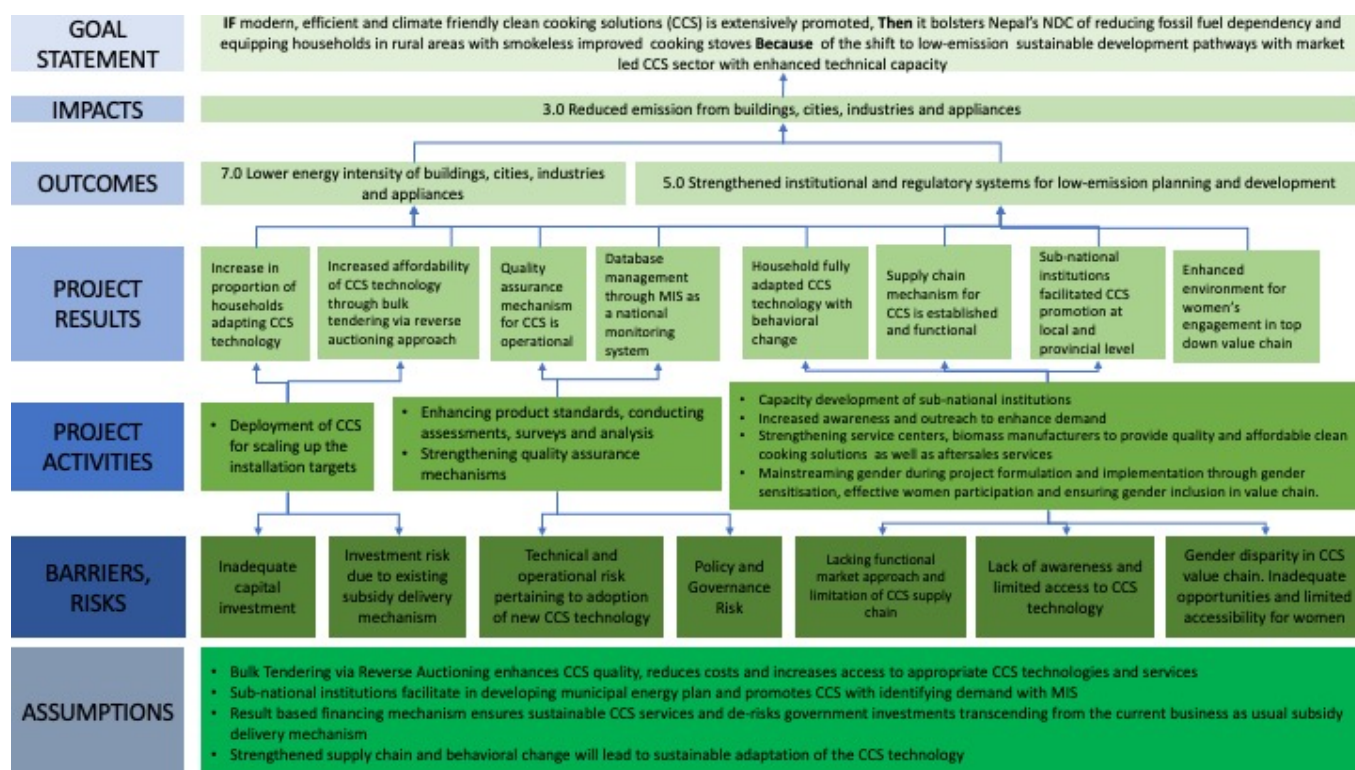
- Technical and financial capacity,
- Awareness on the impact of clean energy,
- Dedicated human resources,
- Expertise and experience required for planning (energy plans), procuring (demand aggregation and bulk tendering approach), implementing and monitoring renewable energy technologies projects and programme.

The proposed project would empower LGs by heavily engaging them throughout the project implementation phase. The project will capacitate LGs and PGs to prioritize CCS promotion in their periodic and annual planning, formulating appropriate policies and mechanisms.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

Describe the theory of change and provide information on how it serves to shift the development pathway towards a low-emission and/or climate resilient direction. Provide the diagram of the theory of change (approximately 1 page).

Figure 3: Theory of Change



The theory of change should include any barriers (social, gender, fiscal, regulatory, technological,

financial, ecological, institutional, etc., as relevant) that need to be addressed. Use a results chain of inputs, activities, outputs, outcomes, and impact statements, and identify the how and why of causal relations to deliver the project's expected results.

The primary problem that this project will address is **Increasing greenhouse emissions and indoor air pollution due to inefficient cooking practices**. This project aims to mitigate the problem by increasing climate resilience, through CCS. This will result in the shift of Nepal towards a low carbon development path with increased climate resilience and sustainable development.

While high dependence on biomass fuel using inefficient TCS and growing use of LPG is in increasing trend, the transition to CCS is challenging. The set of barriers and risks that might prevent the project objectives from being achieved are as following:

B1. Inadequate capital investment and operational cost: Most of the TCS users do not have access to banking facilities to borrow money. In the capital market, banks and financial institutions lack loan products for CCS technologies. Similarly, the CAPEX of tier 3+ ICS and biogas will be higher compared to the technology being replaced i.e. TCS. In case of ECS, the new consumers will need to purchase ECS utensils separately. The additional cost presented by switching over may prevent a user to adopt new CCS technology. The selection criteria for the given technologies and the subsidized rate for each of the technology will prompt users to switch to the technology and in the long run the project will help maneuver through the barrier by creating an enhanced awareness on the use and the benefits of the technology ensuring long lasting usage of the proposed technologies.

B2. Technical and operational barrier pertaining to adoption of new CCS technology: Despite the abundance of electricity, with linkage of many hydropower projects to the national grid, the quality of electricity on supply side is still weak in most areas of the country, which is a major barrier to self-propagate electric cooking in Nepal. As such, the currently available distribution system may not have the capacity to support additional load demand presented by ECS. Additionally, at household level, the electricity system will need to be upgraded for installing ECS; this will require additional time and capital on the user's part.

B3. Lack of technical and project implementation skill at sub-national institutions: Most sub-national levels do not possess the technical capacity to envision and incorporating energy efficiency programmes in their annual energy plans. Likewise, the capacity may not be up to par for implementing projects of this scale.

B4. Lack of awareness and limited access to CCS technology: There are not enough information, marketing and promotion of Tier 3+ ICS and Electric Cook Stoves readily available for general public. Most of the TCS users are not aware of health benefits of Tier 3+ ICS solutions. Meanwhile, most of the LPG users are unaware of cost savings from Electric Cook Stoves. There is an existing subsidy for building domestic biogas, however many households are unaware of the subsidy scheme.

B5. Lacking functional market approach for CCS: Because of their relatively new introduction to the domestic consumers, currently there are not many manufacturers and suppliers of Tier 3+ ICS and Electric Cook Stoves in Nepal. Likewise, repair and maintenance centers, with limited skilled human

resources are sparse, ever more so in the context of rural areas where large portion of Tier 3+ ICS will be deployed. This may present a risk to consumers for adoption of CCS due to limitation of after sales services.

B6. Gender disparity in CCS value chain, and Inadequate opportunities and limited accessibility for women

Barriers to an equal representation up and down in value chain includes traditional social norms biased against women, poverty as a result of limited access to education, early marriage, limited asset ownership. Furthermore, women's participation is limited to decision making and income generating activities, and technical workforce.

R1. Investment risk due to existing subsidy delivery mechanism: Current subsidy delivery mechanism is a very bureaucratic and lengthy process that makes it difficult for users to access it, hence, for the government to achieve big target in a short period.

The project activities are based on three major targets:

Increasing access to CCS by deployment of proposed technologies

- 1. Deployment of CCS for scaling up the installation targets:** This activity involves procurement and deployment of CCS technologies with initial support to users which will directly increase access to CCS. Likewise, the activity will use revised procurement and payment delivery methods in bulk tendering via reverse auctioning and output based payment that will ensure reduced investment risks from vendor's view. The activity will address B1 and R1.

Strengthening quality assurance and enhancing product standards

- 2. Enhancing product standards, conducting assessments, surveys and analysis:** Within this activity national level technical standards and documents will be prepared for CCS technologies proposed in this project. The outputs of the project will address technical and operational issues pertaining to adoption of proposed CCS technology. Likewise, Project Implementation Unit (PIU) will also systematize procurement and subsidy delivery mechanisms as per project requirement. The activity will address B2 and R2.
- 3. Strengthening quality assurance mechanisms:** The activity will mainly focus on strengthening testing facilities and developing quality assurance mechanisms for procurement of CCS technologies. The activity will address B2.

Capacitating sub-national levels, enabling functional CCS market and increasing awareness of CCS at user level

- 4. Capacity development of sub-national institutions:** Through this activity, sub-national institutions will be capacitated to prioritize CCS promotion in their annual plans, formulating appropriate policies and mechanisms to achieve those targets. Likewise, the project will also assist local governments in preparing their energy master plans. The activity will address B3.

- 5. Increasing awareness and outreach to enhance demand:** Multiple awareness campaigns and promotional activities regarding CCS and its benefits will be conducted targeted towards beneficiaries. This activity will address B4.
- 6. Strengthening service centers, biomass manufacturers to provide quality and affordable clean cooking solutions:** The activity will majorly focus on strengthening CCS supply chain as well as capacity development through vocational training of local technology manufacturers, entrepreneurs and beneficiaries. The activity will address B5.

The project results that would be achieved by conducting the aforementioned activities are as following:

- 1. Increase in proportion of households adapting CCS technology:** This indicates the total increase in no. of households shifting from their existing cooking technologies to proposed CCS.
- 2. Increase affordability of CCS technology through bulk tendering via reverse auctioning approach:** Bulk tendering via reverse auctioning reduces the initial price of CCS in comparison to BAU model, which will in turn reduce the initial investment barrier by users in CCS.
- 3. Quality assurance mechanism for CCS is operational:** National level standards on CCS will be strengthened; likewise, one central level testing facilities for CCS technologies will be strengthened.
- 4. Database management through MIS as a national monitoring system:** MIS for real time data collection and demand aggregation will be developed at each LG.
- 5. Household fully adapted CCS technologies with changed behavior:** The project targets 90% of the total households will continue the use of CCS technology.
- 6. A supply chain mechanism for CCS is established and functional:** Local level CCS service centres will be supported for maintenance through dedicated vocational trainings to individuals.
- 7. Sub national institutions facilitated CCS promotion at local and provincial level:** Each LG will be assigned a dedicated CCS facilitator and will be supported in creating dedicated energy units. Likewise, the project will support LGs in preparing energy master plans and assist in developing annual targets on CCS and promotion of other sustainable energy technologies.
- 8. Enhanced environment for women's engagement in top down value chain:** Institutional and managerial arrangement will be made during project formulation implementation, and monitoring & evaluation. Extensive use of GESI specialist during the project implementation period and gender sensitization training for project staff at all levels to maintain desired level of gender awareness will be done. Gender mainstreaming in all local and provincial institutional setup will assure increased employment and volunteering opportunities to women.

These envisioned project results will help in achieving following project level outcomes:

- 5.0 Strengthened institutional and regulatory systems for low-emission planning and development** (Results linked to the outcome: 4, 6, and 7)
- 7.0 Lower energy intensity of buildings, cities, industries and appliances** (Results linked to the outcome: 1, 2, 3 and 5)

The ultimate fund level impact that this project will achieve is:

- 3.0 Reduced emission from buildings, cities, industries and appliances**

In order for the project to derive desired results from the proposed inputs and activities, the project is

based around following fundamental assumptions:

1. Bulk tendering via reverse auctioning enhances CCS quality, reduces costs and increases access to appropriate CCS technologies and services;
2. Local Government leads in developing Municipal Energy Plan (MEP) and promotes CCS with identifying demand linked with online management information system;
3. Output based financing mechanism ensures sustainable CCS services and de-risks government investments transcending from the current business as usual subsidy; and
4. Strengthened supply chain and behavioral change will lead to sustainable adoption of the CCS technology.

B.3. Project/programme description (max. 2000 words, approximately 4 pages)

Define the project/programme. Describe the proposed set of components, outputs and activities that lead to the expected Fund-level impact and outcome results. Components should reflect the project/programme level outcomes.

This should be consistent with the financing by component in section C.2, the results and performance indicators provided in section E.5, and the implementation timetable in annex 5.

The Project “**Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)**”, financed by GCF grant and co-financed by Government of Nepal (via AEPC and LGs) will contribute to mitigate GHG emissions and strengthen the resilience of most vulnerable communities to adapt to climate change. The project will address specific barriers from transitioning traditional inefficient cooking practice to efficient and climate friendly cooking solutions. Scaling up the government initiative on CCS, the proposed project aims at reducing an estimated 6.51 million tons of CO₂eq by bringing transformative change in cooking pattern with wider usage of modern clean cooking solutions via ECS, Tier 3+ ICS and Biogas plants.

The project targets the poor households of the urban, peri-urban and rural area with electric cookstoves for the urban and peri urban poor households and Tier 3+ Improved Coostoves for the poor households in the rural setting. The project follows a bottom up approach in terms of identifying these households and providing the subsidy and clean technology. Clean technology and its access in the country are limited to more affluent households. These cleaner means of cooking stays at the affluent household for a longer period and trickle down slowly towards the households in the low level income category. The major reason for this sluggish transition is the income level of the poorer households and the percentage of their income required for investing in these technologies. Thus, the customer perceived value is at low level when it comes to clean technologies, further along with the subsidy the project will invest in the cost associated with the model which will ensure pre-purchase behavior enforcement and post purchase behavior enforcement among these households.

The project aims to surge use of Clean Cooking Solutions by instigating the concepts of (i) bulk tendering via reverse auctioning (least cost based selection) for cost effectiveness (ii) Output based financing for de-risking of investments and (iii) mainstreaming and capacitating local governments in the renewable energy (RE) sector. The project also intervenes to tackle existing gender related challenges. In order to remove the specific barriers as identified in section B1 and achieve its goal of replacing lower Tier cook stoves and fossil fuel in cooking sector with clean and renewable energy sources. The project focuses on **three core components** as listed below:

COMPONENT 1: SCALING UP THE DEPLOYMENT OF CLEAN COOKING TECHNOLOGIES THROUGH ACCELERATED INVESTMENT AND MARKET DEVELOPMENT

Output 1.1: 500,000 Electric Stoves, 490,000 Tier 3+ ICS and 10,000-biogas plants installed.
Under component 1, the proposed project will support growth of clean cooking solution in *Tera*.

Activity 1.1.1: Development of Annual Procurement and Deployment plan

Based on the demand aggregation of each three CCS technologies, a cluster wise deployment procedure will be prepared at regional or provincial level. Different bidding documents for all three CCS technologies will be prepared to procure the technologies through bulk tendering via reverse auctioning (Least Cost Selection Method). AE being a government entity adheres to the Public Procurement Act, 2007 (PPA) and Public Procurement Regulation, 2007 (PPR). The reverse auctioning model refers to the Least Cost Selection (LCS) Method of the PPR. LCS method ensures that technology is available in the most cost effective way. The reverse auction/LCS mechanism will follow the given structure for procurement of goods as the PPA/PPR.



Figure: Least Cost Selection Process (LCS)

Activity 1.1.2: Procurement and Deployment of annual targeted number of CCS

The project aims to deploy 490,000 Tier 3+ ICS, 10,000 Domestic Biogas Plants and 500,000 ECS over 5 years period. The beneficiary and technology will be selected by a two-step process: 1) eligibility criteria and 2) prioritization. –

Eligibility criteria includes mandatory technical feasibility such as,

- i. Cattle and livestock ownership for biogas,
- ii. 15 Amp meter connection for ECS, and

iii. Fuel accessibility and affordability for ICS.

Prioritization will be made based on socio-economic vulnerability indicators scoring system (such as level of income, disability in the family, female headed households, single women, widows, indigenous people, and affordability factor). Further in terms of the beneficiary selection in case of the respective technologies, the following ecological and economical factors shall be taken in to consideration. The electric cookstoves is targeted towards the urban and peri-urban poor household of the Terai with the average annual income ranging between \$250 to \$2000. The income distribution has been further been presented in section D.4.

Similarly, project location and demand from LG will be subsidiary factors in beneficiary selection. Vendor selection will follow standard Public Procurement Process. Prequalified companies identified by AEPC for specific technologies will be the potential vendors. Least Cost based selection method as per Public Procurement Acts and Regulation will be adapted.

Furthermore, The selection criteria will be defined in detail during the project implementation stage while preparing the standard operating procedure/operational manual for the project.

Following rationales have been linked with the type of technological intervention required:

- 0.5 Million ECS are targeted for households using LPG and fuel wood with reliable electricity access.
- 0.49 Million Tier 3+ ICS are targeted for households using loose biomass, dung cake and fuel wood with no or bad quality electricity access.
- Finally, 0.01 million biogas system are targeted for households having sufficient cattle/livestock and relying on agriculture sector.

Switching from the TCS to Biogas plant for a typical family requires an upfront capital, the subsidized rate, therefore lowers the burden on the household. Further, the longer lifespan of biogas plants adds tremendous value. The annual operational cost of biogas for the households with cattle and live stocks is significantly lower compared to the LPG. In fact our financial and economic analysis has no cost associated with fuel consumption for bio-gas technology as the targeted beneficiaries of biogas technology are envisioned to have cattle at home, therefore providing a free source of animal and agriculture residue. The biogas owners will have the added benefit of fertilizers. The health benefits and the efficiency factor when understood by the beneficiaries will trigger a widespread utilization and replication of the technology.

Similarly, in case TCS and ICS, TCS consumes more fuel (2500-3500 kg per annum) than ICS which consumes (1,200-1500 kg per annum). We have envisioned a higher scale of subsidized rate compared to biogas and electric cook stoves, considering that the targeted beneficiaries will be on the lower strata of financial and economic capabilities. A widespread utilization and replication of the technology will be triggered when beneficiaries understand the fuel wood consumption rate, health benefits and efficiency factor.

Domestic biogas is developed on site locally whereas electric cookstoves, tier 3+ ICS, and accessories of domestic biogas are imported. Pre-tax subsidies are provisioned to import ICS/ raw materials for ICS and Biogas accessories with custom duty discounted to 1%. In 2020, GON has reduced custom charges in induction cookstoves from 15% to 5%, promoting the use of ECS.

Out of the total 284 LGs in the 22 districts, the project targets 150 LGs selected based on demand of CCS. There are about 1.5 million households in the targeted 150 LGs. While selecting the beneficiary households, priority will be given to economically impoverished and socially excluded families, indigenous people, single women, female-headed households and widows.

The annual deployment target is given below:

Technologies	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Electric Cook Stoves	50,000	100,000	135,000	135,000	80,000	500,000
Tier 3+ ICS	50,000	100,000	130,000	130,000	80,000	490,000
Domestic Biogas Plant	2,000	2,000	2,000	2,000	2,000	10,000

In addition to the annual deployment target, the unit cost and the funding proportions of each of the financiers is given below:

Unit cost of the technologies and Cost sharing for technology (Without GCF Support and before Reverse Auctioning)

SN	Technology	Unit Cost (USD)	Cost Sharing			
			GCF	AEPC	LGs	Beneficiaries
1	Tier 3+ ICS	\$ 48.70	-	-	-	100%
2	Biogas	\$ 677.08	-	-	-	100%
3	ECS	\$ 47.92	-	-	-	100%

The Procurement of CCS technologies for this project will follow the following process:

Bulk Tendering via Reverse Auctioning (Least Cost Based Selection Method as per the Public Procurement Act/Public Procurement Regulation of Nepal): The bulk tendering process will be **inclusive and follow bottom up approach** and follow the PPA/PPR. The LG level staffs will collect demand based on type of technologies required at household level and manage data through a Management Information System (MIS)³⁴. MIS will automate the demand aggregation process. This will provide a clear picture of the type and number of technologies required at a particular LG. For the ease of CCS deployment, these demands would be pooled into different clusters, each cluster encompassing different LGs based on geographical convenience. The entire one million CCS deployment target would be broken down into various clusters. Clusters would be aligned to achieve the prescribed annual CCS deployment targets. The project targets have been determined to align with the country priorities. For instance, the Second NDC Nepal has priorities on the clean cooking sector that will be to ensure 15% of the total energy demand is supplied from renewable energy sources; In Residential cooking by 2030,

³⁴ Detail on MIS is mentioned in Activity 2.2.2

ensure 25% of households use electric stoves by 2025, install 500,000 improved cook stoves, specifically in rural areas and by 2025, install an additional 200,000 household biogas plants.

In the event of variances in demands, this will be addressed at the Local Government Level. In case of variances, an informed guess can be made stating the overall demand from the LGs will be on the higher side. Here the beneficiary selection criteria will assess the demands and LGs can offer first come first serve basis with targeted reservations for inclusiveness. Detailed beneficiary selection criteria will be developed as the Standard Operating Procedure is prepared during the project implementation stage.

AEPC will then call for bulk tender via reverse auction as per PPA/PPR to procure the technologies, for each cluster. As there are three different technologies involved, separate tenders would be called for each of the technologies. Based on the demand received, clusters will be formed between LGs in terms of the geographical convenience. The exact formation of these clusters will be decided during the implementation stage and will basically be clusters based on the number of provinces. Further, the quantity for an optimum reverse auctioning price point will be assessed during the procurement. The procurement will be governed by the public procurement act and rules of the Government of Nepal. AEPC will abide by the public procurement rules and regulations to prepare the tender documents. Lowest bidders that comply with all the norms would be awarded the contracts.

The reverse auction approach is expected to bring down the cost of technology significantly, through competition and the economies of scale. It is estimated that the cost of the technologies will further go down by 20% after the proposed intervention. Furthermore, inclusive participation of consumers in choosing the requirement of technology before the auction, ensures quicker adoption and maximum utilization of the deployed CCS technologies.

Unit cost of the technologies and Cost sharing for technology (With GCF Support and after Reverse Auctioning)

SN	Technology	Unit Cost (USD)	Cost Sharing			
			GCF	AEPC	LGs	Beneficiaries
1	Tier 3+ ICS	\$ 38.96	40%	40%	15%	5%
2	Biogas	\$ 541.67	28%	48%	10%	14%
3	ECS	\$ 38.33	10%	48%	19%	23%

Activity 1.1.3: Disbursement of initial 60% payment installment to the vendor upon delivery of CCS to respective municipalities

Mobilization advance amount not exceeding 20% will be provided to the vendors against bank guarantee at this stage. Additionally, the vendor must submit performance guarantee equivalent to at least 5% of the contract value as per the PPA/PPR prior to the contract signing. The performance guarantee will be tied to the quality of the technologies and covers all the technologies ensuring after sales services in the form of guarantees and warranties for the period of one year (defect liability

period). The 20% will be financed by the GCF and AE. A batch verification process will be triggered at this stage. The batch will be tested at the Renewable Energy Testing Station (RETS)

Immediately after the procurement and quality assurance, the technologies will be taken to the respective municipalities. And upon clearance letter from the municipalities, 40 % will be disbursed to the vendors in coordination. A detailed SOP during the later stage can further elaborate on options of financing. The 60% financing from the LGs will be made once the technologies are available at the LG level.

Activity 1.1.4: Monitoring and Verification of Installed CCS units for Output Based Financing (OBF)

Output based financing is another breakthrough approach compare to the BAU process, wherein subsidies are being provided based on installation of the technologies. In the BAU scenario, the vendor was collecting demand and completing installations at households across the country. Post installation, the vendor was then claiming the applicable subsidy amount from AEPC. At this stage, the vendor was released total 90% of the subsidy amount. Post disbursement, AEPC conducted monitoring of the installed systems based on certain sample percentage and released the remaining 10% upon satisfactory assessment or else levied penalties in the cases of non-compliance. In this process, AEPC was minimizing its risk by demanding upfront capital from the vendors. However, it has failed to address the risk of non-utilization of the technology by the households after the installation. This BAU model also had another major ramification. As vendors were required to put upfront capital, they were bearing more risks. As such, they were reluctant to collect more demand, thus resulting in failure for AEPC to achieve their installation targets.

The proposed OBF approach would provide upfront capital for vendors, giving them leverage to service more households. Further, with final disbursement tied to the independent third party monitoring, major chunk of the payment would be released only after the verification of the technologies, which guarantees a quality assurance. This would create a win-win situation for both the vendors and AEPC.

The output based financing approach will have several mechanisms set in place. Firstly, the technical specification of the required technologies will be thoroughly tested. Mobilization advance amount not exceeding 20% will be provided to the vendors against bank guarantee at this stage. Immediately after the procurement, the technologies will be taken to the respective municipalities. And upon clearance letter from the municipalities, 40 % will be disbursed to the vendors. At this stage, total disbursement not exceeding 60% will be provided to the vendor.

The vendors, then in coordination with LGs, will install CCS technologies at the identified households. Simultaneously, a separate third party monitoring process will be conducted parallel to distribution of the technologies wherein the third party will specifically assess appropriate percentage of total beneficiaries in terms of (i) technical parameters of the technologies along with its (ii) mitigation impact through replacement of fuel wood, dung cake and LPG and (iii) adaptation benefits in terms of vulnerable communities and remote areas, among others. The independent verifier will collect disaggregated data based on sub-categories such as gender income status and so on. The third party will be selected via procurement following the PPA/PPR. Request for proposal will be floated to all qualified firms. Once the third party monitoring is completed and confirmation received from the deployed system, final disbursement of 40% will be provided to the vendors. Hence, at this stage, the vendor will be paid 100%

of the total amount. AEPC's experience with various renewable energy technologies including the solar home system, biogas and cookstoves that a substantial change is required in the existing payment approach. The 90-10% approach reduces the ownership of the vendors and does not bolsters the quality and reliability of the technologies, as almost the full payment is disbursed immediately post installation or dissemination. The output based financing approach will shift the paradigm from the existing 90 -10 % payment modality and create ownership of the vendors and engage them throughout the process.

In terms of the beneficiaries' contribution to the project, this will be under the discretion of the LGs. The beneficiaries will either contribute during the installation or during the demand collection. The LGs can also opt for partial or full payments during the various stages. A detailed operational modality will be developed during the implementation of the project.

A pilot study conducted by GIZ/Endev result based financing project³⁵ clearly indicates the need of adequate capacity building activities for successful implementation of CCS project. To address this issue, the project has designed adequate number of capacity building activities. Another scope for improvement is to provide capital investment for the vendors as the limited fund available with them has been limiting their capacity to invest more. Hence, the project will provide up to 60% of advance payment to motivate the vendors.

COMPONENT 2: STRENGTHEN ENABLING ENVIRONMENT THROUGH SECTOR BASED ASSESSMENTS AND QUALITY ASSURANCE OF THE TECHNOLOGIES

Output2.1 Enhancing product standards, conducting assessments, surveys and analysis

- Municipal Energy Plan template prepared
- Standards on technical specifications of all three CCS technologies developed
- A robust quality assurance mechanism is in place

To achieve these outputs following activities will be performed.

Activity 2.1.1: Develop partnership agreement between AEPC, Province Governments, Local Government and other implementation Partners

AEPC will sign a Memorandum of Understanding (MoU) with PGs and LGs. AEPC will facilitate PGs to develop and reinforce policy and strategy in promoting CCS, and as well as support LGs in the project implementation. Similarly, AEPC will also sign MoU with Nepal Academy of Science and Technology (NAST)/Renewable Energy Test Station (RETS) and Nepal Bureau of Standards & Metrology (NBSM) to upgrade existing testing centre at central level, prepare standards and guidelines, and expand after-sales service centre at local level.

³⁵ GIZ/EnDev, 2017. EnDev Result based financing facility <http://www.endev-nepal.org/sites/default/files/publications/2018-02/RBF-project-factsheet.pdf>

Activity 2.1.2: Update existing standards³⁶ and set benchmarks for the technologies required for the project implementation

The existing standards on CCS will be analyzed and benchmarks will be set for the technologies required for project implementation. As standards on induction Stoves³⁷, ICS, and domestic biogas plants have been set previously, these standards will be reviewed and updated.

AEPC have developed an interim benchmark for solid biomass burning cook stoves “**Nepal Interim Benchmark for solid biomass cook stoves (NIBC, 2016)**”³⁸, in a close coordination with the Nepal Alliance for Clean Cookstoves (NACC) members, national and international experts, NBSM and other relevant stakeholders. It categorizes the solid biomass stoves in three different types i.e. chimney stove, chimneyless stove and forced draft stove. This document further needs to **categories the stoves in multitier i.e. Tier 0 to Tier 5 Stoves in terms of type of fuel used in particular stove, their efficiency, total emissions, indoor emissions, safety score, and durability**. As a part of the project activities, this document will be updated if needed, with rigorous interaction and consultative meetings with national stove testing laboratories, NACC, NBSM, and feedback from international practice updates on stove testing.

For ECS, the standard will be developed to cover different types of stoves, including induction, infrared, coil, etc. and their **specific standards on safety and performance, including efficiency, durability, and other performance metrics**. In collaboration with Nepal Electricity Authority (NEA), a further cost effective tariff structure for electric cooking will be introduced analyzing the increment in the revenue due to addition of electric cooking loads to the distribution system during project implementation and facilitation would be done to ensure quality of electric power supply in the terai referring the performance report of delivered technologies extracted from the Management Information System (MIS). Gender inclusive and diversity friendly boards will be formed in the formulation or upgradation of above mentioned normative and standardized tools.

It is to be noted that the NEA will not be involved in any of the procurement, financial arrangement, monitoring and evaluation including project oversight in this project. NEA’s role will be in the area of strengthening of electricity distribution system and improving the quality electricity supply in the areas where electric cookstoves will be deployed. The collaboration in fact will involve coordination between the AE and the NEA will be basically through the exchange of information. The project targets have been determined to align with the country priorities. Nepal’s SDG-7 targets to achieve the electrification rate of 99 percent and the Second NDC aims to strengthen transmission and distribution links to support up scaling of e-cooking, e-heating, e-transport and charging stations. Furthermore, the Second NDC Nepal has priorities on the clean cooking sector that will be to ensure 15% of the total energy demand is supplied from renewable energy sources; In Residential cooking by 2030, ensure 25% of households

³⁶ Standards on Biomass Cookstoves is as per the National Interim Benchmark for Solid Biomass Cook Stoves,(NIBC, 2016) and National Standards for Household electrical appliances has been recently developed in coordination with Nepal Bureau of Standard and Metrology

³⁷http://nbsm.gov.np/uploads/files/2_NS%20xxxx%20Induction%20hobs_Safety_Particular%20Requirements.pdf

³⁸<https://www.aepc.gov.np/uploads/docs/2018-0619Nepal%20Interim%20Benchmark%20for%20solid%20biomass%20CookStoves.%202016.pdf>

use electric stoves by 2025, install 500,000 improved cook stoves, specifically in rural areas and by 2025, install an additional 200,000 household biogas plants. In terms of electric cooking and strengthening transmission, AEPC and NEA being public entities are entrusted by the Government of Nepal to achieve the respective targets. Hence, it is imperative that both the entities fulfill their respective responsibilities to achieve the targets. In 2015, when there was a shortage of cooking gas in the market, many depended on electricity to meet their demand for cooking resulting in an unexpected rise in electricity consumption which has overloaded the distribution system. It is critical to perform regular mapping of distribution systems based on their capacity, and load status in order avoid such situation. Urban and peri-urban areas where electricity consumption is generally low and load centres are often under loaded, are potential start points for the project. This is where the project activities conducted by AEPC will support the NEA. The AE will approach NEA in terms of providing information on upgrading transformers and improving distribution and transmission lines to upgrade and control voltage fluctuations in the project area. This will ensure that NEA are well-informed on the existing capacity and streamline their upgradation at the identified project location.

Activity 2.1.3: Develop Municipal Energy Plan template by analyzing standards and formats

The project will develop a Municipal Energy Plan template by analyzing existing standards and formats.

The MEP will cover in detail the following aspects:

- Energy baseline situation Prerequisites and parameters for electric cooking
- Renewable energy resource inventory
- Opportunities and challenges
- Energy vision, mission and goal
- Strategies to reach the goal
- Energy intervention plan
- Annual target breakdown
- Estimated budget and financing source
- Estimated annual budget
- Time frame

The MEP design takes participatory model with sufficient stakeholder consultation, which primarily includes representatives from Indigenous community, women led enterprises and organizations, key LG officials, service-supply and service-demand stakeholders.

The household situation in terms of adequate infrastructure for ensuring reliable and prolonged application of electric cookstoves will be a key factor taken in to consideration. The existing household capacity is limited to five ampere and it will be critical to upgrade the capacity to 15 ampere to ensure electric cooking is feasible at the households. Furthermore, the additional demand will require enhancement of the existing transformers in the project location. Here, the project proponent will again collaborate with NEA and its regional offices along with LGs to ensure that the required infrastructures are in place. This is critical component of the project which will dismiss the rebound effect to LPG.

Output 2.2 Strengthening quality assurance mechanisms

Outputs:

- A supply chain mechanism for CCS is established and strengthened
- Management Information System database is set up at AEPC

- At least one testing centre is strengthened at central level

Following activities will be performed to achieve the aforementioned outputs.

Activity 2.2.1: Strengthening and establishing testing centre for ensuring the quality of the proposed technologies

AEPC will strengthen the existing testing centre located at Nepal Academy of Science and Technology (NAST)/ Renewable Energy Test Stations (RETS) in Kathmandu. It will also collaborate with Nepal Bureau of Standards & Metrology (NBSM) and other relevant stakeholders to build standards and specifications for Tier 3+ Stoves and Electric Cooking Stoves. The collaboration will ensure gender sensitivity. The sub activities under this includes:

- Lab upgradation (entails equipment purchase and training) and improved testing service as per ISO standards for ECS, and upgradation of National Interim Benchmark of improved Cookstoves 2016 inline to tier based criteria and improved testing service
- Development of Electric Cookstoves and Tier 3+ICS standards on performance and safety in collaboration with National Entities, Local Governments, Universities, Agencies working in Clean Cooking Solutions through Seminars/ webinars/ symposiums. Relevant stakeholders including RETS, AEPC, LG, beneficiaries, sub-national institutions, suppliers, repair and maintenance service providers oriented on the developed standards and specifications of the technologies.

Activity 2.2.2: Development of Management Information System as a national monitoring system with real time data collection from LGs

Development of Management Information System is another critical element of the project wherein the demands from various LGs are recorded and aggregated, and the distribution/installation is regularly updated along with performance of the disseminated technologies. As AEPC is in process of developing a MIS accommodating local and provincial level, a fit gap analysis will be conducted of existing MIS and assess implementation of the modules required for the project implementation. In addition, training will be provided to LG personnel on real time data collection and operationalization of MIS. MIS will mainstream gender and sensitivity in data collection and its design.

COMPONENT 3: EMPOWERING INSTITUTIONS, CAPACITATE SUPPLY CHAIN AND ENSURE INCREASED ACCESS TO CLEAN COOKING SOLUTIONS

After almost three decade-long absence of locally elected representatives, the local level elections were conducted in 2017. Most of the Local Governments, and all of the Provincial Governments, are newly established, and lack skills and resources. They have very limited knowledge on renewable energy.

Output 3.1: Capacity development of sub-national institutions

Outputs:

- Municipal Energy Plan developed by at least 150 LGs
- 4 Energy Unit at Provincial level with 2 Officers (1 Technical and 1 Coordinator) and at least 150 CCS Mobilizer will be allocated at local level
- Demand aggregation and cost-effective reverse auction process introduced in MEP of local government with priority to CCS promotion

To achieve these outputs following activities will be performed.

Activity 3.1.1: Conduct national stakeholder consultation workshop between AEPC, PG and LG

AEPC will conduct gender inclusive national level stakeholder consultation workshop with PGs and LGs to plan their approach in Implementing project activities.

At provincial level, 4 energy units will be set up with 2 energy officers (1 technical officer and 1 CCS coordinator) at each province, preferentially women targeted. The technical officer will oversee the technical issues, support service providers and vendors, and the CCS coordinator will oversee project activities and coordinate planning at province or regional level. At LG level, the project will allocate at least one CCS mobilizer, preferentially women, in each local government and capacitate them to facilitate the project particularly for door to door awareness campaigns on benefits of CCS, collection of demands, type of the technologies required at household level and data management through the MIS. Also, a monitoring plan will be drafted at this stage for LGs to support monitoring CCS in their respective LGs.

Activity 3.1.2: Develop LG Municipal Energy Plan

Municipal Energy Plans would be developed to identify best available resources at local level for energy generation and to assess consumption pattern of households. The MEP design takes participatory model with sufficient stakeholder consultation, which primarily includes representatives from Indigenous community, women led enterprises and organizations, key LG officials, service-supply and service-demand stakeholders. MEPs will also be instrumental in indicating potential energy resources and in selecting appropriate technology options to implement clean cooking initiatives. This will create an environment wherein MEPs will become a critical element of the annual budgeting and planning process and hence ownership of clean cooking initiatives by the local governments. While preparing MEPs, complementarities will be ensured with other ongoing program/projects.

The process to prepare MEP include following nine steps³⁹:

Step 1: Identify Stakeholders to ensure a broad level of engagement from a cross-sector of energy players.

Step 2. Establish an Energy Sub-Committee Team to lead development of energy plan, and to implement the plan and monitor progress. **Step 3. Develop Energy Baseline to provide a holistic and updated picture of the energy situation in a Municipality and assess the prerequisites for electric cooking**

Step 4. Develop Energy Vision, Mission and Goal to reflect overarching commitment and purpose.

Step 5. Develop Activities in collaboration with the Project, NEA with its regional offices and other stakeholders ensuring inclusiveness to achieve the energy vision, mission and goal.

Step 6. Evaluate and Prioritize Activities to remain focused, resource-efficient and address immediate energy needs.

Step 7. Identify Funding Source to ensure smooth implementation of prioritized activities.

Step 8. Compile and Finalize the Plan to compile all information from all planning stages into a single plan document that will be approved by a Municipality Council.

Step 9. Implementation and Monitoring & Evaluation to implement a plan assesses the progress and formulates necessary modifications to ensure maximum impact.

³⁹ AEPC 2020. Guidelines - Preparation of Municipal Energy Plan (MEP)

Activity 3.1.3: Capacitating Local Governments and Provincial Governments

AEPC will capacitate LGs to prioritize CCS promotion in periodic and annual plans and to formulate appropriate policies and mechanisms. The project will coordinate with the Provincial Governments to create an enabling environment for the project at province level and will support the strengthening of capacity of the Provincial Governments to perform their coordination role.

Also, the project will assist LGs in placing dedicated personnel related to environment safeguard and gender in CCS projects and in conducting gender sensitive monitoring and evaluation. Besides conducting awareness campaigns for behavioral change, LGs will be capacitated in use of MIS for recording transaction values from demand generation to implementation. Further, resources supported through the project will be responsible for demand collection, establishing commitment mechanisms on beneficiaries' side and regular support throughout the project implementation process. The training and promotional materials will be ensured to address gender-mainstreaming and will ensure that the information is gender-sensitive.

The project will build capacity of LGs to engage with their local Community based Forest User Groups to procure biomass at local level and process them locally to ensure swift and cheap supply of fuel for Tier 3+ ICS.

Activity 3.1.4: Annual national level review and consultation meeting

AEPC will collaborate with its outreach partners at the district or regional level to review the annual objectives and targets and improve MEP guidelines and project materials as required. Sex-disaggregated and gender relevant data used for correction of measures aiming to hold all partner accountable to gender equality will be collected and analysed. Then the GAP will be revised and reviewed annually.

Output 3.2: Increased awareness and outreach to enhance demand

Outputs:

- A total of 450 clean cooking champions identified, three at each local government
- 750 awareness campaigns conducted (Once in each LG per year for five years)
- Climate resilience of the people and community enhanced with the savings and awareness on climate change adaptation measures

Campaigns and promotional activities in coordination with PGs, LGs, community champions, non-government organizations and volunteers will ensure that the prospective users have increased awareness on the benefits, technology, services and financing of clean cooking solutions. The activities will include:

Activity 3.2.1: Mobilize clean cooking mobilizers in each LG

At least 150 CCS mobilizers, preferentially women will be mobilized in the project area. The CCS mobilizer will be trained to help in raising awareness on benefits of CCS, facilitate in bottom up need assessment and in designing bespoke technical assistance and capacity development programme for respective LGs.

The selection criteria for clean cooking mobilisers includes:

1. Minimum of a year experience in community mobilization/campaign and renewable

- energy will be given priority consideration
2. Experience in advocacy of health and socio-economic benefits with good understanding of the possibilities and tactics for influencing decision
 3. Ability to communicate effectively with a wide range of audiences at local, provincial and national levels
 4. Ability to build and maintain relationships with partner agencies and key contacts in the government, NGO and civil society.
 5. Fluent Local language skills
 6. Basic English language skills
 7. Priority given to women for the role and will have at least equivalent gender representation

Activity 3.2.2: Development of training materials

The consultation workshop will be conducted at national level between corresponding ministries, Agencies, AEPC, LGs, beneficiaries, CCs champions, and local vendors and private sector service providers to identify the need and to develop materials/manual/guidelines on CCS awareness to local communities. Programme documents, training materials, promotional materials will be revised to ensure effective gender-mainstreaming, and that the information is gender-sensitive. This will include:

- I. Development of extensions materials and guidelines for implementing CCS awareness in targeted project areas
- II. Drafting mechanism to effectively address gender equality and supporting vulnerable communities through CCS

Activity 3.2.3: Identifying clean cooking champions

At local level, the project will identify 450 clean cooking champions (60% women targeted) (3 at each local government) who will be trained to provide installation, repair and maintenance services of the technologies and will also be the advocates of technologies. The champions will also provide support to CCS Mobilizers.

The approaches for identifying clean cooking champions includes:

1. Stakeholder mapping exercise will be carried out at local level to identify the relevant stakeholders such as local NGOs, Community Based Organizations (CBOs), Forest User Group (FUG), and Dalit Associations, female community health volunteers, etc. To make the selection process more participatory, an Interaction programme with such relevant stakeholders will be carried out and Terms of Reference (ToR) for clean cooking champions will be developed
2. Some clean cooking champions can be listed out from the MEP prepared by the Local Government.
3. In addition, the potential clean cooking champions can be identified during the demand collection for CCS.
4. Also, some potential clean cooking champions will be identified through the feasibility study.

Activity 3.2.4: Develop materials for awareness on CCS (in national and local languages) with its impact potential on livelihoods and support to vulnerable communities:

The project will develop awareness materials on national and local languages which will underpin the activity 3.2.5. The awareness materials will consist of information, education and communication materials on the benefits of CCS in financial savings, cleanliness, health benefits and drudgery

reduction compared to TCS. Awareness material will be in form of information on CCS through mass media (Local FM, Electronic media, Print Media, Social Media), consultation meetings, and distribution of leaflets, user manuals and door-to-door visit to educate and empower users.

Activity 3.2.5 Campaigns and promotional activities in coordination with the PGs, LGs, community champions, NGOs and volunteers

The project will support and empower users by CCS mobilizers and clean cooking champions for awareness raising on climate change impact, highlighting the benefits of CCS such as fuel saving, cleanliness, health benefits and drudgery reduction compared to TCS., by disseminating information on CCS developed by activity 3.2.4. The awareness programme will mainly focus on marginalized communities and households ensuring at least 60% of female participants. CCS mobilizers and champions will conduct behavioral change programme through awareness activities, Operation and Maintenance manuals and cooking guides using CCS.

AEPC would coordinate with NGO like Renewable Energy Confederation of Nepal (RECON), National Association of Community Electricity Users-Nepal (NACEUN) and Community Rural Electrification Entity (CREE) in electric cook stove awareness raising programme.

Activity 3.2.6: Ameliorate awareness materials and guidelines

AEPC will conduct annual national review and planning meeting with AEPC's outreach partners at the district or regional level. At this stage, the awareness materials and guidelines will be improved if required.

Output 3.3: Strengthening service centres, biomass manufacturers to provide quality and affordable clean cooking solutions

Outputs:

- 24 number of trainings provided to beneficiaries in collaboration with Council for Technical Education and Vocational Training(CTEVT) institutions on repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring (training conducted at province level; one training per province per year from year 2 to year 5)
- 24 Province level trainings provided to service centres or providers in collaboration with CTEVT institutions (trainings conducted at province level; one training per province per year from year 2 to year 5)
- 4 provincial service centres established and strengthened
- 900 local service centres established and strengthened with at least six service centres established and strengthened at each Local Government

One of the major barriers for clean cooking sector growth is **inadequate technology and frail CCS supply chain industry**. The project aims to strengthen each element of CCS supply chain (production, distribution and after sales service), biogas installers, electrical wiring service providers, pellet industries, private investors, entrepreneurs and financial institutions.

Activity 3.3.1: Capacity and training needs assessment

AEPC will conduct national level stakeholder consultation workshops with AEPC/NAST/RETS/NBSM, beneficiaries, local manufacturers, vendors, financial institutions, LGs, training institutions (e.g. CTEVT)

and other relevant stakeholders to assess existing situation, capacity and training needs assessment of service centres for providing CCS services. Similarly, based on consultation with the participants, necessary changes will be made in existing training manuals and new training manuals and guidelines will be drafted to support local manufacturers.

Activity 3.3.2: Ameliorate project materials and guidelines

AEPC will conduct an annual national level coordination and sharing meeting with the project partners and review the progress, draw lesson and planning with AEPC's outreach partners at district or regional level. At this coordination meeting, the project materials and guidelines will be improved.

Activity 3.3.3: Establish and strengthen service centres at provincial and local level

In collaboration with Nepal Academy of Science and Technology (NAST)/ Renewable Energy Test Stations (RETS), AEPC will strengthen the existing provincial service centres and capacitate them to work as AEPC focal point for their respective local service centres. In addition, the project will establish and strengthen 900 local service centres, with at least six service centres at each Local Government, and capacitate them to provide after sales services.

Activity 3.3.4: Conduct trainings to service centre operators and local manufacturers in collaboration with financial and CTEVT institutions

A total of 24 Province level trainings will be provided to service centres/providers in collaboration with CTEVT institutions. At each province (6 provinces), one training per year will be conducted for 4 years (from 2nd to 5th year). At least 50 persons will be trained at each training with total of 1200 persons trained by the end of the project.

Capacity development trainings will be provided at local level to the service providers, local manufacturers, mobile and other electric appliances repair centers and metal workshops to capacitate them in providing after sales service, domestic production and maintenance of CCS and strengthen the supply chain in coordination with CTEVT institutions. Besides, training on business performance and business continuity plan will be provided to manufacturers and service centres. Capacity development programme will be conducted to promote innovation and R&D. In case of ECS, a technical network will be created to minimize accident such as short circuit.

Activity 3.3.5: Conduct skills development training to CCS beneficiaries and vulnerable groups in collaboration with CTEVT at Province level

A total of 24 Province level trainings will be provided to **CCS beneficiaries and vulnerable groups** in collaboration with CTEVT institutions. At each province (6 provinces), one training per year will be conducted for 4 years (from 2nd to 5th year). At least 50 persons will be trained at each training with total of 1200 persons trained by the end of the project.

Training on making fuel for Tier 3+ ICS, repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring will be provided to **CCS beneficiaries and vulnerable and disadvantaged groups (DAG)**. Training will be provided to women (electric cook stove users) on electrical safety and basic technical skills on using electric stoves to increase their confidence and the use of stove. Self-employment skilled training related to CCS and others as per the local demand will be provided to the most vulnerable people such as widows, single women, Female headed households,

minor ethnic groups, lower caste people (*Dalit*) and indigenous group (*Janjati*), ensuring at least 33% female participation.

Activity 3.3.6: Strengthen existing regional level service centres under AEPC's past CCS projects into fully operational province level service centres

The existing regional level service centres developed as part of previous CCS programmes of Nepal, will be assessed and strengthened to function as province level service centres. The province level centres will be the focal technical point of contact for all local level service providers. These service centres will have the capacity to provide maintenance and repair services to CCS users as well as local service providers of all CCS technologies.

Referring to the feasibility study, describe why this set of interventions was selected instead of alternative solutions and how the project/programme can help unlock the needed support in a sustainable manner. Also identify trade-offs of the selected interventions, if applicable.

The proposed activities are demand driven and have been developed through consultations with the project target groups, beneficiaries and other relevant stakeholders.

AEPC has already implemented clean cooking technologies (biogas and ICS) in several areas of *Terai*. The set of proposed activities are defined based on experiences and learnings from the past. With a view of scaling up good practices and the tested models, more weightage have been provided on capacity development of Local Governments and after sales service activities, rather than just focusing on distribution of the technology. Furthermore, the proposed reverse auctioning via bulk tender would reduce the cost of technology in comparison to procurement via existing fixed rate subsidy.

Similarly, after the completion of project time period i.e. withdrawal of GCF funding after five year, at that stage the market for ICS, ECS and Bio gas plant will increase. The population will understand the beneficial aspects of the technology (i.e. both financial and technological benefits) which will increase the intake further. The provincial energy unit is part of the project management component. In the case of the one central level testing centre. Renewable Energy Testing Centre will be enhanced to ensure the quality assurance of the proposed technologies, especially in the case of the ECS. In the case service centers both provincial and local, this will be under the leadership of the private sector. The project will trigger a market enabled approach thus ensuring the sustainability of these service centers post the project interventions and will no further require additional support from other donors and the Government

For Enhanced Direct Access (EDA) proposals and projects/programmes with financial intermediation (loans or on-granting), describe the selection criteria of the sub-project and types.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Provide a description of the project/programme implementation structure, outlining legal, contractual, institutional and financial arrangements from and between the GCF, the Accredited Entity (AE) and/or the Executing Entity (ies) (EE) or any third parties (if applicable) and beneficiaries.

Provide information on governance arrangements (supervisory boards, consultative groups among others) set to oversee and guide project implementation. Provide a composition of the decision-making body and oversight function, particularly for Enhanced Direct Access (EDA) proposals

The institutional framework for this project will engage with a wide range of stakeholders, ensuring their sustained engagement from project design to implementation, and for monitoring and evaluation (M&E). AEPC will implement the project, in close collaboration with a wide range of stakeholders, including Local Governments and Private Sector. The project would be executed by and under the technical and administrative supervision of AEPC, following the Government of Nepal's rules and procedures regarding the acquisition of project services, supplies, and works. The project work plan and budget will be approved annually by the Government. The government's budget process is stipulated by the Constitution of Nepal and has a fixed annual timeline for 15th day of the Nepali month Jestha (generally falls during the final days of the month May). Prior to the national budget speech, rigorous exercise is conducted at all levels of the government and among all the concerned government entities. The Ministry of Finance sets budget ceilings and negotiates with the respective line Ministry. There are no delays in the budgeting process and it will not affect the project implementation. Furthermore, the No Objection Letter from the Government of Nepal ensures that the project credibility and the government commitment.

AEPC will perform both functions of Accredited Entity (AE) and Executing Entity (EE) and will have a separate Project Implementation Unit to execute the project. A dedicated Project Implementation Unit will be set up at AEPC, and will be guided by a Project Management Committee. Four Provincial Project Management Unit will be established with two (one technical and one coordinator) staff at each unit. The associated costs for the activities are reflected under project management cost.

The project will engage with private sector, as a vendor, to provide CCS technologies and services, whereas Local Governments will play an important role in planning, demand collection, awareness raising for behavioral change and monitoring of the project.

The project will ensure a just and inclusive energy transition. It mainstreams the gender and equity aspect by uplifting the women of Terai Region to ensure their involvement in the technical and community outreach activities related to the project. The project aims to provide skill development training to disadvantaged groups, vulnerable communities, women group, single women to capacitate them in repair and maintenance of the cook stoves.

Ministry of Energy, Water Resources and Irrigation (MoEWRI): The MoEWRI is the line ministry of AEPC. It will provide policy guidance and strategic support. During inception phase of the project, MoEWRI will review and approve the Standard Operating Procedure prepared by AEPC for project implementation. It provides strategic guidance and coordination among various partners and stakeholders.

Compliance and Ethics Subcommittee: AEPC has a Compliance and Ethics Subcommittee, which provides complete oversight functions and ensures accountability of the project.

Project Management Committee: The committee will be chaired by AEPC Executive Director and its membership will include the head of each department of AEPC, and representatives from private sectors and technical institutions such as NAST and RETS. Member secretary will be the head of a

particular department of AEPC that manages the project. The management committee will ensure the collaboration among the partners and stakeholders that involve in project implementation.

Project Implementation Unit /AEPC: A dedicated PIU will be established within AEPC to oversee and manage the project. Under strategic guidance of the project management committee, PIU will be responsible for day-to-day management and monitoring of the project activities according to approved annual work plans and budgets. . PIU will ensure that the results of the project are achieved. PIU will engage private service providers to provide services to the project beneficiaries in installation of CCS and after sales services. PIU will be responsible to provide technical and capacity development support to Local Governments. In the inception phase, the PIU will develop a detailed work plan and monitoring framework for the full project duration and get approval from competent authority. Further, PIU will develop guidelines and procedures required to implement the project effectively. These documents will outline the detailed implementation procedure of the project including roles of all responsible institutions and project stakeholders, including in terms of procurement, monitoring, reporting and auditing. The project would be executed by the PIU, following the Government of Nepal's rules and procedures regarding the acquisition of project services, supplies, and works.

Provincial Project Management Unit: The project area touches six Provinces of Nepal, however to optimize the resource allocation, only four Provincial project management unit will be established that shall cover all the local governments. One technical officer and one CCS coordinator would be deployed at each provincial project management unit. The technical officer will look after the technical issues, support service providers and the vendors, and the CCS coordinator will oversee the activities and coordinate planning at province or regional level. The province level staff will not be limited to only one province, but will also support nearby LGs of adjacent provinces. Provincial Governments will be responsible for coordination among provincial level stakeholders in implementing the project. The project will coordinate with the Provincial Governments in creating an enabling environment for the project at province level and will support in strengthening capacity of the Provincial Governments in performing their coordination role.

Local Governments (i.e. Municipalities/Rural Municipalities): Local Government plays a very important role in the project implementation. AEPC will identify at least 150 LGs for the deployment of CCS, where LGs would also contribute partially in funding the technologies to their respective households. Local Government will primarily be responsible for Municipal energy planning, CCS demand collection, raising awareness on benefits of CCS and monitoring of the project progress. At LG level, the project will allocate at least one CCS mobilizer in each local government They will collaborate with their respective LGs for the project implementation, and help LGs to develop their MEPs. Overall, LGs will be instrumental in supporting PIU to implement the project activities at the ground level.

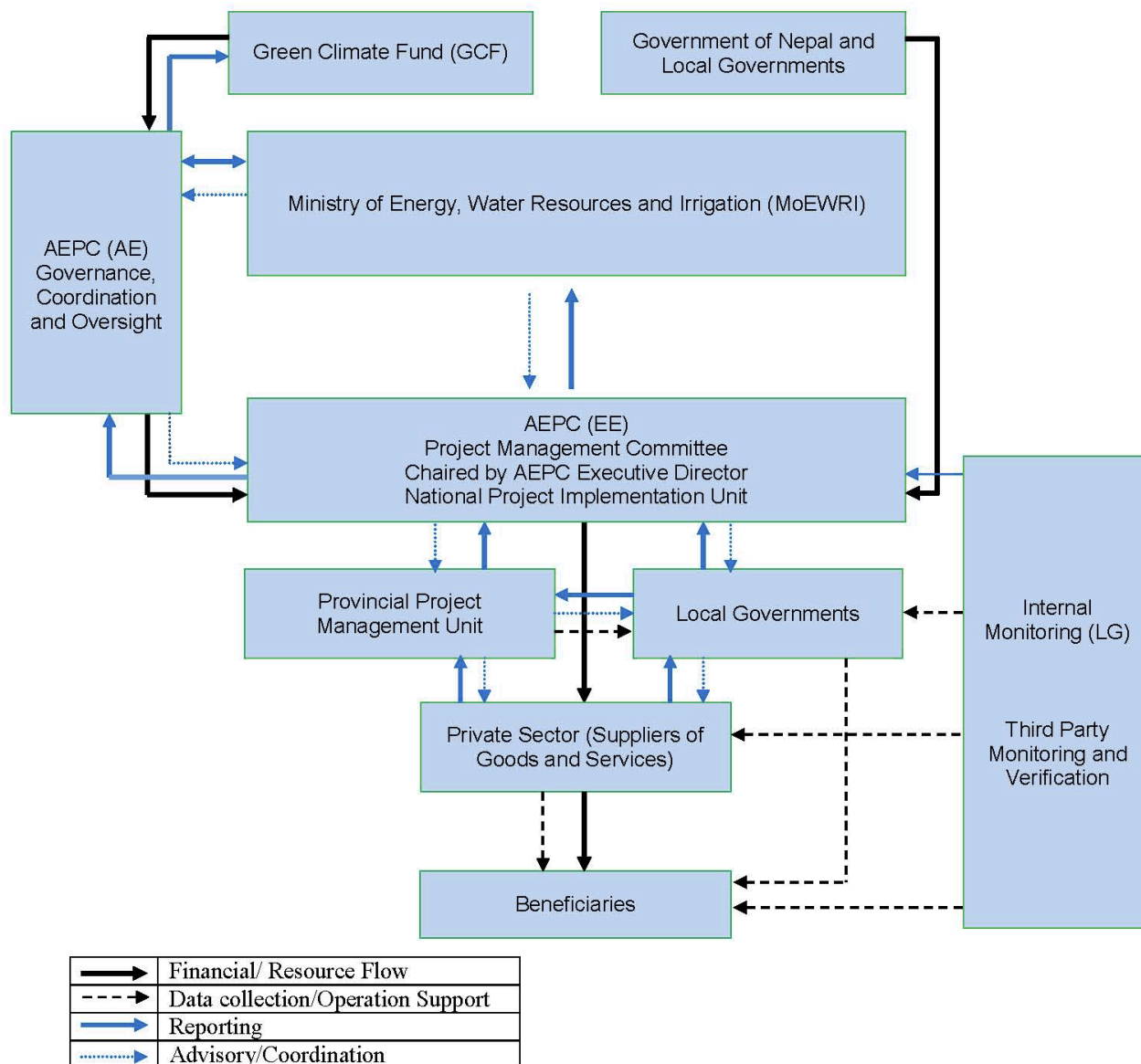
Vendors: Vendors contracted through **bulk tendering and reverse auctioning process as per PPA/PPR**, will be responsible for providing services on CCS, particularly the distribution and installation of the CCS Unit and after sales services to the beneficiaries. The vendors interested to procure CCS technologies in the bulk tendering process will participate in tender calls of AEPC. The selected vendors then in coordination with LGs will set-up distribution and installation of the technologies as per the local demand. The supply chain will be formed in coordination with AEPC and LGs.

Internal and Third Party Monitoring Agency: Direct monitoring of delivery of CCS will be done by an internal monitoring mechanism formed at LGs level. An independent third party monitoring agency will be hired for outcome and impact monitoring. The third party will specifically assess and verify appropriate percentage of the total beneficiaries in terms of:

- The technical parameters of the CCS technologies,
- Quality and standard of the CCS technologies,
- Mitigation impact through replacement of fuel wood, dung cake and LPG and
- Adaptation benefits in terms of vulnerable communities and remote areas, among others.

Project Beneficiaries: The project beneficiaries are the ultimate target groups of the project and will receive services from private service providers and LGs on CCS technologies. Further, the beneficiaries will receive technical assistance and capacity development services that will help to reduce their vulnerability and build resilience to climate change impact through the time saving, health benefits, gender empowerment and increase in agricultural productivity by using cow dung as fertilizer, by replacing its use for cooking fuel. Further details of the project beneficiaries and the selection criteria have been defined in section B.3 Project and Program Description.

Figure 4: Project Implementation Modality

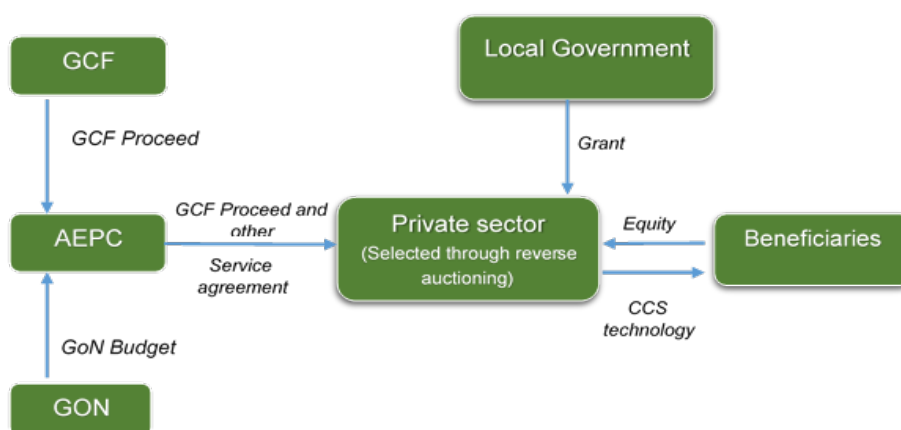


Provide information on the financial flows and implementation arrangements (legal and contractual) between the AE and the EE, between the EE or any third party and beneficiaries. For EEs that will administer GCF funds, indicate if a Capacity Assessment has been carried out. Where applicable, summarize the results of the assessment.

For financing the project, GCF will enter into the **Funded Activity Agreement (FAA)** with AEPC as an AE. As the Executing Entity is internal to AEPC, Legal agreement between AE and EE is not required. Government of Nepal's financial system, public procurement procedure and fund flow mechanism will be adopted for this project. As an accredited entity, AEPC will receive the fund from GCF as per the Nepal government procedures. A separate bank account will be opened for the project purpose and a separate financial management guideline will be developed for the transparent and effective management of the financial resources. The PIU will prepare annual work plan and budget, including a procurement plan, which will be approved by the Project Management Committee. Based on the approval of this plan, PIU will mobilize and release the fund according to the agreements/contracts to the service providers.

AEPC will call tender for each CCS technology and then will select a vendor for bulk purchase. An advance payment, not exceeding 20% of the total cost, will be provided to the vendor against bank guarantee as per PPA/PPR. Additionally, the vendor must submit performance guarantee equivalent to at least 5% of the contract value as per the PPA/PPR prior to the contract signing. The performance guarantee will be tied to the quality of the technologies and covers all the technologies ensuring after

Figure 5: Financial Flow



sales services in the form of guarantees and warranties for the period of one year (defect liability period). The performance bank guarantee has to be provided by the vendor and will be tied to the quality of the technologies and covers all the technologies ensuring after sales services in the form of guarantee and warranty for the period of one year. After the CCS technology will be deployed, the total amount not exceeding 60% will be disbursed to the vendor by AEPC, LG and beneficiary. However, AEPC will hold 40% of the final payment, which will only be paid after the independent third party monitoring and verification. The equity from the beneficiaries will form part of the technology payment. The portion of equity varies according to the technologies as indicated in section B.3, Component 1. The equity portion from the beneficiaries will be provided to the private sector after installation of the given technologies. The equity from the beneficiaries is not reflected in the total budget of the project.

Describe the experience and track record of the AE and EEs with respect to the activities (sector and country/region) that they are expected to undertake in the proposed project/programme

AEPC, under Ministry of Energy, Water, Resource and Irrigation, is the national focal agency for promotion of renewable energy technologies in Nepal. They have been implementing different programmes and projects under renewable energy portfolio with a focus on scaling up renewable energy technologies and achieving the objectives of the United Nations “Sustainable Development Goals” (SDGs) and “Sustainable Energy for All” (SE4All) initiatives while reducing dependence on traditional and imported energy by increasing access to renewable energy to improve the livelihoods of people and create employment opportunities, especially in rural areas.

AEPC is also a technical arm of the Ministry on climate change related activities. Established in 1996, AEPC was set up to promote the use of alternative/renewable energy technologies to meet the energy needs in Nepal. Mandated as the focal agency in the area of renewable promotion and energy

efficiency, AEPC's activities include renewable energy policy formulation and development, and implementation of projects. AEPC has implemented projects and programmes in the area of Climate change mitigation and adaptation, RE promotion and Low carbon economic development worth more than 328 million USD up until 2017.

AEPC has implemented several programmes and projects in coordination various development partners. Some of the projects implemented up until 2017 are

1. Rural Energy Development Programme (REDP) - Phase I, II & III of 33 Million USD in collaboration with UNDP, World Bank
2. Biogas Support Programme (BSP)- Phase III, IV & V of 21 Million USD in collaboration with KfW, SNV
3. Energy Sector Assistance Programme (ESAP) - Phase I&II of 84 Million USD in collaboration with Denmark, Norway, KfW, DFID
4. Renewable Energy Project (REP) of 16 Million USD in collaboration with EU
5. Climate and Carbon Programme- Phase I & II of 1.54 Million USD in collaboration with DFID, SNV
6. Kabeli Transmission-REES of 1.2 Million USD in collaboration with World Bank and
7. National Rural and Renewable Energy Programme (NRREP) of 171 Million USD in collaboration with Denmark, Norway, WB, ADB, UNDP, UNCDF, SNV, GIZ, UNESCAP, DFID, KfW.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Explain why the project/programme requires GCF funding, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?

The GCF funding is crucial in addressing the barriers in CCS sector in the demand and supply side: in terms of awareness, access to finance, regulatory frameworks, quality assurance of electric Stoves, ICS and biogas, strengthening supply chain, and empowering institutions to create an enabling environment for CCS in long-term.

Government of Nepal investment independently cannot address the climate change mitigation needs: Nepal, with very limited resources, has not been effective in expediting finance for low carbon and climate resilient projects and programmes with its fiscal and monetary policy. On one hand, the financial sector of Nepal has not prioritized lending for this sector; on another hand, their hefty interest rates, up to 18% by cooperatives⁴⁰, have been detrimental in expanding CCS market in Nepal.

GCF support will be noteworthy for empowering vulnerable communities: Lower Tier users are willing to switch to Tier 3+ ICS, however they can't afford to pay the capital requirement to switch to more expensive Tier 3+ ICS. They are very poor and most of them are from socially excluded groups. They do not have access to banking facilities to borrow money. In addition, there are no loan products for CCS technologies catered by the financial institutions. Therefore, the project provides financial support for the proposed CCS technologies, which will allow vulnerable user communities to transition to better CCS technologies with high returns.

⁴⁰GIZ, 2018. Final Report on Gauriganj Municipal Energy Situation of Province 1

GCF investment will allow the creation of an inclusive CCS market and a supply chain ecosystem: There is a lack of appropriate financing mechanism to incentivize the private sector and de-risk their initial investment in renewable energy sector. The existing subsidies model, where the vendors (or PQ companies) has the task to collect demands door to door and invest 100% capital upfront, is not financially motivating and therefore, has not achieved good results. Hence, the proposed bulk tendering, which ensures a certain volume of demand, and output based financing that would spread the risk between the vendors and AEPC, would alleviate the confidence of vendors to invest more in the CCS market.

Currently there are not many manufacturers and vendors of Tier 3+ ICS and ECS in Nepal. Moreover, the existing ones are also concentrated in very few urban centres only. There are not many repair and maintenance centres either. There are not many trained skilled human resources. Hence, the GCF support is required to strengthen the supply chain ecosystem, so that the CCS technologies become easily available throughout the country, with well-functioning after sales service centres.

The combined investment support from GCF will help to achieve the GoN national targets in terms of CCS: Current subsidy delivery mechanism is a very bureaucratic and lengthy process, which makes it difficult for the government to achieve big targets in a short time. Furthermore, prequalified companies are responsible for identifying the households who need subsidy, and they have not been able to meet the national targets.

GCF support is instrumental in accelerating climate change mitigation and adaptation needs: Under BAU subsidy model, AEPC targets to install 20,000 ICS and 25,000 domestic biogas installations as per its annual plan⁴¹. Moreover, it has no concrete program for deploying ECS. With this new approach of bulk tendering via reverse auction, CCS can reach 200,000 households annually, which will increase the pace of climate adaptation and mitigation.

Reverse auctioning (RA), ideally, escalates the supply base competition, which leads to lowered price for buyers. For instance, IEC certified, China-manufactured cookstove⁴² has an MRP of 21-22 USDi in China. The product incurs 5% customs duty & 13% VAT (and other logistic expenses) and has an MRP in the range of USD 33-40 landed in the Nepalese market. Further, the retail price for various brands of single hob 2000W induction cook stoves varies between USD 38-90 depending on the brand and quality of the product. Reduction in price for the end-user is expected between 15-20% with no compromise in the quality. Similarly, the Nepalese manufactured TIER 3 ICS having MRP of USD 7743 is reduced to USD 38 for end user in reference to Reverse Auctioning. As such, RA reduces the end-use pricing as winning bid effectively captures required quality and generates supply base competition.

The Project emphasizes increasing **awareness for behavioral change that will lead to an increase in demands for CCS**, which will cater to increase in investment from the private and banking sectors. GoN and beneficiaries will share about 60% of the cost of the project.

⁴¹<https://www.aepc.gov.np/uploads/docs/progress-at-glance-a-year-in-review-fy-207576-2018191572879503.pdf>

⁴²IEC Certified 4 star rated Chinese Manufactured 'AILIPU' Single Hob 2000W Induction Cookstove

⁴³Alternative Energy Promotion Centre Maximum Retail Price on Biomass Technologies

The financing from GCF **not only support the implementation of low carbon and climate resilient technologies in every household cooking of Nepal but set a precedence by introducing an avant-garde implementation mechanism- reverse auctioning, output based financing, renewable energy MIS and mainstreaming LGs role in the renewable energy.** This will create a massive change in the business-as-usual process of financing renewable energy technologies in Nepal and the government will have a tried and tested model that can be replicated all over the country.

The project involves **low risk and yet yields high return** in terms of economic benefits for GCF in the form cost saved for cooking for beneficiaries, reduced carbon emissions, improved health and time saved for beneficiaries, as well as reduction in deforestation.

Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.

With regards to the selection of the financing instrument, during an activity related to the rehabilitation of earthquake affected biogas technology and distribution of small solar home system, AEPC have had a small scale experience with the bulk tendering process. The process though had proven beneficial, it could not be replicated due it's nature associated with disaster risk recovery. Bulk tendering needs to be mainstreamed and the support from GCF will be crucial in instigating the process. The bulk tendering approach is a transition from existing financial mechanism such as subsidy given to beneficiaries and vendors. In terms of reverse auctioning, as mentioned in the FP, the process needs to be aligned with the Public Procurement Act/Regulation (PPA/PPR) of the country. Hence for the alignment with the PPA/PPR, reverse auctioning aligns with the least cost based selection process. Least cost based selection process will ultimately ensure the technology is procured from the lowest bidder. This will create a competitive environment that will result in better valued products attracting more consumers to adopt CCS.

Both of the processes were chosen to ensure creation of a market place for the vendors and better control over the technology in terms of quality assurance. Bulk tendering will mean that the technologies will be procured at a singular cost. The marketing for the vendor is done by the government which means no additional overheads for the vendors. The output based payment approach ensures the cost recovery at the final stage which results in vendor engagement throughout the process and reliability and sustainability of the proposed technologies. Furthermore, the successful implementation of the financing instrument will result in a replication of the process in the other technologies such as solar home systems. This will create an impact at an unprecedented scale.

In the fiscal year 2018/19, AEPC conducted CCS programmes worth \$4.58 million wherein approximately 98% budget was allocated for technology deployment. Through this project, AEPC will address the shortcomings in implementation of CCS programmes of BAU approach by reducing the resource limitation. By conducting activities for accelerating climate change mitigation and adaptation needs, creating a market and supply chain ecosystem, mitigating the risk of implementing an innovative

approach, implementing innovative approaches of demand aggregation and mainstreaming LGs proposed in the project. The remaining GAP identified was \$ 20 million dollars. The internal capacity of AEPC can absorb this grant without stressing the internal working mechanisms.

The co-financing amount is based on the expected annual budget available to AEPC from MOEWRI/GoN for conducting programmes on CCS in BAU scenario, \$4.58 million budget was AEPC's spending in fiscal year 2018/19 and \$5 million was allocated in fiscal year 2019/20 for conducting CCS related programmes and maximum achievable financial leverage possible due to GCF's concessionality and the project.

The current modality of financing of \$20,958,901 by AEPC as co-financing and \$21,127,474 grant by GCF was derived based on the experience of AEPC and its partnership with development partners.

AEPC has allocated 93% of its project budget contribution in component 1, in contrast to around 52% of GCF's project share. Meanwhile, AEPC seeks contribution entirely from the GCF grant on the components 2 and 3, i.e. 100% investment done by GCF for these two components. Therefore, co-financing contribution is maximum for technology deployment whereas, GCF support will be fully utilized for strengthening the supply chain capacity and capacity of the local government to create the enabling environment.

Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?

As of now, the annual budget plan doesn't include any form of activities/programmes for dissemination of electric and Tier 3+ cook stoves, therefore, this project will also help AEPC in leveraging maximum finance required to promote these CCS technologies in Nepal which has never been directly done before by GoN. Without GCF concessionality and the project, CCS promotion in Nepal would follow BAU scenario, i.e. major focus on only deployment of technologies as opposed to minimum efforts on ensuring project sustainability, establishing sound CCS market and supply chain and capacity building at all beneficiary levels.

AEPC has allocated the maximum budget it can contribute to the project as co-financing. Additionally, 14.37% (\$7,065,442) of the project budget is shared by LGs as co financing based on LG's allocated budget for CCS. Therefore, the minimum required financial structure to transition from BAU to currently planned project was identified as \$21.12 million. The pricing structure also falls in line with the concept of minimum concessionality because the identified grant amount from GCF is the minimum required amount that would enable AEPC to assess additional financial leverage for promotion of electric and tier 3+ cook stoves.

Concessionality will stimulate the market since lower tier users can't afford the switch without the subsidy, whereas LPG users need a strong market sign aligned via subsidy to make the switch. Since above 80% of the budget is allocated for the deployment of the technology, this will go directly to the beneficiaries. AEPC and Local Government will benefit by creating an enabling environment and empowered institutions, while the vendors will benefit from supply chain and market development.

In your answer, please consider the risk sharing structure between the public and private sectors, the barriers to investment and the indebtedness of the recipient. Please reference relevant annexes, such as the feasibility study, economic analysis or financial analysis when appropriate.

The project involves low risk and yet yields high return for GCF investments. GoN will be supporting roughly 60% of the total cost as grant and equity. This model ensures the use of grant as a cost reduction and risk reduction instrument to incentivize customers using LPG or TCS to switch to CCS technology. Vendors who bid for bulk tendering will need a bank guarantee that provides a risk reduction cushion for AEPC so quality products are made available and delivered to the customers. The payment method is scheduled in three tranches with one year time frame for vendors who provides the technology and installs them, provides risk management on behalf of AEPC, while ensuring quality assurances and project implementation guarantees.

In terms of beneficiaries, The project will capacitate the local service centers and trained them to ensure sustainability in the long term. Initially a market will be created by the project and subsequently the market will sustain itself due to the initial push. Nepal has witnessed these phenomena in different sectors including hydropower, solar and clean cooking solution. Hydropower development stood still for a prolonged period and has started gaining massive traction in the recent years. These sectors need an initial financing push which has proven to sustain itself in the long run. As per our Financial and Economic Analysis, in case of Traditional Cook Stove and ICS, although NPR 50 and NPR 20 has been taken as the monthly cost fuel wood for TCS and ICS. In practice very few households actually buy firewood in rural areas of Nepal, and most of the households collect it rather than buying from the market. Therefore, a market price for firewood is non-existent for most of the households. TCS and ICS beneficiaries usually source firewood from community forests free of cost. Therefore, a meager figure has been associated to fuel wood costs that accounts for labor for fuel collection. This also indicates the sustainability of the project when beneficiaries are aware of the health and economic benefits of the Tier 3+ in compare to TCS.

The investment from the individuals though not reflected in the total budget of the project, however we are expecting contribution ranging from 7 to 20% from the users based on the type of technologies. The Tier 3+ ICS is where we expect low user contribution as these are targeted towards the vulnerable communities while ECS will be targeted for urban communities and thus demanding higher user contribution. The barriers to investment and indebtedness will not pose a major concern as the cost of technologies Tier 3+ ICS and ECS will be on an average of USD 38 per technology. The cost of utensils for the ECS which shall be borne by the beneficiaries has been considered in the Financial Analysis. An upfront cost of \$16.67 (NPR 2000) has been considered that will include all basic utensils required by a typical household for e-cookstoves. ECS will be disseminated in the urban household of terai region which will have capacity to purchase the utensil.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

Explain how the project/programme sustainability (financial, institutional, social, gender equality, environmental) will be ensured in the long run after project closure, including how the project's results and benefits will be sustained.

In terms of the electric cookstoves targeted to the most vulnerable with an annual income ranging between USD 250 to USD 2000, we have performed sensitivity analysis considering various scenarios based on the requirement and consumption at the beneficiary level. Taking in to the consideration that the households will require minimum level of utensils and electricity consumption stays low, the financial analysis shows an IRR of 39% with payback period of 2.15 years without external interventions. Furthermore, with household opting for additional utensils thus increasing repair and maintenance along with electricity consumption our financial analysis shows IRR of 3% with payback period of 4.57 years without external intervention. Here the cost benefit analysis definitely shows financial sustainability post

project implementation compared to the existing LPG used for cooking. In terms of Tier 3+ICS targeted for the poor rural households of the country, the decrease in fuel-wood consumption by 30% compared to traditional cookstoves and open burning would generate positive cash from the very first year to the beneficiaries. Though the cumulative cash flow remains negative due to the CAPEX requirement for the Tier 3+Stoves, our economic analysis shows that with the additional health benefit, the technology can deliver an Internal Rate of Return of 16% with payback period of 3.30 years to the beneficiaries. The additional project benefits of improved indoor air quality and with more than 50% time saved on fuel collection that mostly affects the women and children will instigate and maintain the transformation towards clean cooking technologies. Furthermore, the project with the implementation of reverse auctioning will reduce the price of the proposed technologies by 20%. As a result, Tier 3+ ICS and ECS will be available in the market at an average rate USD 38.00 per technology and biogas at an average rate of USD 542. The government. As per the Financial Ordinance, 2078 (2021), pre-tax subsidies are provisioned to import ICS/ raw materials/accessories for ICS and Biogas accessories with custom duty discounted to 1%. In 2020, GON has reduced custom charges in induction cookstoves from 15% to 5%, promoting the use of ECS. The custom duty has been further decreased in 2021 and currently stands at 1% in case of ECS. The national budget for fiscal year 2021/2022 indicates addition of 1,629 MW of electricity to the national grid by completing Upper Tamakoshi, Rasuwagadhi, Upper Sanjen, Madhya Bhotekoshi and small and medium projects promoted by private investors. In addition to the import incentives on the clean cooking solutions, the national budget 2021/22 has further provisioned electricity tariff incentives with arrangement of free electricity to households consuming up to 20 units of electricity and 50 percent waiver on households using up to 50 units of electricity per month. These financial incentives and cost benefit analysis ensures the financial sustainability of the project. AEPC will further work with NEA to introduce cost effective tariff structure for electric cooking, which will speed up the transition from LPG to ECS. LPG is the third most imported item by Nepal with about \$300 million dollars spent a year. Hence, switching from LPG to electric cooking will reduce trade deficit and also provide energy security for Nepal, thus, directly contributing to Nepal's sustainable development.

The proposed CCS project is designed with close coordination and consultations with relevant government stakeholders to ensure ownership and sustainability of the project. The project will continue after GCF investments as it aligns with key national policies and is one of the key priority programmes of the Government of Nepal. The Second NDC of Nepal has set up the following quantitative targets in terms of Residential Cooking (i) By 2030, ensure 25% of households use electric stoves as their primary mode of cooking (ii) By 2025, install 500,000 improved cookstoves, specifically in rural areas and (iii) By 2025, install an additional 200,000 household biogas plants and 500 large scale biogas plants (institutional/industrial/ municipal/community). The 15th five-year plan of the National Planning Commission envisions the expansion of renewable energy for multipurpose use. It has prioritized promotion of biogas, ICS, gasifier, bio briquettes and ECS. Likewise, Nepal is a signatory of SDGs, which envisions CCS for all by 2030. The White Paper, 2018 prepared by the Ministry of Energy, Water Resources and Irrigation envisions "Electric Stoves in every house" and likewise aims to generate 15,000 MW of electricity by 2028 44, of which 10,000 MW would be consume internally, justifying government's efforts in promoting electric cooking. NEA has been reforming the tariff structure for its consumer base that will ultimately create a conducive environment for electricity consumption at affordable rates for users. This indicates that with successful intervention of the policies and programmes set by the government, surge in adoption of electric cooking would grow, as reliability and affordability of electricity supply gets better.

The project will support domestic production of Tier 3+ ICS, Biogas, ECS and pellets. The Stoves/fuel will be produced locally using local work force and materials, which is an important factor for long-term sustainability of the project. Besides, the project will prioritize multi fuel type Tier 3+ ICS that uses locally

⁴⁴White Paper 2018; (clause 55, pg. 24); Ministry of Energy, Water Resources and Irrigation, Government of Nepal

available materials. Technical assistance and capacity development component will help beneficiaries to produce the fuel locally and mobilize community managed forest user groups to bring down the cost of raw materials such as biomass, woodchip, husk, etc. The project will also help in reducing production cost of CCS technologies, with a scaled up market.

The project will address existing barriers in promotion of CCS to poorest households by improving access for vulnerable communities and providing support and training on repair and maintenance of proposed technologies to women and marginalized groups to ensure inclusive sustainability of the project. There are several local NGOs working in clean cooking sector. The project shall collaborate to mobilize their resources in awareness campaigns and skill development trainings.

The project would strengthen and increase the number of after sales service centres, which will result in quicker repair and maintenance of CCS technologies. Skill development trainings for local manufacturers for domestic production of Tier 3+ ICS, biogas, repair centres, pellets, operation and maintenance service would build the necessary skilled human resources to run the CCS supply chain and after sales services. Accessible and affordable CCS options, with well-developed supply chain and after sales service would not only increase the CCS usage, but also minimize the rebound effect to TCS and LPG.

One of the major components of the project is to capacitate local and provincial governments to prioritize CCS promotion in their periodic and annual planning, formulating appropriate policies and mechanisms. Provincial Project Management Units will be established with the purpose to capacitate provinces in mainstreaming Renewable Energy Technologies. The PMU will be established within the Provincial Ministry of Physical Infrastructure and Development under its Water Resources, Irrigation and Energy Division's Energy Branch. The project will be a concerted effort of the Government and the GCF. Hence, the project will enable the existing resources at the Energy branches to ensure the sustainability of the interventions of the project and strive towards reaching the national targets.

This will help in building required capacities, expertise, human resources, budget allocation and support directly in development of MEP for prioritizing CCS at local level perpetually.

The project will create a conducive environment in the CCS sector which ensures that bank and other financial institutions will develop appropriate loan products and explore opportunities to invest in the CCS sector during and post the project cycle, enhances long-term sustainability of the project. In the case of biogas, beneficiaries can approach several Micro Financing Institutions. The project will ultimately lead to a market led and enabled approach in the CCS sector coupled with conducive policies, wide spread availabilities of technologies and required resources, well capacitated CCS sector and extensive use of the proposed technologies. In addition, other ongoing energy related development projects, mainly GIZ/RERA⁴⁵ and DFID/NREP⁴⁶ projects, will also support private sector to develop CCS technologies by providing access to finance, e.g. via low interest rate loan and challenge fund. However, these projects are not directly linked with the proposed projects but contribute to the overall national renewable energy framework.

GoN is providing about 60% co-financing for the project, which indicates its strong desire to promote CCS in a longer term. Capacity enhancement of the government institutions, training, development of MEP and training modules on CCS and GHG mitigation will support the government to allocate and mobilize the required budget, and continue investment in clean cooking sector beyond the project closure.

⁴⁵<https://www.giz.de/en/worldwide/73840.html>

⁴⁶<https://www.nrepnepal.com/>

In a nutshell, the project enables creation of user demand with improved reliability by addressing the major issues in terms of quality control and testing centre. The project will further develop local level value chain with targeted campaigns and promotional activities at the community level, thus ensuring that the value chain becomes systemic change and the potential users have increased level of awareness on the proposed technologies. With a strong country ownership and based on the financial and economic benefit of the technologies with additional import incentives both on the CAPEX and OPEX ensures the affordability of the proposed technologies.

Include information pertaining to the longer-term ownership, project/programme exit strategy, operations and maintenance of investments (e.g. key infrastructure, assets, contractual arrangements). In case of private sector, please describe the GCF's financial exit strategy through IPOs, trade sales, etc.

The project beneficiaries and related stakeholders were well consulted during the project development phase to ensure their ownership in the project, which is key for successful implementation of the project and to sustain its outcome in the long run.

The project involves private service providers to deliver the CCS technologies and services through bulk tendering and reverse auctioning approach. With this, the CCS technologies and services will be made available to the beneficiaries and other interested users at an affordable price. Further, easy and affordable access of CCS technologies to users will increase the demand of CCS, which will attract more vendors in the sector, that would create a competitive environment, resulting in better products at cheaper rate, which could again attract more consumers to adopt CCS technology, thus creating a positive circular loop and ultimately ensuring the long term sustainability of the CCS market.

Provide information on additional actions to be undertaken by public and private sector or civil society as a consequence of the project/programme implementation for scaling up and continuing best practices.

As per the Constitution of Nepal, implementation of renewable energy programmes has been mandated to the Local Government. The project will directly work in formation of Municipal Energy Policy formulation, which will be an essential part for the sustainability and replicability of the project. Moreover, capacitating Local Governments will be instrumental in promoting and implementing renewable energy related projects and further set precedence on bulk procurement of renewable energy technologies.

The project will strengthen technical and institutional capacities of the sub-national government, introduce a cost effective mechanism of bulk purchase and reverse auction, set up a stringent technical due diligence process with output-based financing that can provide a governance-working model for other divisions of the governments of Nepal.

In terms to the price point, the proposed technologies ECS and Tier 3+ will be on an average of USD 38. The project will instigate the trend of using technologies with its known benefits. Behavioral change is one of the major proponents of the project. The post project financial sustainability will be secured by the market led approach. Further the consumers will be secured through price point, reliability and usability of the technologies.

In addition, AEPC will work in parallel on creation of loan products for larger technologies and special products for the vulnerable communities. Beside the regular government subsidy mechanism, AEPC has had various experience in implementing loan product including the (i) Micro Hydro Debt Fund (MHDF) with the overall objective to improve the access of the rural population to clean energy solution by providing credit for off-grid MHPs. (ii) Biogas Credit Fund with the objective of financing the loan via intermediate wholesale lending organizations and receiving non-government organizations, saving and credit cooperative and/or rural development banks (NGO/SCO/ RDBs) for granting sub-loans to those

farmers, who are not able to cover the cost of the biogas plant by their own. As of now, more than 300 LFIs/MFIs are involved to channel this credit to the biogas users. (iii) Challenge fund for various Renewable Energy Technologies.

C. FINANCING INFORMATION							
C.1. Total financing							
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency			
		21.12		million USD (\$)			
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	Enter amount	Enter years	Enter years	Enter %		
(ii)	Subordinated loans	Enter amount	Enter years	Enter years	Enter %		
(iii)	Equity	Enter amount			Enter % equity return		
(iv)	Guarantees	Enter amount	Enter years				
(v)	Reimbursable grants	Enter amount					
(vi)	Grants	21.12million USD (\$)					
(vii)	Results-based payments	Enter amount					
(b) Co-financing information		Total amount		Currency			
		28.023		million USD (\$)			
Name of institution		Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
AEPC		Grant	20.958	million USD (\$)	Enter years Enter years	Enter%	Options
Local Governments		Grant	7.065	million USD (\$)	Enter years	Enter%	Options

				Enter years		
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	49.15			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Co- Financing: Local Government: \$7,065,442 The Local Government will contribute 14.37% (\$7,065,442) of the total project budget as co-financing. This amount has been completely allocated as part of the Component 1 budget; the component wise budget share breakdown being 33% from GCF, 49% as co-financing from AEPC and 18% as co financing from LG. A MoU between LG and AEPC will stipulate the co financing disbursement structure. The contributions from both AEPC and LG will be in the form of grant and not in-kind. The co-financing amount will be directly paid out by LG to private technology providers when the installation targets in the respective LGs are verified. These are the same targets that have been identified in Section B.3, Component 1. In terms of the co-financing, the amount of financing will be secured during the project implementation stage wherein Memorandum of Understanding will be signed with the Local governments. Out of the total 287 LGs in the Terai region, the project will be implemented on 150 LGs. AEPC will publicly call for MoUs with the Local Governments on the implementation of the project and financing requirement from LGs will be defined during the project implementation stage. Furthermore, the co financing letter from the LGs will be part of the MoUs and will be made available during the project implementation stage. The LGs will finance the technology portion of the project only. The amount that will be mobilized from the LGs will require the assistance of project management component of the project, thus					

LGs financing is depicted as co- financing that contributes to the overall budget of the project.

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarized cost estimates in the table below and the detailed budget plan as annex 4.

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
COMPONENT 1: SCALING UP THE DEPLOYMENT OF CLEAN COOKING TECHNOLOGIES THROUGH ACCELERATED INVESTMENT AND MARKET DEVELOPMENT	Output 1.1: 500,000 Electric Stoves, 490,000 Tier 3+ ICS and 10,000-biogas plants installed	\$39,815,778.33	\$13,249,026.11	Grants	\$26,566,752.23	Grants	AEPC Co financing: 19.5 Million USD LG co financing: 7.06 Million USD
COMPONENT 2: STRENGTHEN ENABLING ENVIRONMENT THROUGH SECTOR BASED ASSESSMENTS AND	Output 2.1 <u>Enhancing product standards, conducting assessments, surveys and analysis</u>	\$105,000	\$105,000	Grants			
	Output 2.2 <u>Strengthening quality assurance mechanisms</u>	\$395,000	\$395,000	Grants			

QUALITY ASSURANCE OF THE TECHNOLOGIES							
COMPONENT 3: EMPOWERING INSTITUTIONS, CAPACITATE SUPPLY CHAIN AND ENSURE INCREASED ACCESS TO CLEAN COOKING SOLUTIONS	<u>Output 3.1: Capacity development of sub-national institutions</u>	\$650,000	\$650,000	Grants			
	<u>Output 3.2: Increased awareness and outreach to enhance demand</u>	\$2,731,747.44	\$2,731,747.44	Grants			
	<u>Output 3.3: Strengthening service centres, biomass manufacturers to provide quality and affordable clean cooking solutions</u>	\$2,996,700.83	\$2,996,700.83	Grants			
Component: Project Management Component	Establish and Operationalize Project Implementation and Management Units	\$2,457,590.90	\$1,000,000.00	Grants	\$1,456,841	Grants	AEPC
Indicative total cost(USD)		\$49,151,818	\$21,128,224		\$28,023,593		

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

Out of the total \$21.12 million requested as GCF funding, \$6,878,448.27 has been allocated for capacity building and technology transfer to strengthen enabling environment through sector based assessment and quality assurance of the technologies and empower sub-national institutions, capacitate supply chain and ensure increased access to clean cooking solutions. These activities will be conducted at the following levels:

Local Government: The project will build the capacity of LGs in promoting CCS technologies. AEPC will provide technical support in development and implementation of Municipal Energy Plans with focus to CCS. The institutional capacity of the LGs will be enhanced further with deploying CCS mobilizer in selected LGs to facilitate the development and implementation of the MEP. Training and exposure to LGs on demand collection, prioritization and monitoring system using MIS will also be provided by the project.

Vendors: The project will enhance the technical capacity of vendors, mainly for promoting CCS technologies and services including manufacturing and repairing. The project will strengthen capacity of supply chain industry such as, stove manufacturing companies, ECS service centres, biogas installers, electrical wiring service providers, pellet industry along with private investors, entrepreneurs and

financial institutions through guidance, training, and updates. In addition, training will be provided to local mobile and other electrical appliances repair centres, metal workshops, to capacitate them in providing after sales service for CCS. The project will provide training to private service providers on standards throughout the project period. AEPC will strengthen existing testing centre at central level to determine minimum technical standard and process to approve products based on the standard.

Beneficiaries: The project will create awareness amongst beneficiaries in the project areas as well as the country in the area of CCS and climate change adaptation. Skill development training will be provided to DAG, vulnerable community, women group, single women to capacitate them in repair and maintenance of the cook stoves.

Provincial Government: The project will strengthen the capacity of the Provincial government in performing their coordinating roles effectively and in creating an enabling environment for the project at province level.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Describe the potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas. As applicable, describe the envisaged project/programme impact for mitigation and/or adaptation. Provide the impact for mitigation by elaborating on how the project/programme contributes to low-emission sustainable development pathways. Provide the impact for adaptation by elaborating on how the project/programme contributes to increased climate-resilient sustainable development. Calculations should be provided as an annex. This should be consistent with section E.2 reporting GCF's core indicators.

This project is aligned with the GoN's vision to implement renewable energy and energy efficiency projects for climate change mitigation with multiple co-benefits involving vendors, PGs and LGs. Altogether, one million households located in 22 districts of Terai region will adopt CCS technologies- 490,000 Tier 3+ ICS, 10,000 Household Biogas, and 500,000 ECS respectively. As a result, the project will potentially reduce greenhouse gas emission by about 6.51 million tons of CO₂eq⁴⁷ during the lifetime of the technologies implemented by replacing lower tier Stoves with Tier 3+ ICS and Biogas, and by replacing LPG cylinders with ECS. UNFCCC methodologies⁴⁸ AMS I.C, AMS II.G and AMS I.E were used to estimate the amount of CO₂ that will be reduced by the project. The Emission Reduction will be monitored as per the requirements provided in the methodologies.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Describe the degree to which the proposed activity can catalyze impact beyond a one-off project or

⁴⁷ The estimation of the GHG emission reduction is provided in supporting annex (Annex 23).

⁴⁸ <https://cdm.unfccc.int/methodologies/SSCmethodologies/approved>

programme investment. Describe the following, if applicable:

- Potential for scaling up and replication
- Potential for knowledge sharing and learning
- Contribution to the creation of an enabling environment
- Contribution to the regulatory framework and policies
- Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

Paradigm Shift in public procurement process: The project introduces concept of bulk tendering via reverse auctioning, along with an output based financing model. Bulk tendering is a relatively new concept in the context of public procurement of Nepal, especially when the tendering involves demands aggregation from multiple LGs. AEPC will procure the technologies in bulk via reverse auctioning method, thereby significantly reducing the cost of CCS technologies. The output based financing modality will incentivize the vendors by lowering their risk, engender multiple check and balance mechanisms to ensure quality assurance and de-risk project investment.

The bulk tendering with reverse auctioning model, coupled with output based financing, will transcend from the BAU subsidy delivery mechanism of GoN, creating a paradigm shift in the public procurement process that could be adopted by other government agencies too.

Paradigm Shift in implementation of technologies: The specification of these technologies will be precise during project implementation through component 2 and activities under output 2.1, “Enhancing product standards, conducting assessments, surveys and analysis”. The project, during its implementation stage, will identify the right technology. The projects baselines the technology on current available guidelines and standards for the biogas plants, such as the Nepal Interim Benchmark for Solid Biomass Cookstoves (NIBC, 2016). AEPC currently promotes biogas plants of different capacity, such as 2 cu m, 4 cu m, 6 cu m and 8 cu m plants, and the project mainstreams the innovative GGC 2047 and modified GGC 2047 designs.

Additionally, the project will also mainstream the use of Tier 3+cookstoves (e.g. Forced Draft cookstove) and surge its penetration in the market, dominated by TCS. The cookstoves that the project intends to dispense will have high efficiency (greater than 45%), reduced PM emission (ie. High Power PM emission to less than 41mg/MJd) and increased safety factors (greater than 95 Johnsons).

Furthermore, the Nepal Bureau of Standard and Metrology (NBSM) recently released standards for safety and performance for domestic electric appliances, including electric cookstoves. The project will develop and mainstream the standardization in regards to electric cookstoves. For instance, Renewable Energy Testing Station (RETS), the government agency, is aggressively working on enlarging its testing to implement testing procedure set by NBSM and IEC on electric cookstoves. In consequence, this will also incentivize the Nepalese private entities for the local manufacture. The project will extensively support these institutions in preparation of the technical guidelines and standards for the given technologies.

As such, the consequence and spillover the project creates will help paradigm shift in adopting new technologies and novel standardization. This will surge the penetration of electric cookstove in the market, nudge the local manufacture and production, and catalyze the standardization and testing

procedures for quality in check.

Scaling up of CCS technology: The project has huge potential for scaling up as this aligns with the GoN aim of attaining Clean Cooking Solutions for all by 2030. The project meets 98% target of the Second NDC of Nepal which aims to install 500,000 improved cookstoves, specifically in rural areas by 2025. Furthermore, as per the Second NDC of Nepal, currently, around 5% of households use electric induction stoves, either as their primary or secondary mode of cooking and it aims to ensure 25% of households use electric stoves as their primary mode of cooking by 2030. The project contributes significantly to the achieve the target on the use of electric stoves. Strengthened supply chain, expansion of the technologies promoted during the project, will proliferate across the country without external interventions even after the project period.

Graduation from TCS to Tier 3+ ICS is an innovative approach as they are more efficient, are value for money and have a high affordability quotient for the households. Electric cooking is a segment that has remained uncharted and unwieldy at times, as its adoption is very low. The shift towards electric cooking with the government aspiration of generating surplus electricity from hydropower will bring in the next generation cooking into practice.

In the areas of the country where LPG and biomass are being used extensively, it will transform the energy usage pattern of households creating an environment where the future generations will easily adopt CCS as a primary means of cooking. The financial and economic benefits coupled with import incentives on the proposed technologies will ensure that the targeted groups of vulnerable households shift from the traditional way of cooking. The projects enables demand creation and ensures a market led approach with improved reliability of the proposed technologies with standards and guidelines in place. This will also develop and underpin the local level value chain with extensive campaigns and promotional activities at the community level.

Contribution to CCS policies at sub-national level: Mainstreaming the LGs, this project aims to support in developing municipal energy plans and the overall project implementation. This will empower the LGs to develop and implement similar projects in the future, further it will have a domino effect with more LGs reaching out AEPC for technical support to develop MEPs and implement CCS deployment programmes. Major shifts will be seen across the LGs with their capacities enhanced in identifying available local energy resources and implementing projects related to renewable energy with expertise gained on CCS.

Potential for knowledge and learning: The project has potential for knowledge generation and sharing. It will provide empirical knowledge on bulk tendering, reverse auctioning and output based financing. Further, the use of a management information system, where data will be available from the stage of energy plans, demand collection to project implementation, can be utilized to assess the demands and result of such intervention and help plan accordingly.

LGs and local communities will learn a lot from the use of CCS as the project will create awareness on and access to CCS, inform the vulnerable people of benefits in terms of health and efficient systems. The project will hence increase the knowledge of Local Governments and the stakeholders about the impacts that efficient technologies can have on their daily lives.

Developing a model of Green Growth for Nepal: The project will contribute to low carbon development pathways consistent with Nepal's climate change adaptation policies, strategies and plans. Nepal's future economic model has to pursue renewable energy to run its kitchen. This project is a milestone in the direction. A successful execution of this project will be instrumental for Nepal to reinforce its power infrastructure and industrialization in that direction. Since this project is being conducted in *Terai*, which borders Nepal with India, a successful implementation can pave way for Nepalese companies and power producers to expand in neighboring India as well as Bangladesh.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Describe the wider benefits and priorities of the project/programme in relation to the Sustainable Development Goals and provide an estimation of the impact potential in terms of:

- *Environmental co-benefits*
- *Social co-benefits including health impacts*
- *Economic co-benefits*
- *Gender-sensitive development impact*

The use of modern, efficient and clean cooking solutions will result in numerous sustainable development benefits. The project will contribute to several Sustainable Development Goals (SDGs), such as poverty reduction (SDG 1), lessening exposure to indoor air pollution (SDG 3), reducing drudgery for women in collecting wood fuel (SDG 5), improving access to affordable and clean energy (SDG 7) and reducing GHG emissions and strengthening the resilience of vulnerable communities to climate change (SDG 13). Besides, the project will also help in reducing pressure on the forest, thus, conserving the forest resources (SDG 15).

Environmental co-benefits: reduction of indoor air pollution and improve air quality, and reduction of forest degradation

The project will contribute in reduction of indoor pollution and improve air quality. On average, Tier 3+ ICS will reduce fuel wood consumption by 30%, and biogas and ECS will completely replace the use of fuel wood, thus reducing deforestation and forest degradation. This will help to conserve fragile ecosystem of *Churia*, which is a primary source of timber, fuel wood, fodder and other forest products for downstream communities in *Terai*.

In addition, the amount of cow dung replaced through Tier 3+ ICS can be used as organic manure, thus improving soil quality. Besides, using cow dung in domestic biogas plants will help to mitigate carbon emissions by converting methane into carbon dioxide.

Social co-benefits including health impacts: increased resilience through improved health

The use of clean cooking solutions promoted by this project will contribute in reduction of health risk caused by indoor air pollution because of improved household air quality, which will primarily benefit women and children. In total, more than 30% of reduction in indoor air pollution is expected by replacing the TCS with Tier 3+ ICS. WHO estimates the death of 8,700 people per year due to indoor air pollution in Nepal. A recent study shows the prevalence of productive cough, tearing of eyes, difficulty in

breathing, asthma, headache within cooking member of family who are involved in more than five years in kitchen⁴⁹. Introduction of CCS will help in reducing such adverse health impacts and in reducing deaths caused by indoor air pollution.

Economic co-benefits: job creation, increased revenue from cost savings, time gained for income-generating activities and education, and country's savings from LPG replacement

Approximately, 2,400 people will be trained during the project period. A total of 1,200 trainings will be provided to service providers and local manufacturers, mobile and other electric appliances repair centres and metal workshops to capacitate them in providing after sales service, domestic production of CCS technologies, pellets and briquettes, as well as strengthen the supply chain. In addition, 1,200 training will be provided on making fuel for Tier 3+ ICS, repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring to **CCS beneficiaries and vulnerable and disadvantaged groups (DAG)**. The training of stove repair and maintenance, electricians, and installers will create the much needed skilled human resources for the sector. The project would directly employ about 170 staffs, whereas it would create employment opportunities for approximately 3000 persons in the CCS sector.

Cost for fuel wood can be saved in households where fuel wood is purchased rather than collected. These savings can be used in livelihood enhancing activities such as education or for starting small businesses. In households where fuel is collected, the time spent on fuel wood collection can be saved and utilized for other income-generating activities and education. Besides, in cases where fertilizer needs to be purchased, the cost for fertilizers can be saved by using cow dung as fertilizer.

The LPG replacement will nurture the internal consumption of clean energy in line with the hydropower potential of the country and should significantly reduce the government's trade deficit⁵⁰.

Gender-sensitive development impact: reduced workload and exposure to indoor air pollution, time gained for income-generating activities and education, training and employment to women, impacts on revising traditional gender norms

The project enhances women's access and entitlement to benefits. Since the women have been the direct user of the cook stoves, it will benefit women by reducing their exposure to indoor air pollution, thereby improving their health. Also, the document 'Gender responsive electric cooking in Nepal (Energia, 2021)' has survey output of male beginning to participate in cooking and preparing snacks after the introduction of induction cookstoves in their households. This is changing gender norms of our society. Besides, reduction of fuel wood after the installation of more efficient stoves will reduce the workload of women to collect fuel wood, resulting in time saving that they can use for other productive activities such as getting an education, an employment or care for children or socialize, as per their preferences⁵¹. More than 50% of the time will be saved via the proposed CCS. Further, the project

⁴⁹Ranabhat et al., 2015. Consequence of indoor Air Pollution in Rural Area of Nepal: A simplified Measurement Approach. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4306325/>

⁵⁰The trade deficit during the fiscal year 2018/19 amounted to approximately USD 12 billion, wherein the USD 323 million worth of LPG were imported in the same fiscal year.

⁵¹<http://documents1.worldbank.org/curated/pt/772731563431443718/pdf/Have-Improved-Cookstoves-Benefitted-Rural-Kenyans-Findings-from-the-EnDev-Initiative.pdf>

creates equal footing for women by creating employment as CCS mobilizers, clean cooking champions, and technically trained women, who will be skilled enough to work in CCS sector, even after the project duration.

The project will be implemented based on the Gender Policy of AEPC and a gender assessment is provided along with this funding proposal (**Annex 8**).

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Describe the scale and intensity of vulnerability of the country and beneficiary groups and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc.). Describe how the project/programme addresses the following needs:

- *Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)*
- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity*

In Nepal, about 73.5 % of population use biomass (fuel wood, cattle dung or agro waste) as their primary fuel source with traditional cook stoves⁵². This inefficient choice of cook stove has result in increased greenhouse emissions, indoor air pollution and put pressure on the forest. The WHO has estimated 8,700 deaths a year from indoor air pollution in Nepal. Apart from biomass, a large portion of urban residents in *Terai* use LPG, which is derived from fossil fuel and emits greenhouse gas. In addition, LPG based cooking is a huge economic problem for Nepal stressing its budget deficit.

The TCS users are the most vulnerable community as they are among the poorest and socially excluded groups. 23.44 % of the population in the *Terai* region survives under absolute poverty. Against this, 33 % of the populations are disqualified to cross the threshold of human poverty. In addition, 25.9% *Janjati* (Indigenous group) and 38.2% *Dalit* population of *Terai* live under extreme poverty⁵³.

In Nepal, cooking is almost entirely carried out by women, requiring on average about four hours a day time for cooking using TCS. Women in Nepal spend about 142 minutes to collect the fuel wood compared about 50 minutes on average for men⁵⁴.

The per capita income stood at US \$ 1,004 (Rs 103,335) in the fiscal year 2018/19 and the main portion of income goes for sustaining livelihoods. In 2016, Nepal achieved a score of 0.148 according to the

⁵² World Bank, 2019. NEPAL | Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework

⁵³ CBS, 2011. Nepal Living Standard Survey 2010/2011. https://cbs.gov.np/wp-content/uploads/2018/12/Statistical_Report_Vol1.pdf

⁵⁴Venkata et al., 2015. *The State of the Global Clean And Improved Cooking Sector*.

multidimensional poverty index (MPI), with around 22.3% population vulnerable to Multidimensional Poverty (MP), 11.6 in severe MP, and 15 % below income poverty line⁵⁵.

The Government of Nepal aims to graduate from a Least Developed Country to a Developing Country by 2022. To achieve this goal, it is committed to tackle poverty and implement measures to meet the SDGs, in addition to the national targets. The Human Development Index (HDI) ranking for Nepal is 147 out of 189 countries listed⁵⁶, which clearly indicates the socio economic challenges of Nepal.

Economic and social development level of the country affected population

Nepal being a LDC and one of the most vulnerable countries to climate change, the project aims to address the vulnerability and exposure to climate risk and takes in to account the economic and social development level of the beneficiaries. The *Global Climate Risk Index 2021*, *Germanwatch* lists Nepal at the tenth position in the Long-Term Climate Risk Index (CRI) as the country most affected by extreme weather events from 2000-2019 with CRI score of 31.33. Nepal's Terai region is most prone to natural disasters with devastating floods and extreme events triggered by climate change. In terms of economic and social development level, the average monthly income of the household in the Terai region as per the *A Survey of Nepali People in 2018*, *The Asia Foundation*, depict the following information.

Average Monthly Household Income of Terai Region	Average Monthly Income		Annual Income
Less Than NPR. 2500 of 4.5% of sample population	NPR 2,500.00	\$ 20.83	\$ 250.00
NPR. 2501 to NPR. 9,999 of 19% of sample population	NPR 6,250.00	\$ 52.08	\$ 625.00
NPR. 10,000 to NPR.19,999 of 38% of sample population	NPR 14,999.50	\$ 125.00	\$ 1,499.95
NPR. 20,000 to NPR.39,999 of 27.9% of sample population	NPR 29,999.50	\$ 250.00	\$ 2,999.95
More than NPR.40,000 of 10.4 % of sample population	NPR 40,000.00	\$ 333.33	\$ 4,000.00

Absence of alternative sources of financing:

Though the project has high mitigation impact, the current level of funding and irregular nature of development assistance has been a hindrance to promote CCS technologies among the most vulnerable group, coupled with perceived un-bankability of projects like these, as evident by the lack of bank loans and investment in the CCS sector. The sector lacks experience on implementing innovative business models, although the GoN is aware of the drawbacks of existing procedures, it cannot bear the overall risk of introducing a new mechanism in the sector. Hence, GCF funding will secure GoN co-financing for the project, which would not have happened in the business as usual process.

Needs for strengthening institutions and implementation capacity:

⁵⁵ Human Development Index Report, 2019 http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/NPL.pdf

⁵⁶ ibid

After almost three decades of hiatus, the LGs finally have locally elected representatives, after the election conducted in 2017. There are now 753 Local Governments in the country with no or limited knowledge of renewable energy technologies. The project would significantly contribute in strengthening sub-national governments and their implementation capacity. LGs lack the resources and technical skills to develop projects to mitigate climate change and raise awareness on benefits of electric cooking along with other CCS. The proposed project will capacitate the LGs to develop municipal energy plans and implement MIS on demand collection and aggregation, thus empowering them in the renewable energy sector. The proposed project has received support from the GCF in the form of Project development Technical Assistance to alignment of this project with the GCF investment framework.

D.5. Country ownership (max. 500 words, approximately 1 page)

Please describe how the beneficiary country takes ownership of and implements the funded project/programme. Describe the following:

- *Existing national climate strategy*
- *Existing GCF country programme*
- *Alignment with existing policies such as NDCs, NAMAs, and NAPs*
- *Capacity of Accredited Entities or Executing Entities to deliver*
- *Role of National Designated Authority*
- *Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups*

GoN has endorsed Investment Prospectus and Biomass Energy Strategy in 2017, with aim of attaining Clean Cooking Solutions for all in two stages i.e. by improved solutions of cooking by 2022 (no traditional way of cooking with open fire burning) and modern solutions of cooking with at least Tier 3+ level by 2030. The project supports Nepal's Nationally Determined Contributions⁵⁷ of reducing 50% fossil fuel dependency by 2050 and equipping households in rural areas with smokeless kitchen. Approved by the NDA, the project concept aligns with the GoN's *Terai* Clean Cooking Programme and recently adopted Approach Paper for 15th Periodic Plan (2019/20 to 2023/24) that gives high priority to household biogas, ICS, and production of bio-briquette and pellets.

The climate change policy 2019, aims to contribute to socio-economic prosperity of the nation by building a climate resilient society. The policy has been introduced with the objective of providing policy guidance to various levels and thematic area to develop resilient society by reducing risk of climate change impacts. The policy have formulated strategies and working policies on different broad sectoral policy namely: Agriculture and Food security, Forest, Biodiversity and Watershed Conservation, Water Resource and Energy, Rural and Urban Habitats, Industry, Transport and Physical Infrastructure, Tourism and Natural and Cultural heritage, Health, Drinking Water and Sanitation, Disaster Risk Reduction and Management, Inter-thematic areas (Gender Equality and Social Inclusive, Livelihoods and Good Governance, Awareness Raising and Capacity Development, Research, Technology Development and Expansion and Climate Finance Management). In brief, the policy also targets to enhance climate change adaptation capacity of persons, families, groups and communities vulnerable to,

⁵⁷Nationally Determined Contributions; October, 2016; Ministry of Population and Environment, Government of Nepal

and at risk of, climate change; mobilize national and international financial resources for climate change mitigation and adaptation in equitable and just manner; mainstream gender equality and social inclusion (GESI) into climate change mitigation and adaptation programmes. To achieve its goal, low emission technologies and adaptation programmes will be incorporated in settlement development plans, which would mean CCS technology would be a key priority. Clean cooking solutions fits perfectly while designing these integrated settlement plans. The introduction of CCS technology in rural and urban habitat will significantly reduce the GHG emissions with shift from current practice of cooking. With implementation of this project, the project will be replicated in other locations and across other household renewable technologies including solar and bio energy. Nepal has recently submitted its Second NDC in December 2020 depicting its ambitious targets for the next decade to mitigate emissions and increase resilience of vulnerable communities to adapt to the impacts of climate change. The Second NDC Nepal has priorities on the clean cooking sector that will be to ensure 15% of the total energy demand is supplied from renewable energy sources; In Residential cooking by 2030, ensure 25% of households use electric stoves by 2025, install 500,000 improved cook stoves, specifically in rural areas and by 2025, install an additional 200,000 household biogas plants. The project help to change the perception toward CCS technology and create the environment for them to adopt CCS technology. Hence, this project will facilitate to incorporate CCS technology in settlement development plans and meet the target set by the NDC.

AEPC and local government will collaboratively develop framework, guidelines and procedures to call the vendors to provide access to affordable and accessible CCS technologies for beneficiaries. Further AEPC has been conducting several rounds of discussion with LGs and has even signed Memorandum of Understanding (MoUs) with some LGs on the implementation of renewable energy programme including the *Terai* Clean Cooking Programme.

The Ministry of Finance (MoF) is the **National Designated Authority (NDA)** of the Green Climate Fund. Nepal aims to enhance national ownership process beyond NDA; strategically and systematically build the national capacity at different levels to ensure that the knowledge and processes related to GCF mechanisms are transferred to diverse stakeholders and contribute to sustainable capacity building efforts.

The role of Nepal NDA are as follows:

1. Provide broad strategic oversight that is aligned to national priorities for developing the GCF country programme and communicating strategic country priorities to access GCF funding.
2. Convene national stakeholders, including the public and private sectors and civil society, via coordination, consultation, and multi-stakeholder forums to engage in raising awareness on GCF processes, capacity development needs, documenting lessons learnt, and facilitating opportunities for shared learning with other countries.
3. Facilitate the accreditation process for national institutions and provide 'Nomination Letters' to institutions (e.g. government, private sector and civil society organizations) for their application for the accreditation status under the GCF.
4. Facilitate the development of GCF funding proposals and the implementation of GCF funded projects/programmes in Nepal by engaging with accredited entities to align funding proposals with national strategic priorities.
5. Provide leadership on the deployment of readiness and preparatory support funding in Nepal.

This project is in alignment with country priorities as clearly mentioned in the No Objection Letter granted from the NDA for the project. The project is a priority of the GoN and is linked to the Government led national initiative of Terai Clean Cooking Programme. The project will emphasize on adequate engagement of relevant stakeholders, which are aligned with national climate change priorities. This is a priority project for Nepal. Though the country programme is still under development, the Government of Nepal has indicated this project as a priority project for GCF-1. Govt of Nepal is providing about 60% co-financing to this project. The 15th five-year plan of the National Planning Commission envisions the expansion of renewable energy for multipurpose use. It has prioritized promotion of biogas, ICS, gasifier, bio briquettes and ECS. The second NDC which was submitted in December 2020 also highlight clean cooking sector as a priority, alongside, Nepal also a signatory of SDGs, envisions CCS for all by 2030. In addition to promoting clean cooking solutions this project also supports a strong capacity enhancement component to strengthen government institutions, provide training, development of Municipal Energy Plans and training modules on CCS and GHG mitigation; while helping government to allocate and mobilize the required budget, and continue investment in clean cooking sector beyond the project closure.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers, and providing the minimum concessionality to ensure the project is viable without crowding out private and other public investments. Refer to section B.5 on the justification of GCF funding requested as necessary.

Nepal has not been able to effectively expedite climate financing for its low carbon and climate resilient programmes. There is also lack of appropriate financing mechanism to incentivize vendors and de-risk their investment in renewable energy sector.

The existing BAU subsidies model, where in the vendors have been carrying out the task of collecting demands door to door, and taking all financial risk by putting 100% capital upfront, has failed to drive enough participation from the vendors to achieve the CCS installation targets set by AEPC. The fund from GCF would help AEPC to pay 20% amount in advance as per PPA/PPR to vendors against bank guarantee, which would allow them to use leverage financing to install more CCS technologies, thus increasing the CCS deployment than in BAU scenario.

On the institutional side, the project would set precedence by introducing an avant-garde implementation mechanism: reverse auctioning, output based financing, renewable energy MIS and mainstreaming LGs role in the renewable energy. This will create a massive change in the business as usual process of financing renewable energy technologies in Nepal and the government will have a tried and tested model that can be replicated all over the country.

GCF grant will support achieving national targets and Nepal's NDCs⁵⁸ while curbing the usage of fossil based LPG and fuel wood for cooking. For a country which aims to generate surplus electricity in the near future and the residential sector as the primary consumer of energy in the country, proliferation of

⁵⁸ Nationally Determined Contributions; October, 2016; Ministry of Population and Environment, Government of Nepal

ECS will be a tipping point in creating a paradigm shift in the overall clean energy consumption scenario.

Successful implementation of the project could allow the replacement of LPG at a larger scale in the country. Bringing additional ties in the sector, the GCF grant will be utilized to remove barriers seen in the demand and supply side in terms of awareness, access to finance, regulatory frameworks, and quality assurance of Tier 3+ ICS, biogas and ECS.

The project involves low risk and yet yields high return for GCF investments as AEPC and LGs will be supporting approximately 60% of the total cost for promotion of clean cooking solutions as co-financing.

Appropriate implementation models, specially the reverse auctioning will be adopted to ensure the effective and sustainable operation of the system. Partnering with local governments and financial institutions will further strengthen the efficiency and effectiveness of the project.

The project directly ensures increased involvement of private sector as vendors by the creation of big and confirmed demand of CCS through the bulk purchase. Further, rapid awareness among the users will also ensure market growth of CCS, especially due to promotion of electric cooking.

Please describe the efficiency and effectiveness of the proposed project/programme, taking into account the total financing and mitigation/ adaptation impact the project/programme aims to achieve, and explain how this compares to an appropriate benchmark.

The project expects to reduce an estimated 6.51 million tons of CO₂eq with significant co-benefits in the area of health, environment, gender, social, and economy. The total per unit mitigation cost for the project is about 7.54 USD/tCO₂eq and 3.24USD/tCO₂eq of GCF fund. The USD 21.12 million of GCF grants will be combined with USD 28.02 million of co-finance from AEPC and LGs.

The use of the clean cooking solutions promoted by this project will contribute to the reduction of health risk caused by indoor air pollution because of improved household air quality, which will primarily benefit women and children. In total, more than 30% of reduction in indoor air pollution is expected by replacing the TCS by Tier 3+ ICS.

The project will be implemented based on the Gender Policy of AEPC and a gender assessment will be provided during the funding proposal stage. The project enhances women's access and entitlement to benefits by reducing their exposure to indoor air pollution thereby improving their health and increasing earning opportunities.

Please specify the expected economic rate of return based on a comparison of the scenarios with and without the project/programme.

Calculation of Economic Rate of Return (EIRR) is performed both from the project perspective and from the perspective of the beneficiaries and was based on monthly savings in fuel consumption due to transfer of cooking technology, health benefits due to reduction in indoor air pollution and global benefits in carbon savings from reduced GHG. Sensitivity analysis has been performed from the beneficiaries' perspective considering the uncertainty surrounding technology cost and its associated cost that affects making an informed investing decision at the beneficiary level. Cash outflow from the beneficiary perspective in terms of the usage of utensils, cost of repair and maintenance, annual fuel cost will vary

upon households. In terms of the electric cookstoves, the analysis takes into considerations the different types and number of electromagnetic utensils required as per the varying food habit and different types of food items that needs to be cooked, the wholesale and retail price point of these utensils and house wiring, varying annual fuel consumption and repair and maintenance cost with an increased consumption of the technology. In terms of biogas, the analysis has been performed based on the varying frequency of the repair and maintenance required for the biogas plants. Furthermore, a household income percentage analysis has been performed to understand the effect of electric cookstoves (both CAPEX and OPEX) on the household income based on the different level of household income prevalent in the Terai region. In case of the project perspective, two case scenarios were evaluated; with and without GCF support. In the first scenario GCF support is introduced as grant support to the project i.e. to AE's co-financing and LG's co financing, however, without GCF support the model is based on BAU scenario.

BAU is based on the annual installation target achieved by AEPC in FY 2018/19 and assuming the same no. of CCS deployment for the next five-year, i.e. 11,750-biogas plant and 40,000 ICS implemented each year for the next five years. As the CCS projects under AEPC so far do not deploy ECS, the BAU scenario does not include it in the analysis. The total budget expended by AEPC to conduct CCS related activities that are in line with the proposed project was \$4.5million in FY 2018/19.

EIRR in case of inclusion of GCF grant is at 39% as opposed to no returns with a negative NPV of \$18.8million in the BAU scenario.

Parameters	With GCF Funding		Without GCF Funding (BAU Scenario)	
	Financial (FIRR)	Economic (EIRR)	Financial (FIRR)	Economic (EIRR)
Internal Rate of Return	31%	39%	-	-
Net Present Value (NPV)	\$ 19,779,329.68	\$ 43,433,452.05	\$(38,592,365)	\$(18,844,261)
Discounted Payback Period (Years)	5.08	4.59	-	-

In case of the beneficiaries' perspective, primarily three scenarios were taken in to consideration for each of the technologies while conducting the economic and financial analysis. Scenario 1: Without GCF Support and before Reverse Auctioning, Scenario 2: Without GCF Support and with Reverse Auctioning and Scenario 3: With GCF Support and with Reverse Auctioning.

- The following are the outcomes of the EFA for each of the technologies for the three scenarios calculated at a discount rate of 12%.

1. Tier 3+ Cook stoves vs. Traditional Cook stoves

Parameters	Scenario 1		Scenario 2		Scenario 3	
	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR
Internal Rate of Return	-	7%	-	16%	115%	608%
Net Present Value (NPV)	\$(40.3)	\$(5.42)	\$(30.29)	\$4.32	\$ 6.72	\$41.33
Payback Period (Years)	-	4.12	-	3.30	0.88	0.16

2. Electric Cook Stoves Vs. LPG

Given the need for utensils that are pre-requisite to induction cooking and that the market has a range of products with varied price points, the analysis need to consider different scenarios which depicts that the varying cost of utensils, house wiring, and annual fuel cost and repair/maintenance. Hence, a sensitivity analysis has been performed taken into the following considerations

- Sensitivity Analysis Scenario One: (1) Applying a conservative approach wherein households will only purchase a single utensil and cost of house wiring considered when there is a bulk procurement, the cost stands at \$33 that needs to be borne by the beneficiaries with (2) Electricity cost at NPR 1114 per month and (3) Assuming 2% of the CAPEX plus utensils in repair and maintenance
- Sensitivity Analysis Scenario Two: (1) Applying the approach based on the varying food habits that requires different utensils, in this case a stainless steel pot and a 3 litre pressure cooker and the cost of house wiring at retail level, the cost stands at \$66.67 that needs to be borne by the beneficiaries (2) Considering an increase in electricity tariff by 5% with increased usage (3) Assuming 5% of the CAPEX plus utensils in repair and maintenance as a result of increased usage.
- Sensitivity Analysis Scenario Three: (1) Further, expanding the range based on the varying food habits that requires different utensils, in this case a stainless steel pot, a 5 litre pressure cooker and a frying pan along with house wiring cost of \$83.33 at retail level (2) Considering an increase in electricity tariff by 10% with increased usage (3) Assuming 10% of the CAPEX plus utensils in repair and maintenance as a result of increased usage.

Sensitivity Analysis Scenario One

Parameters	Scenario 1	Scenario 2	Scenario 3
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	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR
Internal Rate of Return	32%	36%	39%	44%	76%	83%
Net Present Value (NPV)	\$ 45.69	\$56.07	\$55.28	\$65.66	\$84.82	\$95.20
Payback Period (Years)	2.42	2.23	2.15	1.98	1.28	1.18

Sensitivity Analysis Scenario Two

Parameters	Scenario 1		Scenario 2		Scenario 3	
	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR
Internal Rate of Return	3%	7%	6%	10%	19%	24%
Net Present Value (NPV)	\$ (24.29)	\$(13.91)	\$(14.71)	\$(4.32)	\$14.84	\$25.22
Payback Period (Years)	4.57	4.14	4.22	3.81	3.11	2.80

Sensitivity Analysis Scenario Three

Parameters	Scenario 1		Scenario 2		Scenario 3	
	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR

Internal Rate of Return	-22%	-17%	-20%	-15%	-13%	-7%
Net Present Value (NPV)	\$ (89.71)	\$(79.33)	\$(80.13)	\$(69.75)	\$(50.58)	\$(40.02)
Payback Period (Years)	-	-	-	-	-	-

3. Biogas Vs. TCS

Given that the variation on the timeline required for repair and maintenance cost, a sensitivity analysis has been performed taken into the following considerations. The qualitative factor that relates to the availability of organic fertilizer and time saved on fuel collection due to the usage of biogas plant ensures further socio-economic benefits.

- Sensitivity Analysis Scenario One: (1) Annual Repair and maintenance.
- Sensitivity Analysis Scenario Two: (1) Repair and maintenance every fifth year.
- Sensitivity Analysis Scenario Three: (1) Repair and maintenance every seventh year

Sensitivity Analysis Scenario One:

Parameters	Scenario 1		Scenario 2		Scenario 3	
	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR
Internal Rate of Return	-	-	-	-	-	7%
Net Present Value (NPV)	\$(727.36)	\$(618.66)	\$(591.94)	\$(483.24)	\$(124.27)	\$(15.57)
Payback Period (Years)	-	-	-	-	-	8.3

Sensitivity Analysis Scenario Two:

Parameters	Scenario 1		Scenario 2		Scenario 3	
	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR
Internal Rate of Return	-	-	-	-	-	-
Net Present Value (NPV)	\$(727.36)	\$(618.66)	\$(591.94)	\$(483.24)	\$(124.27)	\$(15.57)
Payback Period (Years)	-	-	-	-	-	-

Internal Rate of Return	-	-	-	-	-	26%
Net Present Value (NPV)	\$(650.17)	\$(541.47)	\$(514.75)	\$(406.05)	\$(47.08)	\$61.63
Payback Period (Years)	-	-	-	-	-	3.48

Sensitivity Analysis Scenario Three:

Parameters	Scenario 1		Scenario 2		Scenario 3	
	FIRR	EIRR	FIRR	EIRR	FIRR	EIRR
Internal Rate of Return	-	-	-	-	-	27%
Net Present Value (NPV)	\$(644.30)	\$(535.60)	\$(508.88)	\$(400.18)	\$(41.21)	\$67.50
Payback Period (Years)	-	-	-	-	-	3.48

Tier3 + ICS definitely require the support as indicated by the result seen in scenario 3 compared to both the other scenarios. Both the scenarios without GCF funding questions the feasibility of the investment from the beneficiaries. Depending on the sensitivity factors, ECS shows varied level of return, thus it becomes important to assess these practical aspects of electric cooking while making justification regarding the financial incentive required for the beneficiaries. Furthermore, as household income plays an important role in making the investment decision, this has been further analysed from the lens of household income level and the percentage of capital and operational expenditure of electric cooking on the various ranges on income level at the Terai Region. The Household Income Percentage analysis has been presented as separate annex and the highlights of the analysis is presented in the following section related to FIRR. Hence, grant support is justified which will kick start the process and ECS will ultimately be the technology on which the country will cook in the coming days.

In terms of the biogas, the comparison has been done with both LPG and traditional cook stoves. Please see the Annex 3 for the comparison between LPG and Biogas. Given that the beneficiaries targeted by the project for biogas plant are the TCS users. Hence the emphasis is given on the Biogas vs. TCS category in this section. The EFA clearly substantiate the request for grant for biogas. The CAPEX is on the higher side which seeks for grant support for its feasibility. Scenario 3 with the GCF contribution still shows a deficit for investing on the technology from the end user perspective. Ultimately, on the OPEX with no fuel cost and longer duration biogas will create a long lasting impact.

In terms of the cost of disposal or revenue from the sale from the old cook stoves, we do not foresee neither the sale of the existing cookstoves nor any cost for disposal. TCS will be dismantled and the

proposed technologies will replace the existing cook-stoves. However, some level of stacking is expected in reality especially when it comes to LPG. This will be gradual process as electricity becomes cheaper and convenient in the years to come. This has the potential to drive growth of the country in the direction as envisioned by the national commitments and plans.

Please specify the expected financial rate of return with and without the Fund's support to illustrate the need for GCF funding to illustrate overall cost effectiveness.

The financial rate of return (FIRR) from a project perspective for the scenario with GCF is 31% with the NPV of \$19,779,329.68 whereas the FIRR for scenarios for BAU has no returns with a negative NPV value of \$38.6million. The financial rate of return from the beneficiary perspective has been presented in the earlier section.

The reduced returns in both financial and economic analysis for BAU scenario can be cited to the following reasons:

- I. Procurement of technologies is done by traditional bidding methods or on a retail basis by the users. This in turn increases the initial CAPEX of CCS projects in the BAU scenario.
- II. In BAU scenario, AEPC implements 11,750 domestic biogas plants and 40,000 ICS each year. Although domestic biogas plants provide higher economic benefits in terms of reduced GHG emissions and health benefits in comparison to both the technology it replaces and the technology it is being implemented together with; TCS and ICS respectively, the OPEX associated with domestic biogas is approximately 5 and 11 times higher than the latter technologies. This increase in financial OPEX renders negative financial returns for BAU scenario.
- III. For almost the same amount of funding as in this proposed project, BAU fails to implement electric cooking in CCS promotion; the fuel savings and economic impacts from electric cooking is larger compared to other CC technologies.

Meanwhile, effectiveness of the project without GCF support is an important factor to consider. Due to the resource limitation of AEPC, the BAU scenario would fail to invest in activities for accelerating climate change mitigation and adaptation needs, creating a market and supply chain ecosystem, mitigating the risk of implementing an innovative approach, implementing innovative approaches of demand aggregation and mainstreaming LGs proposed in the project. Likewise, GCF's support allows promotion of higher Tier 3+ ICS and ECS, which otherwise BAU fails to do.

In case of ECS as mentioned earlier, the household level of income becomes a critical factor while making an investing decision. Especially to justify the level incentive to be provided to the beneficiaries, the analysis below presents the percentage of annual household income required to meet the capital and current expenditure of ECS. Since the targeted beneficiaries are the poor household of the urban and peri-urban areas, annual household income level between \$250 to \$ 2000 has been presented below based on the provision set out by the sensitivity analysis while conducting the economic and financial analysis (ECS Vs. LPG) from the beneficiary perspective: The sensitivity scenario with requirement of two utensils and wiring that leads to 5% increased in electricity cost and repair/maintenance has been presented below. The overall analysis has been presented as a separate annex Household Income % analysis

CAPEX and OPEX % in terms of Average Household Income

Economic and Financial Scenario 1: Without GCF Support and before Reverse Auctioning with Sensitivity Analysis Scenario One							
	Year	0	1	2	3	4	5
Household Cumulative Net Cash Flow (CAPEX)		\$ (81.25)	\$ (48.74)	\$ (14.77)	\$ 20.72	\$ 57.82	\$ 96.58
		% of Annual Household Income					
Households with income less Than NPR. 2500 per month		32.50%	19.50%	5.91%	8.29%	23.13%	38.63%
Households with income between NPR. 2501 to NPR. 9,999		13.00%	7.80%	2.36%	3.32%	9.25%	15.45%
Households with income between NPR. 10,000 to NPR.19,999		5.42%	3.25%	0.98%	1.38%	3.85%	6.44%
Household Cumulative Net Cash Flow (OPEX)			\$ 112.79	\$ 117.87	\$123.17	\$128.72	\$134.51
		% of Annual Household Income					
Households with income less Than NPR. 2500 per month		0.00%	45.12%	47.15%	49.27%	51.49%	53.80%
Households with income between NPR. 2501 to NPR. 9,999		0.00%	18.05%	18.86%	19.71%	20.59%	21.52%
Households with income between NPR. 10,000 to NPR.19,999		0.00%	7.52%	7.86%	8.21%	8.58%	8.97%

CAPEX and OPEX % in terms of Average Household Income

Economic and Financial Scenario 3: With GCF Support and with Reverse Auctioning with Sensitivity Analysis Scenario One							
	Year	0	1	2	3	4	5
Household Cumulative Net Cash Flow (CAPEX)		\$ (42.12)	\$ (9.62)	\$ 24.35	\$ 59.85	\$ 96.95	\$ 135.71
		% of Annual Household Income					
Households with income less Than NPR. 2500 per month		16.85%	3.85%	9.74%	23.94%	38.78%	54.28%
Households with income between NPR. 2501 to NPR. 9,999		6.74%	1.54%	3.90%	9.58%	15.51%	21.17%
Households with income between NPR. 10,000 to NPR.19,999		2.81%	0.64%	1.62%	3.99%	6.46%	9.05%
Household Cumulative Net Cash Flow (OPEX)			\$ 112.79	\$ 117.87	\$ 123.17	\$ 128.72	\$ 134.51
		% of Annual Household Income					

Households with income less Than NPR. 2500 per month		0%	45.12%	47.15%	49.27%	51.49%	53.80%
Households with income between NPR. 2501 to NPR. 9,999		0%	18.05%	18.86%	19.71%	20.59%	21.52%
Households with income between NPR. 10,000 to NPR.19,999		0%	7.52%	7.86%	8.21%	8.58%	8.97%

The income level and percentage of CAPEX and OPEX on the income level presents a stark picture that beneficiaries will require financial incentive to pursue clean cooking technology. Furthermore, as the requirement of utensils, annual electricity cost and repair/maintenance cost increases as per the different sensitivity analysis, the need to provide financial incentive to the beneficiaries becomes even more imperative.

It is to be further note that several qualitative factors exist that affects the investing decision from the beneficiary perspective. The following factors need to be taken in to consideration:

Vulnerability to disruptions and Energy Security: As Nepal has no internal supplies of petroleum or natural gas, its' vulnerability to disruptions in its supplies of imported fossil fuels is the principal reason why the project looks into ECS as a way of increasing national energy independence and ensuring national energy security. Energy demand in Nepal has increased for a decades, It has major impact in its economic, social and political developments. This energy situation will get worse in future, if actions are not taken in time. Hence, major intervention is required to overcome this present crisis. As per the BBC News on Nepal's forest under threat over fuel crisis⁵⁹, during the supply constraint of the petroleum product and natural gas with the shortage in cooking gas and kerosene in Nepalese kitchens, this has suddenly increased the demand of firewood. Conservationist says that there was no choice left for the people to cut down trees despite having tradition of protecting the forest. This also led to a situation where in the price of LPG and petroleum product had increased to over 500% during the crisis.

Poverty and Income Level: The technology targets the poor households of urban and peri-urban area, The per capita income of the country was USD 1,085 in FY 2019/20 (MoF, 2020). Province 2 houses the largest number of multidimensional poor. A separate annex has been prepared taking in to consideration the portion of their income on the CAPEX and OPEX. As per the 2018 survey⁶⁰, about 38.1%of household income of Terai region range between NRP 10,000 to 19,999 per month, followed by 27.9% income between NRP 20,000to 39,999 per month, 19% income between NRP 2,501 to 9,999 per month, 10.4% income more than NRP 40,000 per month.

Remittance: It is expected that the employment sector will be negatively impacted by the COVID-19 pandemic, and the return of Nepali workers, mainly from Gulf countries, is projected to cause a 20% decline of remittances (CBS, 2020). About 20%–25% of the estimated 3 million Nepalese workers abroad are likely to return home, mainly because of two reasons: first, the work tenure of about 60 % of

⁵⁹<https://www.bbc.com/news/science-environment-34468821>

⁶⁰ A Survey of the Nepal People in 2018, School of Art, Kathmandu University, Interdisciplinary Analysts and The Asia Foundation 2019

the Nepalese workers is over and, second, a large number of workers have lost their jobs due to the pandemic (Mandal, May 21, 2020). This will affect the inflow of remittance in the country.

Please explain how best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices

In collaboration with Nepal Academy of Science and Technology, Renewable Energy Test Stations, AEPC will strengthen the existing testing centers and expand its services at provincial level to determine minimum technical standard and process to approve the products based on the standard. This will ensure dissemination of pre-qualified technologies with internationally accepted standards and parameters.

The project will also provide training to private service providers on standards throughout the project period, set up MIS on testing centre, place and operationalize independent third party monitoring agencies and set up coordination mechanisms at Province and LG level.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's [Performance Measurement Frameworks](#) under the [Results Management Framework](#) to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

Please select the appropriated expected result. For cross-cutting proposals, tick both.

☒ Shift to low-emission sustainable development pathways

☐ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq.) to be reduced or avoided (mitigation and cross-cutting only)	Annual	271,401 t CO ₂ eq.
	Lifetime	6,513,629 t CO ₂ eq.
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	<u>49,151,817</u> USD
	(b) Requested GCF amount	<u>21,128,224</u> USD
	(c) Expected lifetime emission reductions	6,513,629 t CO ₂ eq
	(d) Estimated cost per t CO ₂ eq (d = a / c)	<u>7.54</u> USD / t CO ₂ eq
	(e) Estimated GCF cost per t CO ₂ eq removed (e = b / c)	<u>3.24</u> USD / t CO ₂ eq
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	<u>28,024,593</u> USD
	(g) Public source co-financed	<u>28,024,593</u> USD
	(h) Private source finance leveraged	<u>0</u>
	(i) Total Leverage ratio (i = f / b)	1.33
	(j) Public source co-financing ratio (j = g / b)	1.33
	(k) Private source leverage ratio (k = h / b)	<u>0</u>
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	
	Indirect	
	For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.	
E.2.5. Number of beneficiaries relative to total population (disaggregated)	Direct	
	Indirect	.

by sex)	For a multi-country proposal, leave blank and provide the data per country in annex 17.
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E.3. Fund-level impacts

Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
M3.0 Reduced emissions from buildings, cities, industries and appliances	M3.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq.) reduced or avoided - buildings, cities, industries, and appliances	Monitoring of verification reports from Independent Designated Operational Entity(DoE) Monitoring Report generated by the MIS on the installation of technologies Final evaluation report Monitoring of secondary data from the reports on baseline and targets published by the GoN	0	897,604 t CO ₂ eq	3,537,473 t CO ₂ eq	Applied methodologies⁶¹: UNFCCC methodologies⁶² AMS I.C, AMS II.G and AMS I.E are used for emission reduction calculation. Project lifetime: 24 yrs.⁶³ Annual emission reductions: 271,401 t CO₂ eq. Lifetime emission reductions: 6,513,629 t CO₂ eq.

E.4. Fund-level outcomes

⁶¹For electric cookstoves: AMS I.C. Thermal energy production with or without electricity, version 21, For Tier 3+ biomass based ICS: AMS II.G. - Energy Efficiency measures in thermal applications of non-renewable biomass (Version 11), For Biogas: AMS I.E. Switch from non-renewable biomass for thermal applications by the user (Version 10.1)

⁶²<https://cdm.unfccc.int/methodologies/SSCmethodologies/approved>

⁶³ Emission reduction for the CCS technology implemented during the project period is calculated based on the life of the technology which is considered as 5 years for electric stoves and Tier 3+ ICS, and 20 years for Biogas. It is also accounted that the electric stoves and Tier 3+ ICS installed in 5th years of the project period will last till 9th year and biogas installed in 5th year of the project will last till 24th year. So, the emission reduction is calculated within the life span of 24 years considering the downtime factor as 0.8 conservatively.

Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
M5.0 Strengthened institutional and regulatory systems	M5.1 Institutional and regulatory systems that improve incentives for low-emission planning and development and their effective implementation	Final evaluation report and Mid-term evaluation consisting of Monitoring and evaluation of Annual Project reports and Monitoring and evaluation of Annual Municipal reports	# LG don't have dedicated human resource for energy/climate change # All of the LGs in the target area lacks MEP	# All 150 CCS implemented LGs have dedicated CCS mobilizer # MEP developed in at least 38 CCS implemented LGs.	# All 150 CCS implemented LGs have dedicated CCS mobilizer. # MEP developed in all 150 CCS implemented LGs. An increase in access to CCS by 70 % as a result of development of MEP and deployment of CCS mobilizer at Local level.	# The CCS mobilizer will make users aware on benefits of using CCS which increases the willingness of users to use CCS. # LGs are willing to implement the municipal energy plans with focus on the CCS
M7.0 Lower energy intensity of buildings, cities, industries and appliances	M7.1 Energy intensity/improved efficiency of buildings, cities, industries and appliances as a result of Fund support	Impact Assessment Report including Monitoring of verification reports from Independent Designated Operational Entity(DoE) Monitoring	0	# 5,002.80 TJ saved Biogas:280.80 TJ saved Tier 3+ ICS: 3931.20TJ saved Electric cookstoves: 790.80 TJ saved	# 16,179.92 TJ saved Biogas:702 TJ saved Tier 3+ ICS :12841.92 TJ saved Electric cookstoves: 2636 TJ saved	# Target HH will completely use new CCS provided for their cooking purposes

		Report generated by the MIS on the installation of technologies				
		Final evaluation report				
		Monitoring of secondary data from the reports on baseline and targets published by the GoN				

E.5. Project/programme performance indicators

The performance indicators for progress reporting during implementation should seek to measure pre-existing conditions, progress and results at the most relevant level for ease of GCF monitoring and AE reporting. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Database management through MIS as a national monitoring system	#No. of (Management Information System) MIS with real time data collection developed at LGs	Monitoring report on the status of MIS implementation at LGs. Final evaluation report	0	# All the 150 CCS implemented LGs have MIS database system for sex disaggregated data collection	# All the 150 CCS implemented LGs have MIS database system for sex disaggregated data collection and as a result a reliable data source is accessible at municipal	# LG take the ownership of the project and are capacitated to use MIS for data collection.

					level.	
Increase affordability of CCS technology through bulk tendering via reverse auctioning approach	# Percentage price reduction of CCS as compared to business as usual model	Annual Project Report and Mid term evaluation report	0	# Bulk tendering via reverse auctioning is in place	# Bulk tendering via reverse auctioning is in place	# Bulk tendering will reduce the cost of CCS technologies
Increase in proportion of households adapting CCS technology	# No. of HH with LPG shifted to electric stoves # No. of HH with TCS shifted to biogas and Tier 3+ ICS	Final evaluation report and Midterm evaluation report and Impact Assessment Report Monitoring of secondary data from the reports on baseline and targets published by the GoN such as Nepal Living Standard Survey	# 1,108,004 HH LPG users # 11626 HH ECS users # 1,600,221 HH TCS users # 89,983 HH biogas users	# 958,004 HH LPG users # 161,626 HH ECS users # 1,446,221 HH TCS users # 93,983 HH biogas users	# 608,004 HH LPG users # 511,626 HH ECS users # 1,100,221 HH TCS users # 99,983 HH biogas users	# Demand side of the CCS is strengthened with increased awareness of the CCS # Reliable supply of grid is ensured # Behavior change of HH towards the CCS. # Easy access to service centers for repair and maintenance of stoves # Users are oriented on safety measures on operating the stoves
Quality assurance mechanism for CCS is	# Standards on CCS strengthened	Final evaluation report, Midterm evaluation	0	# National standards on CCS updated	# Updated national standards on CCS implemented	# Government willingness to implement CCS standards

operation al	# No. of testing centres strengthened # Independent verification and third party monitoring is in place and operational	report and Impact Assessment Report		#1 CCS testing centre strengthened at national level # A gender inclusive, independent verification and third party monitoring agency is in place	#1 CCS testing centre strengthened at national level that provides additional and regular service to the users as per the demand. # At least 10% of the deployed CCS technologies verified through independent monitoring verification	# Strengthened testing centre will provide additional and regular service as per the demand # Third party monitoring will be conducted annually for the HH where installation was conducted preceding year
A supply chain mechanism for CCS is established and functional	# No. of service centre strengthened at provincial level ensuring participation of women technicians # No. of after sales service providers established and strengthened ensuring participation of women technicians	Annual Project Report Mid-term evaluation report and Impact Assessment Report Final evaluation report	0	# 4 service centre strengthened at provincial level # 275 after sales service providers established and capacitated to provide continued after sales service	# 4 service centre strengthened at provincial level # 900 after sales service providers established and capacitated to provide continued after sales service.	#The province level service centres will coordinate and support in project implementation # Users will receive regular after sales service as per their requirement # Demand side of the CCS is strengthened with increased awareness of the CCS

						#Increased willingness to pay among the beneficiaries
Sub national institutions facilitated CCS promotion at local and provincial level	# No. of LGs with MEP # No. of LGs with dedicated CCS facilitator # No. of PGs with dedicated energy units # No. of clean cooking champions	Project annual report and Midterm evaluation report and Impact Assessment Report Final evaluation report	0	# 38 LGs with MEP # 150 LGs with dedicated CCS mobilizers (50% women) # 4 PGs with dedicated provincial project management units (50% women) # 450 clean cooking champions (60% women) mobilized at LGs	# 150 LGs with MEP # 150 LGs with dedicated CCS mobilizers (50% women) # 4 PGs with dedicated provincial project management units (50% women) # 450 clean cooking champions (60% women) mobilized at 150 LGs	# LGs and PGs provides supports in Project implementation # LGs are willing to develop and implement MEP # Concerned stakeholders proactively participate in the project
Household fully adapted CCS technologies with changed behavior	# % of HH with continued use of CCS technology # No. of awareness campaigns and promotional activities	Project annual reports Mid-term evaluation report and Impact assessment report Final evaluation report	0	# 90% of HH with continued use of CCS technology # 120 gender friendly awareness campaigns and promotional	# 90% of HH with continued use of CCS technology # 300 gender friendly awareness campaigns and promotional activities	# Users proactively participate in the awareness campaigns # Awareness campaigns and training programs (gender

				activities		sensitized) will lead to behavior change of the users
Economic , social and environmental co-benefit	# New Job creation in CCS sector	Project annual report, Final evaluation report and midterm evaluation report	0	# 760 new jobs (50% women)	# 3225 new jobs (50% women)	# At least 90% of the trained persons will work in CCS sector
	# Cost savings from the use of CCS	Project annual reports Mid-term evaluation report and Impact assessment report Final evaluation report Monitoring of secondary data from the reports on baseline and targets published by the GoN	0	# Households 39.26% expenses reduced for fuel wood compared to baseline	# Households 39.41% expenses reduced for fuel wood compared to baseline	# CCS technologies implemented are more efficient and emits less/negligible indoor smoke than TCS and LPG
	# Improvement in health and safety as a result of	Mid-term evaluation report and Impact assessment	Baseline to be defined within the first year	# %Decrease in health and safety issues from the use of	# % Decrease in health and safety issues from the use of CCS	

	reduced indoor smoke	report Final evaluation report Monitoring of secondary data from the reports on baseline and targets published by the GoN	of project implementation	CCS		
	# Reduction in the use of traditional biomass as cooking fuel	Mid-term evaluation report and Impact assessment report Final evaluation report Monitoring of secondary data from the reports on baseline and targets published by the GoN	5.04 tonnes/H H/year ⁶⁴	# 30% decrease in the use of fuel wood	# 30% decrease in the use of fuel wood	

E.6. Activities

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.

Activity	Description	Sub-activities	Deliverables
Component 1:	Scaling up the deployment of clean cooking technologies through accelerated investment and market development		
Activity 1.1.1: Development of Annual Procurement and Deployment plan	Based on the demand aggregation of each three CCS technologies, a cluster wise deployment procedure will be prepared at regional or provincial level. Different bidding documents for all three CCS technologies will be prepared to procure the technologies through bulk tendering via reverse auctioning. The tendering process will be inclusive and bottom up that would allow variety of options for consumers, not limited to only one type of stove. Work plan for output based financing and third party monitoring and verification will be prepared prior to deployment of CCS technology.	# Development of cluster wise deployment procedure in relation to demand aggregation # Preparing bidding documents for procurement through Bulk Tendering via Reverse Auctioning # Preparing work plan for Output based financing and third party monitoring and verification	# At least 3 bids for each technology will be called each year.
Activity 1.1.2: Procurement and Deployment of annual targeted number of CCS	The project aims to deploy 490,000 Tier 3+ Improved cookstoves (ICS), 10,000 Domestic Biogas Plants and 500,000 (Electric cookstoves) ECS over 5 years' time. The Procurement of CCS technologies for this project will follow the Bulk tendering via	# Procurement of CCS through Bulk Tendering via Reverse Auctioning and Testing of parameters adhering to prescribed standards # Cluster Based Deployment and Installation of CCS and after sales and repair & maintenance service	# 500,000 electric cook stoves # 490,000 Tier3+ ICS # 10,000 domestic biogas plant

	reverse auctioning approach.		
Activity 1.1.3: Disbursement of initial 60% payment installment to the vendor upon delivery of CCS to respective municipalities	After signing contract with the vendor, an amount not exceeding 20% will be provided to the vendor against a bank guarantee as a mobilization advance. After the local government has received the CCS technology, the total amount, not exceeding 60%, will be disbursed to the vendor by AEPC, LG and beneficiary.	# Independent field verification # LGs Verification	# 1 Installation and commissioning report for the deployed CCS technologies
Activity 1.1.4: Monitoring and Verification of Installed CCS units for Output Based Financing (OBF)	After the installation of the CCS technologies at households, third party will monitor and verify the technical parameters of the installed technologies. After the verification, AEPC will provide 40% of the remaining amount to the vendor through output based payment approach.	# Assessment of beneficiaries in terms of (i) Technical parameters of installed technologies (ii) Mitigation Impact (iii) Adaptation Benefits through independent third party verification (iv) sex disaggregated socio-economic information	# Monitoring and verification reports # At least 19, 19 and 5 third party monitoring and verification reports for ICS, Electric cookstoves and biogas respectively.
Component 2:	Strengthen enabling environment through sector based assessments and quality assurance of the technologies		
Output 2.1	Enhancing product standards, conducting assessments, surveys and analysis		
Activity 2.1.1: Develop partnership agreement between AEPC, Province Governments, Local Government and other implementation Partners	AEPC will sign GESI integrated-Memorandum of Understanding (MoU) with PGs and LGs and facilitate PGs to develop and reinforce policy and strategy in promoting CCS and support LGs in implementation of the project. Similarly, AEPC	# Place MoU with the PGs and facilitate PGs to develop and reinforce policy and strategy in promoting CCS # Place MoU with all the LGs in promotion of CCS/implementation of the project # Place MoU with	# MoU with major project stakeholders # 150 MoUs placed with LGs # 6 MoUs placed with PGs #At least 904MoUs

	will also sign MoU with Nepal Academy of Science and Technology (NAST)/Renewable Energy Test Station (RETS) and Nepal Bureau of Standards & Metrology (NBSM) to upgrade existing testing centre at central level, prepare standards and guidelines, and expand after sale service centres at local level.	NAST/RETS, NBSM, financial Institutions and private service providers in promotion of CCS	with private service providers
Activity 2.1.2: Update existing standards⁶⁵ and set benchmarks for the technologies required for the project implementation	The existing standards on CCS will be analyzed and benchmarks will be set for the technologies required for project implementation. As standards on induction Stoves, ICS, and domestic biogas plants and have been set previously, these standards will be reviewed and updated, if needed. In collaboration with Nepal Electricity Authority (NEA), the project will facilitate to introduce cost effective tariff structure for electric cooking and facilitation would be done to ensure quality of electric power supply in the Terai.	# Strengthen/develop necessary standards on ICS, domestic biogas plants and electric cook stoves # Develop manual on CCS standards; Annually update all standards and manuals # Collaborate with NEA to introduce cost effective tariff structure and to upgrade quality of electric power supply	# One (1) National level standard and guidelines developed #Three (3) Manuals with standards on CCS (One (1) each on electric cook-stoves, Tier 3+ ICS and Biogas) # One (1) Feasibility report on upgradation of electric power supply referring the performance report of delivered technologies extracted from the MIS and cost effective tariff structure for electric cooking in the terai analyzing the incremental revenue of NEA due to addition of electric cooking

⁶⁵ Standards on Biomass Cookstoves is as per the National Interim Benchmark for Solid Biomass Cook Stoves,(NIBC, 2016) and National Standards for Household electrical appliances has been recently developed in coordination with Nepal Bureau of Standard and Metrology

			loads
Activity 2.1.3: Develop Municipal Energy Plan template by analyzing standards and formats	The project will develop MEP template based on participatory approach (includes representation from diverse representative groups) by analyzing standards and formats.	# Prepare energy baseline situation # Conduct renewable energy resource inventory # Identify opportunities and challenges and energy intervention Plan	# 1 MEP template
Output 2.2	Strengthening quality assurance mechanisms		
Activity 2.2.1: Strengthening and establishing testing centre for ensuring the quality of the proposed technologies	AEPC will strengthen the existing testing centre located at Nepal Academy of Science and Technology (NAST)/ Renewable Energy Test Stations (RETS) in Kathmandu. It will also collaborate with Nepal Bureau of Standards & Metrology (NBSM) and other relevant stakeholders to build standards and specifications for Tier 3+ Stoves and Electric Cooking Stoves	Lab upgradation and improved testing service as per ISO standards for ECS, and upgradation of National Interim Benchmark of improved Cookstoves 2016 inline to tier based criteria and improved testing service Training and capacity building to all relevant stakeholders (RETS, AEPC, LG, beneficiaries, sub-national institutions, suppliers, repair and maintenance service providers) Network building workshops with Local Governments, universities, Agencies working in Clean Cooking Solutions Knowledge development and dissemination through seminars/ webinars/	#One (1) Testing centre upgraded with facilities to test electric cook stoves and technologies proposed by the project. # 1 Central Level testing centre upgraded to accommodate the proposed technologies # Need assessment reports for providing testing facilities from center level in the seven (7) provinces # One (1) Standard on electric cook-stoves with safety and performance

		symposiums	measures # One (1) Standard on Tier 3+ ICS with safety and performance measures
Activity 2.2.2: Development of Management Information System as a national monitoring system with real time data collection from LGs	Development of Management Information System (MIS) is another critical element of the project wherein the demands from various LGs are recorded, aggregated and distribution/installation is regularly updated along with performance of the disseminated technologies. As AEPC is in the process of developing a MIS accommodating local and provincial level, a fit gap analysis will be conducted of existing MIS and assess the implementation of the modules required for the project implementation. MIS will ensure sex disaggregated socio economic data collection. In addition, training will be provided to LG level personnel on real time data collection and operationalization of MIS.	# Conduct fit-gap analysis of existing MIS # Design and implementation of new MIS at central and sub-national level # Provide training to LG personnel on real time data collection and operationalization of MIS # Database management at LG level on demand aggregation; new installation, and performance parameters	# Fit-gap analysis report of existing MIS # MIS installation at 150 LGs and capacitated LGs for using MIS
Component 3:	Empowering institutions, capacitate supply chain and ensure increased access to clean cooking solutions		
Output 3.1:	Capacity development of sub-national institutions		
Activity 3.1.1: Conduct national	AEPC will conduct a gender inclusive national level stakeholder	# Planning LG's approach in setting up an energy unit	# 4 energy unit at province level

stakeholder consultation workshop between AEPC, PG and LG	consultation workshop with PGs and LGs to plan their approach in setting up an energy unit at regional/provincial and local level. The consultation workshop follows participatory approach with GESI integration and inclusion of diverse representative groups. At provincial level, 4 energy units will be set up with 2 energy officers. At LG level, the project will allocate at least one CCS facilitator in each local government and capacitate them to facilitate the project particularly for door to door awareness campaigns, collection of demands, type of the technologies required at the household level and data management through the MIS.	# Drafting monitoring plan for LGs to support monitoring the CCS in their respective LGs	# Monitoring plan for LGs
<u>Activity 3.1.2: Develop LG Municipal Energy Plan</u>	This will involve support in the development of municipal energy plans for identifying best available resources at local level for energy generation and consumption pattern of households. With the standards and formats developed during component 2, actual MEP will be prepared during this stage. The MEPs will also be instrumental in indicating potential energy resources that could be	# Identify Stakeholders to ensure a broad level of engagement from a cross-sector of energy players # Establish Energy Sub-Committee Team to lead the development implementation and monitoring progress of energy plan # Develop energy baseline, vision, mission and goal and associated activities # Identify funding source # Compile and finalize MEP	# MEP at 150 LGs

	harnessed in the future. This will create an environment wherein MEPs will become a critical element of the annual budgeting and planning process.		
Activity 3.1.3: Capacitating Local and Provincial Governments	AEPC will capacitate LGs to prioritize CCS promotion in periodic and annual plans, formulating appropriate policies and mechanisms.	# Coordinate with the Provincial Governments in creating an enabling environment for the project at province level	# 4 capacity assessment and strengthening report
Activity 3.1.4: Annual national level review and consultation meeting	AEPC will collaborate with its outreach partners at the district or regional level to review the annual objectives and targets and ameliorate MEP guidelines and project materials as required.	# Collaborate with AEPC's outreach partners at the district or regional level to review the annual objectives and targets # Improve MEP guidelines and project materials as required	# 1 Improved MEP guidelines
Output 3.2:	Increased awareness and outreach to enhance demand		
Activity 3.2.1: Mobilize clean cooking mobilizers in each LG	At least 150 CCSmobilizers(50% women) will be mobilized in the project area. They will facilitate the project particularly for door to door awareness campaigns, collection of demands, type of the technologies required at the household level and data management through the MIS.	# Prepare gender friendly ToR for clean cooking mobilizer	# 1 CCS mobilization report of 150 CCS mobilizer at the LGs
Activity 3.2.2: Development of training materials	The consultation workshop will be conducted at national level between AEPC, LGs, beneficiaries, CCs champions, and local vendors and private sector service providers	# Develop gender sensitive extensions materials and guidelines for implementing CCS awareness in targeted project areas # Drafting mechanism to	# At least 3 Training manual on CCS

	to identify the need and develop materials/manual/guidelines on CCS awareness to local communities.	effectively address gender equality and supporting vulnerable communities through CCS	
Activity 3.2.3: Identifying clean cooking champions	At local level, the project will identify 450 clean cooking champions (3 at each local government) who will be trained to advocate on the benefits of adopting clean cooking technology, provide training on CCS installment and use, and contribute to the local level decision-making process. Besides, they will also be trained to facilitate in bottom up need assessment and in designing bespoke technical assistance and capacity development programme for respective LGs.	# Carry out stakeholder mapping exercise at local level to identify the relevant stakeholders and conduct inclusive interaction programme with such relevant stakeholders to develop Terms of Reference (ToR) for clean cooking champions.	# 1 ToR of the clean cooking champions # 1 Report on campaign identification and mobilization of 450 Clean cooking champions at LGs
Activity 3.2.4: Develop materials for awareness on CCS (in national and local languages) with its impact potential on livelihoods and support vulnerable communities:	The project will develop awareness materials on national and local languages which will underpin the activity 3.2.5. The awareness materials will consist of information, education and communication materials on the benefits of CCS in financial savings, cleanliness, health benefits and drudgery reduction compared to TCS. Awareness material will be in form of information on CCS through mass media (Local FM, Electronic media, Print Media, Social Media),	# Disseminate information on CCS through mass media (Local FM, Electronic media, Print Media, Social Media) # Printing and publication of gender sensitive leaflets, user manuals and usage of various avenues of media for promotion of CCS	# Information, education and communication materials highlighting the benefits of the CCS such as Leaflets, user manuals, local FM, electronic media and print media.

	consultation meetings, and distribution of leaflets, user manuals and door-to-door visit to educate and empower users.		
Activity 3.2.5 Campaigns and promotional activities in coordination with the PGs, LGs, community champions, NGOs and volunteers	The project will support and empower users by CCS mobilizers and clean cooking champions for awareness raising on climate change impact, highlighting the benefits of CCS such as fuel saving, cleanliness, health benefits and drudgery reduction compared to TCS by disseminating information on CCS developed by activity 3.2.4 The awareness programme will mainly focus on marginalized communities and households ensuring at least 60% of female participants. CCS mobilizers and champions will conduct behavioral change programme through awareness activities, Operation and Maintenance manuals and cooking guides using CCS.	# Disseminate gender sensitive information on CCS through mass media (Local FM, Electronic media, Print Media, Social Media) # Conduct awareness campaign in each LG on CCS and climate change impact and mitigation measures with focusing women beneficiaries (at least 60 % women) # Distribution of leaflets, user manuals and usage of various avenues of media for promotion of CCS (HH level) # Conduct door to door visit to educate and empower CCS users on benefits of CCS by CCS Facilitators	# At least 5 reports on awareness campaigns conducted at LG # At least 1 CCS user guidelines/ manual for each technology # Leaflets and user manuals for promotion of CCS
Activity 3.2.6: Ameliorate awareness materials and guidelines	AEPC will conduct annual national review and planning meeting with AEPC's outreach partners at the district or regional level. At this stage, the awareness materials and guidelines	# Review and planning meeting with AEPC's outreach partners at the district or regional level # Ameliorate awareness materials and guidelines if required	# Annually published revised awareness materials

	will be ameliorated if required.		
Output 3.3:	<u>Strengthening service centres, biomass manufacturers to provide quality and affordable clean cooking solutions</u>		
Activity 3.3.1: Capacity and training needs assessment	AEPC will conduct the national level stakeholder consultation workshop with AEPC/NAST/RETS/NBS M, beneficiaries, local manufacturers, vendors, financial institutions, LGs, training institutions (e.g. CTEVT) and other relevant stakeholders to assess the existing situation, capacity and training needs assessment of the service centres for providing CCS services. Similarly, based on consultation with the participants, necessary changes will be made in existing training manuals and new training manuals and guidelines will be drafted to support local manufacturers	# Assess the existing situation, capacity and training needs assessment (TNA) of the service centres in providing services for CCS # Drafting gender sensitive training guidelines centered to local manufacturers and guidelines for providing skills training to beneficiaries # Familiarizing with existing training manual and making necessary changes based on consultation with the participants	# At least 4 Training needs assessment report # At least 6 training guideline/manuals
Activity 3.3.2: Ameliorate project materials and guidelines	AEPC will conduct an annual national level coordination and sharing meeting with the project partners and review the progress, draw lesson and planning with the AEPC's outreach partners at the district or regional level. At this coordination meeting, the project materials and guidelines will be ameliorated if required.	# Review the progress, draw lesson and planning with the AEPC's outreach partners at the district or regional level # Ameliorate project materials and guidelines if required	# One (1) Annually published revised project materials and guidelines

Activity 3.3.3: Establish and strengthen service centres at provincial and local level	In collaboration with Nepal Academy of Science and Technology (NAST)/ Renewable Energy Test Stations (RETS); AEPC will strengthen the existing provincial service centre and capacitate them to work as AEPC focal point for their respective local service centres; Local service centres will provide services to the CCS users.	# Develop subsidy model to support installing equipment for service centres # Identify potential local CCS service providers # Support in strengthening technical and institutional capacity of local service providers	# At least 4 province level service centre sand 900 local service centres established in project area
Activity 3.3.4: Conduct trainings to service centre operators and local manufacturers in collaboration with financial and Council for Technical Education and Vocational Training(CTEVT) institutions	Capacity development trainings will be provided to the service providers and local manufacturers, mobile and other electric appliances repair centers, metal workshops, at local level to capacitate them in providing after sales service, domestic production and maintenance of CCS and strengthen the supply chain in coordination with CTEVT institutions. Besides, training on business performance and business continuity plan will be provided to manufacturers and service centres. Capacity development programme will be conducted to promote innovation and R&D. In case of ECS, a technical network will be created to minimize accident such as short circuit.	# Gender inclusive training on the domestic production and maintenance of CCS; reuse and safe disposal of damaged/dysfunctional CCS units # Gender inclusive training on business performance, business continuation plans and others of local partners and suppliers # Conduct match making consultation sessions between local manufacturers, financial institutions, private investors, entrepreneurs	# 24 Training reports for service centre operators and local manufacturers in 5 years # Contract agreement with CTEVT institutions

Activity 3.3.5: Conduct skills development training to CCS beneficiaries and vulnerable groups in collaboration with CTEVT at Province level	Training on making fuel for Tier 3+ ICS, repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring will be provided to CCS beneficiaries and vulnerable and disadvantaged groups (DAG). Training will be provided to women (electric cook stove users) on electrical safety and basic technical skills on using electric stoves to increase their confidence and the use of stove. Self-employment skilled training related to CCS and others as per the local demand will be provided to the most vulnerable people such as widows, single women, Female headed households, minor ethnic groups, lower caste people (Dalit) and indigenous group (Janjati), ensuring at least 33% female participation.	# Training on biomass fuel processing, biogas installation, electrical safety and house wiring, production of pellets and use of bio slurry # Self-employment skilled training to most vulnerable people (at least 33% female) related to CCS and others as per the local demand	# 24 training reports for CCS beneficiaries and vulnerable groups in 5 years
Activity 3.3.6: Strengthen existing regional level service centres under AEPC's past CCS projects into fully operational province level service centres	The existing regional level service centres developed as part of previous CCS programmes of Nepal, will be assessed and strengthened to function as province level service centres. The province level centres will be the focal technical point of	# Support existing regional service centres	# 4 province level service centres established

	contact for all local level service providers. These service centres will have the capacity to provide maintenance and repair services to CCS users as well as local service providers of all CCS technologies.		

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements (e.g. annual performance reports) laid out in AMA, please give a summary of the project/programme specific arrangements for monitoring and evaluation. Please provide the types of interim and final evaluations. Describe Accredited Entity (AE) project reporting relationships, including to the NDA/Focal Point and between AE and Executing Entity (EE) as relevant, identifying reporting obligations from the EE to the AE. This should relate to the frequency of reporting on project indicators, implementation challenges and financial status.

AEPC has set up a Monitoring and Quality Assurance (MQA) Unit within the Technical Support Component, which is responsible for setting up an output-based M & E system. One of the major tasks of the unit is to make regular assessment of progress against the set results and annual plans. It will also cover the monitoring of energy-related climate change as well as socio-economic impacts including Gender and Social Inclusion (GESI).

The MQA unit has been setting up necessary M&E systems and tools such as baseline, MIS, verification, reporting, review & feedback, progress tracking etc. toward instituting an output based M&E system. The output based M&E system will therefore be designed for detailing the operation of feedback mechanism including how it provides management information to different levels of PIU. AEPC also envisages developing an integrated web-based MIS as another vital instrument so that everyone interested will have easy access to the required information.



The basic building blocks of AEPC monitoring system are (a) quality-controlled tools to collect, process and analyze data (b) processes to structure monitoring cycles in every semester (c) roles and responsibilities describing who does what (d) knowledge management, for example, description of minimum data requirements, FAQs, etc.

The Operations Committee of AEPC having representation of AEPC management and Local Government representatives have monthly meetings to discuss the results and issues involved in the CCS programme. For the household energy component, AEPC will be responsible for day-to-day management and monitoring of the three components' activities and results, which are categorized into following five areas:

I. Aggregate Demand Collection Verification and Monitoring

1. AEPC will support Local Governments to strengthen their own demand collection from households, monitoring and evaluation systems to facilitate reporting and quality control. Aggregate demand collection will be collected and categorized for each of the 22 districts and their respective Local Governments.

II. Installation Verification and Monitoring

2. AEPC will conduct lab tests on the quality of Tier 3+ ICS, ECS and biogas during the bulk tendering via reverse auctioning process.
3. AEPC ensures that installation service providers are conducting those activities as per AEPC guideline through inspection by the monitoring team. During installation of CCS, installation service providers are required to maintain prescribed dimensions of CCS, maintain serial numbers, follow CCS installation guidelines and ensure proper usage of CCS. After installing CCS, installation service providers will have an agreement paper with GPS information of location of the household and a one year after-sales service warranty card to customers for repair and maintenance service.
4. Installation service providers are required to send one copy of customer agreement paper to AEPC regional offices. Then the installation service providers will enter the CCS installation data

using a web based software and submit disbursement requests to AEPC. We will have a GPS designated system with phone number backup in place to track daily installation data of various fields. The option of introducing this system would allow for automatic update of installation data in the database maintained at AEPC to avoid false claims. Using this technology, customer satisfaction feedback via text messaging would allow for an easy and cost-effective method for collecting customer feedback, thus ensuring enhanced accountability of the installation service providers for proper service delivery. To better assess customer feedback from clean cooking solutions, several IT based options for enhanced reporting and feedback will be explored for implementation for the project. After receiving disbursement requests from the installation service providers, AEPC inspects at least 10% of CCS submitted under each disbursement request. In case of 20% or more discrepancy in a particular cluster, AEPC excludes all CCS submitted under the disbursement request for that cluster.

5. **Monthly sales sheets** will then be communicated to AEPC regional offices. Monitoring officers will revise the sales information, verify and cross-check information, using plausibility checks and random spot checks using GPS tracker and mobile number. After accepting the data, they will be compiled and transmitted to the AEPC head office, which performs another set of verification and crosschecks. The data then become part of AEPC's six-monthly progress reports. These reports inform AEPC's multi-donor governing board meetings about the current status of target achievement and about areas that need attention.
6. At the province/regional level, the project will deploy at least two staffs, 1- technical and 1- coordinator. The technical staff will look after technical issues and support service centres and vendors and the coordinator will coordinate and monitor the project at province level. Quality Control under the programme, AEPC will establish an inspection team, a call center and a web-based software to keep track of each CCS installed under the programme.

III. Project Midway Implementation Verification and Evaluation

AEPC will provide implementation status report (ISR), audit reports by AEPC, interim evaluation and project evaluation and operational committee reports of AEPC as part of monitoring, aide memoirs as part of the monitoring and evaluation reports after 24 months of the implementation of the project.

1. **Mid-term evaluation/review:** an independent mid-term review will be conducted in two and half year of the project implementation. It will focus on an overall review of AEPC's programmatic approach, of project progress, performance and impact (environmental, social, gender), and of finances. The review will also focus on institutional, administrative and organizational set-up of the project. The mid-term review will be based on desk reviews and interviews with key staff and partners of all projects, and on field visits to two projects. This will evaluate the result of project in the areas of (i) reduction in the use of traditional biomass as cooking fuel (ii) improvement in health and safety as a result of reduced indoor smoke (iii) cost savings from the use of CCS (iv) new job creation in CCS sector (v) % of HH with continued use of CCS technology (vi) no. of awareness campaigns and promotional activities and its impact (vii) effectiveness of the Municipal energy plans and institutional arrangement to oversee energy at the provincial level (viii) switch to the proposed technologies, (ix) reduction in price of CCS with bulk tendering and reverse auctioning and (x) impact of MIS on the overall knowledge management system
2. The mid-term review will include recommendations for (1) any corrective actions if necessary (2) updates to the project design and (3) changes to the project monitoring approach. There will be

individual per-project recommendations and recommendations for the AEPC/GCF project as a whole. The recommendations will be addressed in the second half of the intended project duration. The results of the mid-term review will be presented to the stakeholders.

IV. Impact assessment

1. The project will assess **the quality of access provided applying scientific methodology by developing multivariable impact assessment tools** taking into account the transitional character as well as the complexity of improving access to Clean Cooking Solutions. Improving the cooking situation in the sense of CCS means: considering fuel quality or even switching fuel, improving cooking devices and cooking equipment, adjusting user behavior and cooking practices as well as increasing ventilation and modifying the kitchen. While some internationally discussed measurement approaches are mainly based on stoves and laboratory test results, the AEPC will evaluate quality and effects of access to cooking energy based on field-based approximations and household surveys.
2. Further impact studies will be conducted to assess CCS adoption and daily usage, and the subsequent **actual reductions of fuel-wood use and LPG**. This will be measured by developing and applying a kitchen performance tool kit pertaining to Nepali culinary culture of *Terai*, which is a field test used to evaluate CCS performance amongst the beneficiaries. It will be designed to assess actual impacts on household fuel consumption. Based on the fuel saving results, the real resulting GHG emission reduction can be calculated with the methodology applied in the GCF Impact Model. The emission reduction will be monitored annually.
3. AEPC will conduct market development/making scorecards to calculate sustainability of the project.
4. Project impact monitoring will be performed at the final year of the project period to assess appropriate percentage of the total beneficiaries in terms of mitigation impact through replacement of fuel wood, dung cake and LPG and adaptation co-benefit in terms of fuel and time savings
5. Regular internal evaluation of the PIU targets against completed work as well as yearly administrative tasks will be performed throughout the project period.
6. Further **socio-economic co-benefits** (including gender, income, job creation, or health aspects), as well as potential adaptation impacts will be assessed by household survey.

V. Independent Third Party Monitoring and Verification

AEPC will hire a third party independent agency to monitor and verify CCS technology installed at the household. This third party monitoring and verification would be conducted throughout the project period. They will do their monitoring and verification of the CCS technology against the already defined standard and procedure.

VI. Final evaluation

An independent final evaluation would be conducted within six months after the end of project period. It will focus on overall performance vs. indicators, impacts and finances, and on implementation and

effects of the recommendations given by the mid-term review. This will further present a final evaluation in the areas of (i) reduction in the use of traditional biomass as cooking fuel (ii) improvement in health and safety as a result of reduced indoor smoke (iii) cost savings from the use of CCS (iv) new job creation in CCS sector (v) % of HH with continued use of CCS technology (vi) no. of awareness campaigns and promotional activities and its impact (vii) effectiveness of the Municipal energy plans and institutional arrangement to oversee energy at the provincial level (viii) switch to the proposed technologies, (ix) reduction in price of CCS with bulk tendering and reverse auctioning and (x) impact of MIS on the overall knowledge management system The results of final evaluation will be presented to the stakeholders after finalization of the project.

E. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
Governance	Medium	Medium

Description

Establishing clarity vis-à-vis GoN subsidies, GoN's *Tera*i Clean Cooking Programme and GCF Grant

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Clear demarcation on GoN subsidy support and GCF grant. The GCF project will have separate mechanism through bulk tendering and reverse financing options. A Special Bank Account will be opened in a bank to channel GCF and AEPC fund. The fund will ensure synergy, no duplication of subsidy using monitoring matrix such as name, address, house GPS coordinates, phone numbers and visual confirmation. The fund will adhere to GCF Project Guidelines.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Adherence to the Public Procurement Act and Regulation (PPR/PPA) while proceeding with Reverse Auctioning		
Mitigation Measure(s)		
AEPC being a public entity needs to align its procurements with the Public Procurement Act and Regulation. Careful consideration will be taken while instigating bulk tendering and reverse auctioning. The auctioning will ensure technical quality standards and product specification standards of biogas, Tier 3+ ICS, and ECS are met via NAST/RETS within the lowest price of bid offer.		
Selected Risk Factor 3		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Lack of awareness on adoption of electric cook-stove amongst users/beneficiary perspectives		
Mitigation Measure(s)		
Awareness campaigns for the behavioral change for widespread use of electric cooking technologies will be provided. Furthermore, ECS with multiple cooking plates/burner options will be provided for beneficiaries willing to pay.		
Selected Risk Factor 4		
Category	Probability	Impact
Human Capital	Medium	Medium
Description		
Lack of technical and project implementation skill related to Renewable Energy at the LGs.		
Mitigation Measure(s)		
Local Governments will be heavily engaged throughout the project process with demand generation, use of MIS, project implementation support and monitoring and evaluation. AEPC will coordinate with the municipality for the formation of MEP to enhance the skills of their energy officers		
Selected Risk Factor 5		
Category	Probability	Impact

Policy	Medium	Medium
Description		
Lack of proper guidance for implementation of energy efficiency program		
Mitigation Measure(s)		
Municipal Energy Plan will be designed as such to incorporate energy efficiency programs with enhanced skills of municipal energy officers. Furthermore, energy efficiency guidelines will be incorporated within the AEPC working mechanisms and policy formulation at the centre.		

Selected Risk Factor 6		
Category	Probability	Impact
Technical	Medium	Medium
Description		
Quality of electricity supply in different parts of the country is still poor. Numerous distribution transformers were damaged, and the distribution network could not handle the additional load effectively.		
Mitigation Measure(s)		
The power system reliability and electrification information will be assessed with Nepal Electric Authority and AEPC.		
Selected Risk Factor 7		
Category	Probability	Impact
Technical	Medium	Medium
Description		
Supply Chain Risk		
Mitigation Measure(s)		
Biogas and ICS repair and maintenance service providers are available in the project areas however; there are limited services for Tier 3+ ICS and ECS. Capacity development trainings will be provided to the manufacturers, mobile and other electric appliances repair centers, metal workshops, at local level to capacitate them in providing after sales service and strengthen the supply chain in coordination with CTEVT institutions. The operation and maintenance expenses cost will be borne by the beneficiaries.		
Selected Risk Factor 8		
Category	Probability	Impact

Cost of fuel for tier3+ stove	High	High
Description		
Tier 3+ ICS requires pellets or very small size wood, meaning the households would have to further process their regular fuel wood to fit them into the Tier 3+ ICS, or they may have to buy pellets or wood chips, instead of using the fuel wood they usually collect for free. In either case, this may not result in cost saving or time saving, though emissions saving is there.		
Mitigation Measure(s)		
The project will focus on deploying the Tier 3+ ICS that uses locally available resources, wood chips and multiple fuel source. There will be no separate costs for woodchips, however may require users' time to chip the firewood. For processed fuel, monthly minimum fuel costs are assumed to be lower in tier 3+ ICS than that required for Traditional Cookstoves. Since most of the potential beneficiaries of this project are members/users of community-based forestry, the project will facilitate to provide free or subsidized raw materials for Tier 3+ ICS in particular cases. In addition, the project will provide trainings and technical assistance to lower the fuel production cost.		
Selected Risk Factor 9		
Category	Probability	Impact
Cost of fuel for tier3+ stove	High	High
Description		
The electric Stoves requires specific types of cooking utensils. Consumers will have to buy the utensils themselves, which may affect the switch from LPG to ECS.		
Mitigation Measure(s)		
The estimated cost of e-cooking technology i.e. Rs 4,600 (\$38) is for cost of stove and basic house wiring upgradation. Although this estimation is on lower side in comparison to existing market rate, but because of bulk numbers we are expecting this cost on that level. The costs of utensils have not been considered in this estimation and it is expected as users' contribution, means that users have to manage the utensils as per their own requirements.		
Selected Risk Factor 10		
Category	Probability	Impact
Cost of fuel for tier3+ stove	Medium	Medium
Description		
Power Intake: Cost of upgrading the electricity system at the household, e.g. from 5 amp to 15 amp to run the ECS		

Mitigation Measure(s)		
Basic upgradation costs e.g. for wiring, fuse change, miniature circuit breaker (MCB), have been incorporated in the estimated cost.		
Selected Risk Factor 11		
Category	Probability	Impact
Money laundering/terrorist financing (ML/TF)	Low	Low
Description		
Minimal risk of AML/CFT will be on the area of identification/details of the vendors		
Mitigation Measure (s)		
The fund will be mobilized and monitored through government organization. AEPC adheres to the (Asset (Money) Laundering Act 2008. AEPC will duly identify the relevant private parties through the KYC (Know your Customer) forms. Further as mentioned in the Act, AEPC will maintain records of amount transacted beyond the limit prescribed by Nepal Rastra Bank (NRB) at a single or in a series of transactions by a person, investigate and inquire any transactions which seem to be doubtful or transacted with the motive of assets laundering and inform the Financial Information Unit about the suspicious transactions		
Selected Risk Factor 12		
Category	Probability	Impact
Prohibited Practices	Low	High
Description		
Risk related to prohibited practices including abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices. This also includes risk associated with unauthorized, illicit and improper uses of the technologies.		
Mitigation Measures		
In terms of matters related to integrity and prohibited, AEPC has Compliance and Ethics Sub-Committee and a Compliance Unit to oversee these matters. Trainings are planned on annual basis by the Unit to ensure that AEPC staffs have adequate knowledge on matters related to the area. Trainings related to procurement compliance and public integrity have been conducted in the past by the Unit. Furthermore, AEPC being a government entity is governed by the various national level acts and regulations (Asset (Money) Laundering Act 2008 specifically deals with money laundering and terrorist financing) and also internal regulation on Financial Discipline and Good Governance and Financial Mismanagement and Corruption Prevention Resource Book. In addition, the procedure for fraud reporting and investigation and grievance handling mechanism is duly followed by the Compliance and Ethics Sub-Committee and the Internal Grievance Handling Committee. The system allows report/register frauds and allegations through various means, procedures to conduct investigations, concluding investigations and findings. Similarly, the training will be provided to the project proponents by the Compliance Unit on the aforementioned Acts and Regulations on annual basis. The content of the training will be as per the national acts/regulations and also internal regulations and resource book that have been developed. The internal regulation and resource book comprises of procurement procedures, auditing procedures, internal control mechanism, disclosures, grievance management, fraud reporting and investigation procedure and risk management, <i>inter-alia</i> .		
Furthermore, to avoid the illicit, improper and unauthorized use of any materials or procured under this		

project, the project will have a dedicated accounting system with a special bank account to channel GCF and AEPC fund. The fund will adhere to GCF Project Guidelines and annual auditing procedures will be instilled to ensure that the project assets are used only for the intended purposes.

Selected Risk Factor 13

Category	Probability	Impact
Risk of rebounds	Medium	High

Description

Chances of the beneficiaries of the stoves choose to prefer using old methods over new or underutilization of the stoves.

Mitigation Measure(s)

Nepal's vulnerability to disruptions in its supplies of imported fossil fuels is the principal reason why the government have looked into ways of increasing national energy independence. Since electric cooking is a high priority for the Government and the electricity generation in the country is showing a steady growth further incentives are bound to be rolled out for the promotion of electric cooking. Furthermore, the rebound to old methods will be mitigated due to the low operation and fuel cost of ECS compared to LPG. As the country is moving towards a steady supply of electricity from the hydropower projects, the consumers will prefer a cost effective alternative to cooking. In term of the TCS and Tier 3+ ICS, Tier 3+ is beneficial in terms of annual fuel cost compared to TCS. In addition, the socio-economic and health benefit of Tier 3+ are critical factor that lays as strong foundation for the continuity of the system. The project will address the issue by bringing behavioral change among users by raising awareness about the cost and other benefits of the technologies with campaigns and promotional activities, The project duration of five years also ensures that the change is instigated and maintained after project intervention. In addition, scaling up the community based outreach activities through the locals especially women and disadvantaged group as CCS mobilisers and behavior change activities including online/social media and radio messages, community events, publications (brochures and new articles), and direct communication, with beneficiaries will further mitigate the rebound effect. The capacity development of the supply chain through market strengthening have been embedded as different activities in the project. This will further ensure that there remains no gap at the local level including sales, operation and maintenance.

Selected Risk Factor 14

Category	Probability	Impact
Risk of electrocution/fire incidents	Low	High

Description

Safety issue with potential electrocution/fire incidents with the introduction of electric cookstoves at the household level.

Mitigation Measure(s)

Quality assurance and testing of the proposed technologies are critical proponent of the project. The technologies are required to adhere by the National Standards prepared by the National Bureau of Standard and Metrology that includes standards on General safety requirements on household and electrical appliances, specific safety requirements on induction hobs and specific safety requirements on hotplates. The technical specification provided as annex on the proposed technologies also includes installation requirement that includes standards on power socket, MCB, wiring and cord cover. Furthermore, the project has embedded activities that aim to provide training and awareness related to the electrical safety and house wiring to CCS beneficiaries as well as the supply chain.

F. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Provide the environmental and social risk category assigned to the proposal as a result of screening and the rationale for assigning such category. Present also the environmental and social assessment and management instruments developed for the proposal (for example, ESIA, ESMP, ESMF, ESMS, environmental and social audits, etc.). Provide a summary of the main outcomes of these instruments. Present the key environmental and social risks and impacts and the measures on how the project/programme will avoid, minimize and mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with GCF's ESS standards. If the proposed project or programme involves investments through financial intermediations, describe the due diligence and management plans by the Executing Entities (EEs) and the oversight and supervision arrangements. Describe the capacity of the EEs to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting. Include a description of the project/programme-level grievance redress mechanism, a summary of the extent of multi-stakeholder consultations undertaken for the project/programme, the plan of the Accredited Entity (AE) and EEs to continue to engage the stakeholders throughout project implementation, and the manner and timing of disclosure of the applicable safeguards reports following the requirements of the GCF [Information Disclosure Policy](#) and [Environmental and Social Policy](#).

Describe any potential impacts on indigenous peoples and the measures to address these impacts including the development of an Indigenous Peoples Plan and the process for meaningful consultation leading to free, prior and informed consent, pursuant to the GCF [Indigenous Peoples Policy](#).

Attach the appropriate assessment and management instruments or other applicable studies, depending on the environmental and social risk category as annex 6.

Environmental and social risk category: AEPC did not identify any significant adverse environmental and social risks and impacts of the project. The direct impacts are expected to be minor and would come mainly from site-specific impacts related to domestic biogas plants such as management of bio slurry, availability of water resources to operate the biogas plant and the disposal of damaged and obsolete cook stoves. The Environmental and Social Screening of the project based on the GCF and AEPC's Environment and Social Policy, the project in principle is categorized as C, as the implementation of the project would have only minor and manageable negative impacts.

Environmental and social management plan: The accredited entity has provided an Environmental and Social Management Plan (ESMP) that identifies the potential unintended negative environmental impacts and risks of implementing CCS. The ESMP Institutional arrangement involved in safeguards related issues, their responsibilities, grievance redress mechanism and E & S monitoring. The ESMP also provides a brief description of negative impacts and risks from the project, and have suggested potential mitigation measures.

Environmental and social risks and impacts: The project focuses on "triple benefits" i.e. they reduce emissions contributing to mitigate global climate change, provide improved health and time savings for households and preserve forest and associated ecosystems. Introduction of CCS has a number of social benefits particularly benefitting women. Potential negative environmental impacts from the project implementation are limited to effective management of bio slurry from biogas plants.

The direct environmental and social risks from the project implementation is limited to effective management of bio slurry from biogas plants, disposal of damaged and obsolete cook stoves and availability of water resources to run the biogas plant. If the project is implemented in areas of low water availability, women have to travel longer distances to fetch water to run the biogas plant that has a direct impact on women's health and imposes high time burden particularly to women and girls. Measures to avoid the risk are included in ESMP.

Indigenous people: Ten indigenous nationalities are considered endangered in Nepal, of which 50 % are in *Terai*. Diverse indigenous groups are present in *Terai*, and Tharu is the biggest among them, representing 12 % of the *Terai* population. In *Terai*, there are different groups and speak Maithili, Bajjika, Bhojpuri, Awadhi, closely to north Indian dialects and refer themselves as Madhesi, which also makes up the majority of population in *Terai* region.

The major beneficiaries of the project are indigenous people as the project aims to enhance their adaptive capacity. Thus, the project is not expected to adversely affect the indigenous peoples or ethnic communities. Instead, the project will bring about positive impacts as it will contribute to resource conservation and reduce potential conflicts over resources.

Community health and safety: The project contributes to community health safety by reducing air pollution from using CCS as compared to traditional cooking systems. Use of clean energy sources will reduce the workload of women in collecting fuel wood and leads to better life and improved health conditions. End-user exposure risk, however needs to be monitored through field-testing to ensure that

exposure at the household level will not get worse.

Indirect risks and impacts: There are potential indirect risks and impacts of the project across the industry. The popularity and ease of the use of ICS may lead to more fuel and therefore contribute to increasing pressure on fuel wood. This will need to be monitored through the project's impact evaluation that will regularly assess any changes in the fuel wood use in households.

Stakeholder consultations: In the course of developing the project, stakeholder consultations were held with the identified national and local stakeholders including sectoral ministries, Supply chain stakeholders, Gender and climate experts, private sector groups, academic and research centers. Throughout the process, these stakeholders expressed their views, expectations, concerns and recommendations for the proposed project. Over 150 people have been consulted through various workshops at national and sub national level. Further Stakeholder engagement activities are embedded in the project activities including stakeholder engagement plans and are provided in ESMP (Annex 4).

A **grievance redress mechanism** for the CCS project has been developed to ensure that complaints are adequately addressed in an effective and timely manner. Anyone who has a complaint or who is negatively impacted by the project is able to communicate their complaints or problems through the official mechanism. The Project GRC will be formed at Central/AEPC level. A detail of the grievance redress mechanism for the project is provided in ESMP (Annex 4).

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

The proposal contains a comprehensive and thorough gender assessment, and therefore complies with the operational guidelines of GCF Gender Policy and Action Plan. It also includes a gender action plan to support implementation of issues identified in the gender assessment.

The gender assessment was done based on literature reviews, field survey and stakeholder consultations. This assessment provides in-depth information on the cooking sector with respect to gender. The gender assessment shows Nepal as a country with gender equality issues, however has policies made and in different phases of implementation. Cooking is almost entirely carried out by women, requiring on average about 4 hours a day for cooking using a traditional cooking Stoves⁶⁶.

Women in Nepal spend about 142 minutes to collect the fuel wood whereas men spend only about 50 minutes⁶⁷. Likewise, the study shows the involvement of women in collecting agricultural residue and cow dung is 70.2% whereas involvement of men is only 29.8%, this shows that in collecting of both fuel wood and biomass fuel involvement of women is much more than men⁶⁸.

In Nepal, on average, there are 18.3 *bharis* (head loads of about 30 kg on average) of fuel wood

⁶⁶ Practical Action, 2014. *Gender and Livelihoods Impacts of Clean Cookstoves in South Asia*.

⁶⁷ VenkataRamana Putti, Michael Tsan, Sumi Mehta, & Kammila, S. (2015). *The State of the Global Clean And Improved Cooking Sector*.

⁶⁸ Practical Action, 2014. *Gender and Livelihoods Impacts of Clean Cookstoves in South Asia*.

collection per capita per year in Nepal. On average, a household spends 5.01 hours collecting one *bhari* of fuel wood⁶⁹. Although, there are act and policies to maintain gender equality but social norms still need to be addressed to achieve equality. AEPC has its own Gender equality and Social Inclusion Policy to improve living standard, increase employment as well as productivity of rural women and men, reduce dependency on traditional energy and attain sustainable development through integrating the alternative energy with the socio-economic activities of women and men in the rural communities but implementation is still a challenge.

Gender issues has been inline in most of the energy sector related programmes from the past. Financial privileges are supportive from the policy perspectives, but they haven't been tapped efficiently. Interventions are required to change existing social norms. Political participations are progressive but limited. GESI mainstreaming has been completed in Provincial level. Process of GESI mainstreaming is ongoing in local level and will come into implementation in the subsequent years.

Energy-efficient modern clean cooking solutions (Tier 3+ ICS, biogas, Electric Stoves) have a potential to reduce exposure to smoke related health hazards, reduce the burden on women and children to collect fuel wood, and help the poor save money (when the fuel needs to be purchased).

The gender assessment indicates that women's household work is a barrier to spending much more time engaging other income generating activities, spending more time studying or helping with school work and on entertainment and socializing or engage in the activities they prefer. The project is expected to have major positive impacts on addressing challenges faced by women by ensuring access to, use of Tier 3+ ICS to reduce the burden in collecting fuel wood, and provide women with income-earning opportunities. Also, it identifies employment opportunities for women as Clean Cook Stove mobilizers, Clean Cooking Champions and technically trained women to support maintenance and supply chain, and they will be skilled enough to work in CCS sector even after the project completion. Looking at this, the project has a potential to uplift women in different fronts.

The action plan provides corresponding actions to address the issues that have been identified. The gender focal point will be appointed to coordinate the implementation of gender action plan.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Describe the project/programme's financial management including the financial monitoring systems, financial accounting, auditing, and disbursement structure and methods. Refer to section B.4 on implementation arrangements as necessary.

Articulate any procurement issues that may require attention, e.g. procurement implementation arrangements and the role of the AE under the respective proposal, articulation of procurement risk assessment undertaken and how that will be managed by the AE or the implementing agency. Provide a detailed procurement plan as annex 10.

GoN financial system and procedures will be adopted for this project. In exercise of the power conferred by Section 13 of the Alternative Energy Promotion Centre (Development Committee) Formation Order, 2053 Alternative Energy Promotion Development Board has framed the framed Alternative Energy Promotion Centre Financial Discipline and Good Governance Regulation 2018. This regulation assigns

⁶⁹ ibid

responsibility of operation and execution of financial and administrative duties and functions of the AEPC to the Executive Director. This regulation provides authority for AEPC to include grants in its budget.

The regulation assigns duty and responsibility to all the employees of the AEPC to support the Executive Director in the operation and execution of financial and administrative functions. The budgeting and planning of programmes and activities for the fiscal year shall be responsibility of the Executive Director.

Financial and other provisions related to security deposits on activities, projects and programs, fees, service charges, advance and settlement, preservation of current and fixed assets, auctions procedures, and granting pardons will be based on relevant guidelines and procedures of AEPC under this regulatory framework. In accordance with law, all the income received by AEPC will be deposited into the respective bank accounts. Director of finance will maintain the books of accounts of received incomes, submission of statement of income, ensuring audit and recording of the transactions.

Audit

AEPC will conduct quarterly internal audits in accordance with the Financial General Comptroller Office. The Office of Auditor General shall conduct the final audit of the project. If the findings of the internal and external audit depict loss to AEPC due to mislaid cash and kinds or an act of evasion, the Executive Director or the respective section head upon receiving such information shall recover the amount and based on the severity of the case may take departmental action.

Being government entity the annual audit of the AEPC is under the jurisdiction of the OAG. In addition, the annual external audits of the project will be carried out in accordance with Clause 17 of the Accreditation Master Agreement with GCF and internationally recognized accounting standard will be complied to.

Internal Control

The Compliance Unit of AEPC will serve the oversight function to ensure internal control at the institutional level. Compliance and Ethics Sub-committee shall do the oversight of the unit.

Procurement

APEC will follow the procurement system and procedure of the Government of Nepal. Application of Public Procurement Act (PPA) 2063 and Public Procurement Regulation (PPR) 2064. PPA/PPR shall be enforced in all aspects of the procurement processes with regards to securing products and services from any third parties by the Executing Entities (EEs) for funds received from GCF and other sources.

Procurement of goods services for the EEs generally includes assessing and appointing vendors, contracting and creating purchase orders and making payments. The EEs will procure goods and services from responsible vendors or contractors registered in VAT who possess the ability to provide the goods and services on time, quality product and a competitive price. In the case of the project, AEPC will fulfill the role of both AE and EE with distinct institutional setup. This implies that the procurement

procedure of the EE will also be governed by the PPA 2063 and PPR 2064. AEPC has also prepared a procurement guideline for executing entities⁷⁰, 2018 which aligns with PPA/PPR and provides step by step procedure required for procurement of goods, services and works and acts as a simple procedure with following objectives:

- Maximize the value for money.
- Maximum Returns of public expenditures in an Economical and Rational manner by promoting competition, Fairness, Honesty, Accountability and Reliability in procurement processes.
- Ensure Good Governance by enhancing the managerial capacity of procurement of entities in procuring, or causing to be procured, construction work and procuring goods, consultancy services and other services by such entities and by ensuring the equal opportunity for producers, sellers, vendors, construction entrepreneurs or service providers to participate in public procurement processes without any discrimination.

As the Executing Entity is internal to AEPC, Legal agreement between AE and EE is not required. Government of Nepal's financial system, public procurement procedure and fund flow mechanism will be adopted for this project. As an accredited entity, AEPC will receive the fund from GCF as per the Nepal government procedures. A separate bank account will be opened for the project purpose and a separate financial management guideline will be developed for the transparent and effective management of the financial resources. The Project implementation unit will prepare annual work plan and budget, including a procurement plan, which will be approved by the Ministry. Based on the approval of this plan, PIU will mobilize and release the fund according to the agreements/contracts to the service providers.

The procurements will be undertaken by the EE. In terms of the oversight of the procurements. The preventive measures are taken with the applicability of the PPA/PPR for all the procurements made by the EE. The oversight of the planned procurement and risk assessment will be done by the Compliance and Ethics Sub-Committee and Compliance Unit of the AEPC. Compliance and Ethics Sub-committee is the oversight mechanism instilled at AEPC which directly report to the AEPC Board.

Independent Third Party Monitoring and Verification

AEPC will hire third party independent evaluators after annual deployment target is completed, to do the evaluation of demand, installation, implementation, impact, activities, output and outcome. Independent third party evaluators will do their evaluation against the already established standard and procedures which will be linked with the payment system to the service providers. This third party monitoring would be conducted throughout the project period.

Output Based Financing

The output based financing approach will have several mechanisms set in place. Firstly, the technical specification of the required technologies will be thoroughly tested with the testing centre as entailed by Component 1 of the project. Mobilization advance amount not exceeding 20% will be provided to the vendor against bank guarantee at this stage.

Immediately after the procurement, the technologies will be taken to the respective municipalities. Once the technologies reach the premises of the respective municipalities and upon the clearance letter from

⁷⁰<https://www.aepc.gov.np/uploads/docs/procurement-guideline-for-executing-entities-1571136080.pdf>

the municipalities, disbursement not exceeding 60% will be provided to the vendor. This disbursement also settles the advance amount provided previously. Hence, at this stage, the vendor will be paid a total amount not exceeding 60%.

Once the third party monitoring is completed and confirmation received from the deployed system, final disbursement of 40% will be provided to the vendor. Hence, at this stage, the vendor will be paid the 100% of the total amount.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- € Full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- € Redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

G. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People’s Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☐ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)

- | | | |
|-------------------------------------|----------|---|
| ☐ | Annex 18 | Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project |
| ☐ | Annex 19 | Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity |
| ☐ | Annex 20 | First level AML/CFT (KYC) assessment |
| ☐ | Annex 21 | Operations manual (Operations and maintenance) |
| ☒ | Annex x | Other references |

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)



Government of Nepal
MINISTRY OF FINANCE

(International Economic Cooperation Coordination Division)

Ref: IECCD/MoF/GCF/2020.21/895

SINGHADURBAR
KATHMANDU, NEPAL

To: The Green Climate Fund ("GCF")
Songdo International Business District
175, Art Center-daero
Yeonsu-gu, Icheon 406840
Republic of Korea

21 December 2020

Re: Funding proposal for the GCF by **Alternative Energy Promotion Center** regarding **"Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)"**

Dear Madam, Sir,

We refer to the project **"Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions"** in Nepal as included in the funding proposal submitted by **Alternative Energy Promotion Center** to us on 13 December 2020.

The undersigned is the duly authorized representative of **Ministry of Finance, International Economic Cooperation Coordination Division**, the National Designated Authority of Nepal.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Nepal has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Nepal's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme incorporating in national budget system of Government of Nepal

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Shreekrishna Nepal
Joint Secretary

Secretariat's assessment of FP172

Proposal name:	Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)
Accredited entity:	Alternative Energy Promotion Centre (AEPC)
Country/(ies):	Nepal
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project shows a high cost-effectiveness in terms of USD/tCO ₂ removed compared with other GCF approved projects and International Energy Agency (IEA) average.	The project activities will be scaled up in the other regions of Nepal after the project ends, providing continuous government support of replication is in place. Additional funding sources in partnership with the private sector will need to be identified during project implementation.
The project has high country ownership with support from the Government of Nepal, in addition to directly engaging a minimum of 150 local government units in the region.	
The project has high co-benefits including its support to the most vulnerable households through the use of different technologies, as well as an increase in health benefits resulting from improved indoor air quality.	
This is the first project the direct access entity (DAE) has submitted to the GCF Board for consideration.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVIII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Nepal is highly vulnerable to climate change and is ranked as ninth in the Global Climate Risk Index 2020 list of countries most affected by climate change. The Terai (plains) region of Nepal is the region most affected, in terms of death toll due to disasters, compared with the

other two ecological regions: the Hills and the Mountains. Terai is bordered by the Indo-Gangetic plains in the south and the Churia physiographic region in the north.

4. There is very little use of clean cooking technologies in the Terai. According to the 2011 census, around 75 per cent of the entire country's population used traditional biomass as a fuel source (63 per cent fuelwood, and 10 per cent cow dung (dung cake)) and 21 per cent used liquefied petroleum gas (LPG), whereas, during the same period, 56 per cent of households in the Terai used fuelwood, 22 per cent cow dung and 15 per cent LPG. Burning of solid fuels using traditional cooking stoves causes health problems and emits a large amount of black carbon and greenhouse gases such as carbon dioxide (CO₂) and methane. Even though fuelwood consumption in the Terai has remained fairly constant, LPG consumption is increasing: around 40 per cent of households are using it now. According to the Nepal Oil Corporation import and sales report for financial year 2019/20, Nepal imported and consumed 449,063 metric tonnes of LPG. LPG has to be imported. This has a negative macroeconomic impact in the country because of the widening trade deficit with India, and also makes the country very susceptible to supply risk.

5. Nepal relies heavily on hydropower for its electricity generation, which is a clean and renewable energy source with no CO₂ emissions. Currently it has installed capacity of around 1,278 MW electricity of which 95.7 per cent is from hydroelectricity. Nepal has enormous potential for hydroelectricity, estimated to be around 83,000 MW, and 80 per cent of its population are connected to the national grid. The Terai region has a relatively better electricity access rate than the Hills and Mountains. However, use of electricity for cooking has yet to pick up, as merely 0.08 per cent of the target population relies on it.

6. The total project budget is USD 49.15 million, of which USD 21.1 million is requested from GCF in the form of grants. The accredited entity (AE) is financing USD 35.4 million in the form of grants. The environmental and social safeguards (ESS) categorization of this proposal is assessed to be Category C.

2.2 Component-by-component analysis

Component 1: Scaling up the deployment of clean cooking technologies through accelerated investment and market development (total cost: USD 39.8 million; GCF cost: USD 13.3 million)

7. This component involves procurement and deployment of clean cooking solutions (CCS) technologies with initial support to users, which will directly increase access to CCS. Under this component, this project will introduce procurement and payment delivery methods involving bulk tendering via reverse auctioning and output-based payment that will ensure reduced investment risks for the vendor. The interventions under this component will target 500,000 electric stoves, 490,000 "tier 3+" improved cookstoves and 10,000 domestic biogas plants to be installed.

Table 1. Annual deployment plan

Technologies	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Electric cookstoves	50,000	100,000	135,000	135,000	80,000	500,000
Tier 3+ improved cookstoves	50,000	100,000	130,000	130,000	80,000	490,000
Domestic biogas plant	2,000	2,000	2,000	2,000	2,000	10,000

Component 2: Strengthen enabling environment through sector based assessments and quality assurance of the technologies (total cost: USD 0.5 million; GCF cost: USD 0.5 million)

8. Under this component, national-level technical standards and documents will be prepared for the CCS technologies proposed in this project. The outputs of this component will address technical and operational issues pertaining to adoption of the proposed CCS technology. The project implementation unit (PIU) will systematize procurement and subsidy delivery mechanisms as necessary. Furthermore, this component will focus on strengthening testing facilities and developing quality assurance mechanisms for procuring CCS technologies.
9. The component will include the following activities:
 - (a) Enhancing product standards, conducting assessments, surveys and analysis; and
 - (b) Strengthening quality assurance mechanisms including establishing a testing centre and management information system database.

Component 3: Empowering institutions, capacitate supply chain and ensure increased access to clean cooking solutions (total cost: USD 6.4 million; GCF cost: USD 6.4 million)

10. This component involves capacity-building activities for a number of stakeholders. It will build the capacity of subnational institutions to prioritize CCS promotion in their annual plans, formulating appropriate policies and mechanisms to achieve those targets. The project will also assist local governments in preparing their energy master plans. Multiple awareness campaigns and promotional activities regarding CCS and related benefits will be conducted, targeted towards beneficiaries. The project will also focus on strengthening the CCS supply chain as well as capacity development through vocational training of local technology manufacturers, entrepreneurs and beneficiaries.

Project management (total cost: USD 2.5 million; GCF cost: USD 1 million)

11. GCF grant financing represents 40 per cent of the total project management costs. The requested project management costs amount to 4.7 per cent of the total grant requested from GCF.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

12. The project has a mitigation impact potential of 6.5 megatonnes of carbon dioxide equivalent (MtCO₂eq) reduction (lifetime) from 1 million households in 22 districts adopting CCS technologies. This will lead to estimation of 17,179.92 terajoule (TJ) saved.
13. The project will select a total of 150 local governments out of 287 local governments in Terai and provide support to create enabling environment and increase capacity. the project will support the local governments to develop and implement Municipal Energy Plans (MEPs) and identifying demand linked with online management information system which allows real time data collection and usages of on CCS deployment and usages.

3.2 Paradigm shift potential

Scale: N/A

14. Nepal has been making efforts to shift towards clean fuels and cooking technologies. For the safer and healthier use of biomass, ICS were first introduced in the 1950s and the dissemination of biogas began in the 1980¹. While there has been progress in the usage of clean fuels in both urban and rural areas, the share of population with access to clean cooking solutions remain at 1/3 of the population (IEA, 2019). Use of LPG for clean cooking is higher

¹ Dipti Paudel, Marc Jeuland, Sunil Prasad Lohani, Cooking-energy transition in Nepal: trend review, *Clean Energy*, Volume 5, Issue 1, March 2021, Pages 1–9, <https://doi.org/10.1093/ce/zkaa022>

than use of electricity, showing more than 54.1 per cent of urban households and 16.5 per cent of rural households relying on LPG for cooking. In 2015/16, the use of electricity for cooking in the urban population was 37.9 per cent, only showing a slight increase from that of previous year (33 per cent)². According to ESMAP, 95 out of 100 households in Nepal has access to electricity, while 72 out of 100 have access to reliable, affordable and uninterrupted access to electricity³. Overall, about 71.7 percent of households drew electricity from the national grid, while 23 percent were connected to off-grid sources like micro or mini-hydro and solar power. Nepal government sets a high priority on self-sufficient energy production, particularly hydropower generation. The government of Nepal aims to reach 99 per cent of population with access to electricity to support electricity consumption of 1,500 kWh per capita by year 2030 (MOF, 2020). Taking in the context, the project sets target with a mixed application of clean cooking solutions, electric stoves, “tier 3+” improved cookstoves and domestic biogas plants to better support the adoption and continued use of clean cooking solutions introduced. The project will eventually support shifts towards more “cleaner” fuels and practices for cooking as primary mode of cooking.

15. The project introduces the concept of bulk tendering via reverse auctioning, which is a first for Nepal, along with a results-based financing model for subsidy deployment. The project estimates 20 per cent reduction of the costs and 15 per cent of import tax waiver for e-cookstoves which will remove the barriers by having taxing regime developed. The proposed Results based financing shifts away from the business-as-usual subsidy model and would create a market upfront, provide certain percentage of financial resources up front to the vendors, giving them leverage to service more households. The final disbursement of 40 per cent that is tied to the independent third-party monitoring would be released only after the verification of the technologies, which guarantees a quality assurance.

16. The project will increase the reliability of the technology by improving product standards and establishing measures for quality control and testing centres. The CCS sector is evolving with new entrants in the market. The mini moto Tier 3 stove is a Renewable Energy Test Stations (RETS) approved model and the provision exists for other models to be tested by RETS and implemented in the country. The best available technology will be selected based on the bulk tendering process and the technical specifications of the CCS indicated in the project Operations Manual.

17. Moreover, the project concentrates on the strengthening of the value chain ensuring reliability, ease of access to repair, maintenance, access to fuel and policy interventions related to tariff reduction will further underpin the switch to market led approach. Development of local value chain will enable the user to go to community groups and have issues resolved, whereas in the status-quo, there is no channel available at a local level to address these issues.

18. The project will scale up the Terai local government’s initiative on CCS, which will later expand beyond the Terai region into other regions of Nepal. The project is expected to lead to a market-led CCS sector in the country, and together with the government’s policy support to which will enable behaviour change towards clean cooking. Adoptions of CCS at both rural and urban settings have been a major barrier. The project targets the beneficiaries in the rural areas, where the technology adoption has been one of the major barriers, while LPG drives the cooking in the urban and peri urban areas. The technologies proposed by the project will have proven economic, social and health benefits for the downstream or the demand side. This will instigate the switch to clean and energy efficient technologies and their widespread. Beneficiaries especially in the peri urban and urban areas will switch to the e-cookstoves of their own accord due the efficiency and effectiveness, similar to the trends in increase in usages of LPG. Behaviour change in adoption of clean cooking solutions will be key factor that will lead to expansion and

² https://www.mofe.gov.np/downloadfile/E-cooking%20Assmnt%20NDC%202020_1623998059.pdf

³ <https://www.worldbank.org/en/news/press-release/2019/11/19/first-of-its-kind-world-bank-survey-on-quality-of-electricity-access-in-nepal-shows-remarkable-progress-challenges-persist-on-clean-cooking>

continuous use of CCS without additional subsidy in the future, which this project is to expedite its course.

3.3 Sustainable development potential

Scale: N/A

19. This project contributes to the Sustainable development goal (SDG) target 7.1, by 2030 ensure universal access to affordable, reliable, and modern energy service, through directly addressing one of the indicators, share of the population using modern cooking solutions (by urban/rural). According to IEA (2019)⁴, data shows that as of 2018, the proportion of the world population with access to clean cooking is 65 per cent. As of the same year, the proportion of the population in developing countries with access to clean cooking was 56 per cent. Nepal remains at 30 per cent despite its continuous increase since the baseline year of 2000. This project not only targets to increase access to clean cooking solutions, but also aims to ensure continued use of these solutions at least 90 percent of targeted households.

20. The project has a high potential to generate health benefits by reducing indoor air pollution (IAP) while reducing fuelwood consumption. As of 2019, according to the World Health Organization⁵, 69 per cent of population in Nepal rely primarily on polluting fuels and technologies for cooking. The use of solid fuels and kerosene in households is associated with increased mortality from acute lower respiratory, chronic obstructive pulmonary disease, stroke, ischaemic heart disease, and lung cancer. IAP has been identified as the top-fourth risk factor driving overall morbidity and mortality in developing countries⁶. In Nepal, approximately 8,700 people die each year from exposure to IAP⁷. Studies have shown that the adoption of cleaner fuels and levels of IAP are dependent on household's socioeconomic status and geographical regions. A study shows burning biomass fuel for daily activities such as cooking and heating is common in Nepal, especially among rural households, and that the majority Nepalese households use cheaper, solid (unclean) fuels (such as wood, dung and crop residues) which are cheaper and more widely available. The study also shows that the provinces, where poverty intensity and the percentage of the population in severe poverty are high, also showed high level of exposure to IAP⁸.

21. Women and children will be primarily benefitting from the reduction in IAP that this project will achieve. It is estimated that Tier 3+ improved cookstoves can reduce fuel wood consumption by 30 per cent on average and will reduce more than 30 per cent of IAP. The project will monitor and report on the improvement in health and safety as a result of reduced indoor smoke.

22. The project has a high potential to generate socio-economic benefits also. Reducing firewood consumption for traditional cookstoves means that women will spend less time for collecting firewood and cooking, which in turn increase their ability to participate in income-generating activities. The project aims to reduce 39.41 per cent of expenses spent for fuel woods by the end of the project, in comparison to the baseline identified during project year 1. Increase in local employment in CCS sector is expected, especially in rural areas. One of the project activities is to conduct skills development training to CCS beneficiaries and vulnerable and disadvantaged groups. In total of 1,200 persons will receiving the skill training on making fuel for Tier 3+ ICS, repair and maintenance, biomass fuel processing, biogas installation, electrical

⁴ <https://www.iea.org/reports/sdg7-data-and-projections/access-to-clean-cooking> Accessed on 8 September 2021.

⁵ <https://www.who.int/data/gho/data/indicators/indicator-details/GHO/gho-phe-population-with-primary-reliance-on-polluting-fuels-and-technologies-for-cooking-proportion> Accessed on 9 September 2021.

⁶ https://www.who.int/healthinfo/global_burden_disease/GlobalHealthRisks_report_full.pdf

⁷ Microsoft Word - SEARO Country Profiles_22 Jun 09_v2.doc (who.int)

⁸ Ghimire, S., Mishra, S.R., Sharma, A. et al. Geographic and socio-economic variation in markers of indoor air pollution in Nepal: evidence from nationally-representative data. BMC Public Health 19, 195 (2019). <https://doi.org/10.1186/s12889-019-6512-z>

safety and house wiring, as well as electric safety and basic technical skills on using electric stoves. The project targets to create 3,225 new jobs.

23. Another economic benefit at a national level is in the reduction of trade deficit by reducing LPG imports. The project is expected to reduce LPG imports by about 30,000 tonnes, and saving foreign currency amounting to USD 17.5 million each year.

3.4 Needs of the recipient

Scale: N/A

24. Nepal is a least developed country. Terai, the project location, has a very high population density, accounting for 50 per cent of the total population, with extreme pressure on forests. Previous to this project, the AE distributed mud ICS which had immediate positive impacts, but the durability of mud ICS did not provide a long-lasting impact. Promotion of clean cooking solutions with increased reliability and durability is essential. Training opportunities on CCS technologies would be extended to primarily women and vulnerable and disadvantage groups, including widows, single women, women headed households, minor ethnic groups, low caste people (Dalit) and indigenous group (Janiati).

25. The project has defined beneficiary selection criteria at a high level. The criteria include consideration of income category, affordability factor, access to electricity, demands from the local governments, and quality assurance of technologies. The selection criteria will be defined in detail during the project implementation stage while preparing the standard operating procedure/operational manual for the project.

26. Under the new Constitution of 2015, many government functions were devolved to the (newly established) provinces and local government bodies. These lower levels of government are generally lacking in capacity to apply integrated approaches into regional planning, including climate change forecasts. This project will target 150 local governments, out of the total of 287 local governments in the Terai region. The AE will sign Memorandum of Understanding (MOU) with the provincial governments and the local governments to facilitate development and reinforcement of CCS policies and strategies. The participating local governments will be selected based on demands received during the project implementation stage.

3.5 Country ownership

Scale: N/A

27. In its second Nationally Determined Contribution (NDC) submitted in December 2020⁹, Nepal had indicated its priority in residential cooking and biogas as the following:

- (a) By 2030, ensure 25% of households use electric stoves as their primary mode of cooking;
- (b) By 2025, install 500,000 improved cookstoves, specifically in rural areas; and
- (c) By 2025, install an additional 200,000 household biogas plants and 500 large scale biogas plants (institutional/industrial/municipal/community).

28. These three targets will contribute to emissions reduction of 11% from cooking sector by 2025, from 1,999 GgCO₂eq in BAU (baseline set in 2011) to approx., 1,774 GgCO₂eq.

29. The project is in strategic alignment with the NDC. It meets 90 per cent of the target for improved cookstoves and 36 per cent of electricity goals. With increased generation of electricity, having electricity as a main avenue of cooking is in alignment with the Government's

⁹[https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Nepal%20Second/Second%20Nationally%20Determined%20Contribution%20\(NDC\)%20-%202020.pdf](https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Nepal%20Second/Second%20Nationally%20Determined%20Contribution%20(NDC)%20-%202020.pdf)

energy security strategy. Whilst GCF Country Programme for Nepal has not been yet approved, the project has been identified as a priority of the Government given its strong linkages with the Government led larger Terai Clean Cooking Programme, which was announced in the first year (2019/20) of the Government's 15th five-year plan, with an aim to replace traditional fuels with tier 3 or better cook stoves, modern biogas cookstoves in 22 districts of the Terai region.

30. The funding proposal was developed with full consultation among all relevant stakeholders, including representatives of indigenous peoples.

3.6 Efficiency and effectiveness

Scale: N/A

31. The project exhibits high efficiency per USD of GCF funding in comparison with the relevant benchmarks. The project's mitigation cost-effectiveness is at USD 3.24/Mt CO₂ which is well below comparable GCF portfolio projects, well below DAEs' average, and well below international benchmarks for clean cooking projects (USD 40.92/t CO₂ according to the IEA 2020). The funding amount requested by the AE is thoroughly justified. One of the main strengths of the proposal is a granular analysis of both financing gaps and economic and financial considerations for the interventions under consideration, where the AE provided a detailed description of (1) the allocation of each funding component to each technology, and (2) the unit cost of each technology in the Economic and Financial Analysis.

32. The economic analysis shows that the project will deliver high returns at society level with an estimated 31 per cent Economic Internal Rate of Return (IRR) well above the 9 per cent social discount rate used in the analysis. The economic benefits are balanced between emissions reductions, health benefits and cost savings. These results are robust to the sensitivity analysis developed by the AE.

33. The profitability analysis shows that investment in the clean cooking solutions under consideration would not happen without the support of GCF. The AE developed three scenarios: (1) a scenario where investment takes place without the reverse auction mechanism and without GCF; (2) a scenario where only the reverse auction scenario is considered and (3) a third scenario including GCF support.

34. First, improved cookstove technology shows a negative net present value in the absence of GCF support and in the reverse auction case; the technology becomes profitable with GCF support with not only a good IRR of 115 per cent, but also a reasonable one year payback period suitable for a household investment time horizon. For the electric cookstove technology, the picture is more nuanced. The IRR of the technology is high enough without the GCF's intervention in most of the cases considered by the AE (combinations of assumptions); however, the payback period is too high for a household to invest in the technology. With GCF's support, the technology's payback period drops from an initial 2.4 years to a more suitable 1.3 years.

35. In addition to a thorough profitability analysis, the AE also performed a complementary analysis exploring the impact of the investments on household income. Without GCF support, investing in improved cookstoves or electric cookstoves would account for between 7 and 120 per cent of household income.

36. The AE will deliver the support through local governments and stakeholders. Such delivery model will contribute to the development of local value chains for clean cooking. Combining the GCF grant with this choice of delivery model, the project seems well positioned not only to deliver the intended impact, but also induce a real paradigm shift that will unlock private sector financing.

37. Based on the financial analysis, the GCF intervention is not crowding out the private sector. It will ultimately lead to a reduction of costs of technologies as well as an increased

awareness on the benefits and technology service of clean cooking solutions. In addition, increasing electric cookstove reliability and developing local value chains will further increase the adoption of the technology with opportunities for private sector financing.

38. The capacity-building components will cement the impact of the project and increase the likelihood of improved access to finance for clean cooking solutions in Nepal.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

39. The project is classified as Category C by the AE which the Secretariat confirms and that it is within the AE's accreditation level. While the project is geared towards having positive outcomes, potential minor site-specific negative environmental impacts may arise from improper disposal of used cookstove components as well as that of the bioslurry from the operation of the domestic biogas plants. A management plan is put in place to use the bioslurry as organic fertilizer and is codified in an Environmental and Social Action Plan. The project will likewise promote the reuse and recycling of used/damaged cookstove components while non-recyclables will be disposed of in accordance with existing municipal waste management requirements. Labour issues concerning the vendors/suppliers of the improved cookstoves in relation to avoiding the potential for child and forced labour and occupational health and safety risks to workers will be indicated in contractual agreements. The AE will ensure that these risks are avoided and appropriately mitigated.

40. The environmental co-benefits of the project include reduced indoor air pollution, improved air quality and reduced forest degradation as the improved cookstoves/electric cookstoves and the biogas systems are expected to replace the use of fuelwood and cow dung. The use of cow dung as organic manure is also expected to improve soil quality. The social co-benefits include increased resilience through improved health as a result of decreased indoor air pollution and improved household air quality.

41. The project documents state that the major beneficiaries include the Tharu people of Terai and indicate that there will be no adverse impacts to Janajati or indigenous peoples communities. The gender assessment and action plan states that Janajati will be prioritized in the selection of beneficiary households, among economically impoverished and socially excluded families.

42. The project has documented and provided a summary of the consultations conducted. The AE notes limited initial outreach due to the COVID-19 pandemic and consultations with indigenous and Dalit organizations were conducted via email and telephone. A Stakeholder Engagement Plan (SEP) is also developed which lays down the approach for stakeholder engagement during implementation. It identifies the resources required and the corresponding timing and responsibilities. The SEP provides for the engagement principles, requirements and regulations pertaining to stakeholder consultations and includes establishment of a grievance redress mechanism that allows for complaints to be raised both at the entity and activity levels in addition to having access to the GCF Independent Redress Mechanism. In line with the GCF Indigenous Peoples Policy, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

43. The AE has provided a gender assessment and gender action plan and therefore complies with the requirements of the Gender Policy of the GCF.

44. The gender assessment describes the enabling environment for the pursuit of gender equity and gender mainstreaming in Nepal. The Constitution of Nepal commits to ending gender discrimination and supports equal rights to property, reproductive health rights, and proportional representation at all levels of governance as well as the right to enjoy fundamental human rights for all women. The country has ratified the Convention on the Elimination of All forms of Discrimination against Women (CEDAW) and the Optional Protocol to CEDAW. In 2015, Nepal introduced the Sexual Harassment at Workplace Prevention Act and in 2017 the Criminal Code introduced measures to protect women, children and minorities. In 1980 Nepal created a Women in Development office to support an equitable and prosperous society with strategies to mainstream gender, empower women and facilitate inclusion in governance and development. Over the years this has been supported through the creation of the Ministry of Women, Children and Senior Citizens and a National Commission for Women, and a number of national programmes and development strategies. Gender mainstreaming policies and support in the energy sector has lagged behind until more recently. In 2013, the Renewable Energy Subsidy Policy and Rural Energy Subsidy Delivery Mechanism were developed and have specific subsidies targeted to women and the socially excluded groups of households with single women, disaster victims, people affected by conflict and endangered groups. The National Rural Renewable Energy Program (2012–2017) mandated affirmative action aimed at building capacity and empowering women and marginalized groups in the energy sector. There are also community-level groups and networks that are supporting gender equity, such as FECOFUN a network of Community Forest User Groups, and the Women's Network of Energy and Environment.

45. Despite the existence of the enabling environment women still have less access to information, training and capacity-building than men. The implementation of policy commitments is lagging, and the leadership role of women is undermined in the energy sector as well as across other sectors by prevalent gender norms and discriminatory practices. Women's access to finance is also inhibited by cultural norms, limited access to land and the inability of women to present assets for collateral purposes. The barriers exist not only in the policy and legal framework but also in the institutions, programmes and implementation of interventions in the energy and water sectors.

46. The gender assessment was undertaken on the basis of a desk review while also engaging stakeholders such as government agencies, civil society representatives, members of networks of women rights groups and gender experts. It also has consulted with the private sector in both the supply and demand side, including those led by women.

47. The gender assessment identifies that the engagement and involvement of women in technical and expertise levels in the energy sector is very limited. Access to information and training opportunities, in addition to lower educational attainment and prevalent discriminatory gender roles, contribute towards the limited number of women experts in the sector as well as their inability to be in decision-making positions. On the other hand, the assessment also identifies that women are one of the most vulnerable groups, socially expected to be involved more around household activities, spending 25–50 per cent more time on household tasks, economic and agricultural activities than men. Land and house ownership by females is very low compared with men despite the existence of registration fee waivers.

48. There is a relatively high rate of gender-based violence in Nepal and this is particularly high in Province 2, which is a focal area of the project, although the project does not anticipate any escalation as a result of project intervention. Barriers to equal representation up and down the value chains in the energy sector include traditional social norms biased against women, poverty as a result of limited access to education, early marriage, and limited asset ownership. Furthermore, women's participation is limited in decision-making and income generating activities, and the technical workforce.

49. It is noted that women play a significant role in cooking, and bear the negative effects of fuel collection (142 minutes per day spent on collecting firewood compared with 52 minutes per day by men), transportation, air pollution in the home, time consumption and unsafe cooking technologies. Despite the central role played by women, they are not generally involved in selecting cookstoves. Often, the poorest and most vulnerable communities cannot purchase the tier 3+ improved cookstoves. Even if the stoves are distributed free of cost, fuelwood is another major challenge. Most of the communities use cow dung as fuel, which is not compatible with the available clean stove technologies. Clean technologies compatible with available local fuel sources are essential so that the most vulnerable people can assess the technologies, reducing their exposure to indoor air pollution, improving health and reducing the workload of women.

50. The AE has provided a gender action plan and therefore complies with the requirements of the Gender Policy of the GCF. The gender action plan includes activities that address challenges faced by women and includes a baseline, indicators, targets, timelines and budgets. The project will hire a gender expert to ensure gender issues are integrated and addressed throughout the project and implementation of the gender action plan. The AE will also be conducting further consultations and collecting additional baseline information in the first phase of the project. The activities included in the gender action plan address the supply and demand side of the gender issues discussed. It includes activities to raise awareness on gender issues in the sector as well as to create an inclusive enabling environment. The project is also expected to create employment and volunteering opportunities for women. Skills training and opportunities to participate in technical training will be made available to women and women owned and managed enterprises, and at the same time the AE will be taking measures to ensure that the barriers to women's participation will be minimized. Gender mainstreaming expertise will be made available and integrated into the project's decision-making bodies, training materials, publications, communication tools, policies, management and monitoring and evaluation. The AE is also taking specific measures to ensure gender inclusiveness in the design of normative policies and plans, with strengthened representation from women. Gender mainstreaming training will be provided to local and provincial institutions to change policies and practices to provide more equity in employment opportunities and benefit sharing.

51. While the AE believes there is a low risk of increased gender-based violence, it is strongly encouraged to develop and articulate the specific mechanism(s) that the project will put in place to provide women with easy and safe access to grievance and redress mechanisms in case instances of gender-based violence do occur.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

52. The main objective of the project is to increase the use of clean cooking solutions by: (i) bulk tendering via reverse auctioning for cost effectiveness; (ii) output-based financing for cookstoves; and (iii) mainstreaming and capacitating local governments in the renewable energy (RE) sector. The total project cost is USD 49.1 million, of which GCF will contribute USD 21.1 million, AEPC USD 20.9 million and local governments USD 7 million.

53. The majority of the project costs (USD 39.8 million) will be for the procurement and installation of 500,000 electric cookstoves, 490,000 Tier 3+ improved cookstoves and 10,000 biogas plants. For strengthening the enabling environment USD 0.5 million is allocated, about US 6 million for institution and supply chain capacity development and USD 2.4 million for project management costs. Though Nepal is an LDC, the relatively high IRR for electric cookstoves can support the partial use of a reimbursable grant. However, the accreditation scope of the AE constrains the use of a non-reimbursable grant in the funding proposal.

4.3.2. **Accredited entity/executing entity capability to execute the current project (medium to high risk)**

54. AEPC is the AE and the EE for the project. AEPC, under the Ministry of Energy, Water Resources and Irrigation, is the national focal agency for the promotion of renewable energy technologies in Nepal. AEPC has implemented projects and programmes in the area of climate change mitigation and adaptation, RE promotion and low-carbon economic development worth more than USD 328 million up until 2017. This project is AEPC's first project with GCF. AEPC has provided no information on projects implemented with donor funds since 2017.

55. AEPC will engage with commercial parties (vendors) through bulk tendering and reverse auctioning for services for the clean cookstoves. AEPC has created a project governance with regional project management units and a project implementation unit within AEPC. The implementation unit is guided by the project management committee chaired by the Executive Director of AEPC. To achieve stable adoption of the cookstoves by the end users, it is advised that the AE does a quality check on the vendors' supplies along with focusing on price reduction.

4.3.3. **Project-specific execution risks (medium risk)**

56. Demand risk: below-expected offtake of cookstoves would proportionally reduce the impact of the project. For the electric cookstoves and Tier 3 improved cookstoves, significant cost savings combined with reduced costs as the result of subsidies are expected to support demand. For biogas plants, cost savings for the target beneficiaries are low, which could limit offtake. Also, non-financial behavioural aspects could reduce demand as well as logistical challenges to supply remote communities. Components 2 and 3 are expected to support demand via awareness-raising and supply chain support.

57. Use risk: the impact of the project depends on the stoves being fully utilized during their life. The project does not require households to discard old stoves and therefore households will potentially cook on both systems. The need for special fuel or requirements to process fuel for Tier 3+ improved cookstoves could limit day-to-day adoption. Furthermore, insufficient grid capacity could hamper continued use of electric cookstoves. It is advised to consider requiring that the old stoves be retired to support continued use of the appliances.

58. Sustainability risk: the attractiveness of the cooking solutions is supported by a high level of subsidy of 96 per cent, 89 per cent and 82 per cent of the costs of the Tier 3+ improved cookstoves, biogas and electric cookstoves, respectively. This subsidy will halt at the end of the implementation period leading to an approximately fivefold increase in cookstove prices. This could have demand implications. It is advised to consider a gradual phaseout of the level of subsidy to allow for the gradual transition to a market with lower levels of subsidy.

59. Vendor/supply chain risk: the project presumes significant cost savings as a result of reverse auctioning. Failed auctions, for example due to high efficiency requirements, could increase the price of the units and reduce the number of beneficiaries. Underdeveloped supply chains form a risk for the sustainability of the intervention and successful execution of components 2 and 3 and is therefore an important driver of success of the project.

60. Co-financing risk: the co-financing commitment of local governments (USD 7 million) is pending due to budgetary reasons. Over 150 local governments will provide funds. Co-financing commitment letters will be a condition for disbursement but a delay in providing such letters could delay project implementation.

61. **Project viability and concessionality:** Projected financial IRRs for the electric cookstoves are high at 32 per cent without GCF support and 76 per cent with GCF support and a payback period of less than two years. This IRR calculation includes costs for utensils and costs for new wiring in the house. The AE considers that this high level of concessionality is necessary

to break behavioural barriers. It is advised that the AE considers tapering the subsidy amount over time as acceptance of the technology increases. The projected financial IRR for the Tier 3 cookstoves is high at 115 per cent; partial funding through reimbursable grant can be supported based on the needs of the recipients.

62. Projected IRRs for the biogas plants are negative for the target beneficiaries that will switch from traditional cookstoves, but highly positive (182 per cent) for recipients that already have an LPG installation. Biogas installations ending up in households that already have an LPG installation would result in a reduction of expected impact as impact calculations are based on a shift from traditional cookstoves.

63. The financial and economic analysis is highly sensitive to costs and savings assumptions as provided by the AE.

4.3.4. Compliance risk (high risk)

64. The beneficiary country, Nepal, is not subject to United Nations Security Council restrictive measures.

65. The AE will act as the executing entity (EE) for this project, through a dedicated PIU. Four additional PIUs will be established at the provincial level.

66. The AE has assessed the money laundering, terrorist financing and prohibited practices risks and deemed them to be of low probability and high impact. To mitigate the risks, the AE intends to implement measures required by the national law, including those that pertain to “know-your-customer” procedures. Additionally, the AE places great emphasis on its Compliance and Ethics Subcommittee, which is responsible for overseeing the implementation activities during the project’s lifecycle.

67. The Office of Risk Management and Compliance (ORMC)/Compliance team notes that the AE as well as one of its implementing partners, Nepal Electricity Authority (NEA), have encountered difficulties with similar initiatives in the past. These predominantly concerned project oversight and integrity matters, and on some occasions, allegations of misconduct on the part of the NEA’s employees. These allegations were investigated by the Commission for Investigation of Abuse of Authority (CIAA). In particular, ORMC would like to highlight one instance where the former managing director of NEA was sentenced in 2017 for irregularities regarding procurement procedures; therefore ORMC/Compliance team recommends performing enhanced due diligence for all aspects of the project that may require coordination with the said entity.

68. ORMC/Compliance has conducted a review of the project in accordance with relevant GCF Board approved policies and, based on available information for this funding proposal, has determined a risk rating of “High” and has no objection to this request proceeding to the next steps for processing.

4.3.5. GCF portfolio concentration risk (low risk)

69. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

70. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	

Accredited entity (AE)/executing entity (EE) capability	Medium to high	AE/EE capability is qualified as medium to high as a result of limited information on the track record provided in the last four years. Project risk is medium and is mostly driven by behavioural factors such as continued use and demand for the clean cooking technologies. Concessionality for the project is high as shown by high IRRs and short payback period for some of the end beneficiaries.
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	High	

4.4 Fiduciary

71. AEPC will act as both the AE and the EE and will have a separate PIU to execute the project. As an AE, AEPC will receive the funds from GCF. A separate bank account will be opened for the project, and separate financial management guidelines will be developed for the transparent and effective management of the financial resources.

72. In its capacity as EE, AEPC will set up a dedicated PIU to oversee and manage the project implementation under the strategic guidance of the project management committee. The PIU will prepare an annual work plan and budget, including a procurement plan, which will be approved by the project management committee. Based on the approval of this plan, PIU will mobilize and release the funds according to the agreements/contracts to the service providers. Additionally, the PIU will be responsible for the day-to-day management and monitoring of the project activities according to approved annual work plans and budgets. Finally, the PIU will execute the project following the Government of Nepal's rules and procedures regarding acquiring project services, supplies and works.

73. AEPC will conduct quarterly internal audits in accordance with the Office of the Financial General Comptroller. Because AEPC is a government entity, the annual audit of AEPC is under the jurisdiction of the Office of the Auditor General; as such, the Office of the Auditor General shall conduct the final audit of the project. The Compliance Unit of AEPC will serve the oversight function to ensure internal control at the institutional level, and the Compliance and Ethics Subcommittee will provide the oversight of the unit.

74. APEC will follow the procurement system and procedures of the Government of Nepal. Application of Public Procurement Act (PPA) 2063 and Public Procurement Regulation (PPR) 2064 shall be enforced in all aspects of the procurement processes regarding securing products and services from any third parties by the EE for funds received from GCF and other sources.

4.5 Results monitoring and reporting

75. This project is a mitigation activity aimed at the dissemination of CCS in Nepal. Through the establishment of various facilities and incentive mechanisms, the project aims at inducing behavioural change and supporting household and other users in Nepal in switching from non-renewable biomass and fossil fuel-based cooking solutions to clean cooking options.

76. The project is expected to prevent GHG emissions amounting to 6.5 MtCO₂eq over the project lifetime of 24 years, or an average of 271,000 tCO₂eq per year. The emission reductions have been estimated by closely following approved methodologies under the clean development mechanism (CDM), as well as the information on the share of non-renewable biomass in Nepal reported by the country's designated national authority to the CDM secretariat. The additionality analysis has also been conducted, demonstrating the existence of insurmountable common practice and financial barriers to the dissemination of CCS in Nepal.

77. The theory of change of the project clearly articulates the linkages between project activities, results, outcomes and impacts and demonstrates the transformational pathways of the project in line with the GCF requirements.
78. The logical framework has been designed following the GCF results management framework and performance measurement frameworks, and the correct impact and outcome indicators with appropriate means of verification, baselines and targets have been selected. The results indicators have been designed with consideration of the measurement of the behavioural change induced by the project, thus providing the GCF with an opportunity to better monitor the behavioural change and paradigm shifting nature of its investments.
79. Annex 5 of the funding proposal has been designed in line with GCF requirements, and clear milestones and deliverables under each activity have been identified.
80. Annex 11 has been prepared reflecting the indicators in the logical framework and clearly stating the monitoring budget for each indicator, as well as for the interim and final evaluations.

4.6 Legal assessment

81. The Accreditation Master Agreement was signed with the AE on 7 January 2020 (the “AMA”), and it became effective on 19 June 2020.
82. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project. It is recommended that, prior to submission of the funding proposal to the Board (a) the AE has obtained all its internal approvals and (b) GCF has received a certificate or legal opinion from the AE in form and substance satisfactory to the Fund confirming that all final internal approvals by the AE have been obtained and that it has the authority and capacity to implement the project.
83. The proposed project will be implemented in Nepal, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The Secretariat provided a draft agreement on privileges and immunities to the national designated authority on 15 November 2019.
84. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

85. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:
 - (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed project;
 - (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the

Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and

- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP172

Proposal name:	Mitigating GHG emission through modern, efficient and climate-friendly clean cooking solutions (CCS)
Accredited entity:	Alternative Energy Promotion Centre (AEPC)
Country/(ies):	Nepal
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

Scale: High

1. Nepal's contribution to global historical and current greenhouse gas (GHG) emission is very marginal. The country is however among the most vulnerable countries to climate change. The Global Climate Risk Index 2020 ranked Nepal as the ninth most affected country to climate change. Despite its marginal contribution to global GHG emissions, as reflected in its second nationally determined contribution (NDC-2) Nepal is committed to acting on climate change in line with the Paris Agreement. According to the country's NDC-2 which was submitted to the United Nation Framework Convention for Climate Change (UNFCCC) on 8 December 2020, the Government of Nepal (GoN) believes that efforts to limit global average temperature rise to 1.5 °C would result in significantly lower risks for Nepal compared to an increase of 2 °C or more. Nepal, therefore, calls on all Parties to increase their ambitions and move collectively onto emission reduction pathways consistent with the '1.5 °C warming limit in the Paris Agreement. According to information in Nepal's NDC-2, the country is also committed to delivering its own contributions to global efforts to GHG emissions reduction in line with the international principle of "common but differentiated responsibilities and respective capabilities". The NDC-2 lists areas within Nepal's economy where it intends to conditionally reduce its domestic GHG emissions. The intention to reduce sectoral GHG emissions within the domestic activities in Nepal conditional to having international support were spelled out in the NDC-2 and will cover the following sectors: energy; agriculture, forestry, and land use (AFOLU). A significant percentage of the GHG emission reductions that will be delivered by this intervention will come from the energy sector and in particular: the increased greening of the power generation sector; aggressive introduction of clean cooking solutions (CCS) in residential and commercial sectors; and introduction of e-transport.

2. The proposed project submitted to GCF by the Alternative Energy Promotion Centre (AEPC) and reviewed by the independent Technical Advisory Panel (TAP) involves an aggressive implementation of CCS in Nepal. The 2011 census in Nepal gave the following statistics on cooking fuel use within households: 63 per cent use fuelwood; 10.38 per cent use cow dung, while 21 per cent uses liquefied petroleum gas (LPG). Burning of solid fuels (fuelwood and cow dung) to produce energy required for cooking in Nepal occurs in traditional cookstoves (TCS).¹ Burning of these traditional biomass fuels in TCS is not only very energy inefficient, but also releases particulates and noxious gases that are known to cause various

¹ TCS have an efficiency of <10%. These stoves emit >18.3 g/MJ carbon monoxide and >1031 mg/MJ fine particulate matter. Example: three-stone fire.

health problems in people (mostly women) who use these appliances to cook (mostly in indoor areas). The combustion of that occurs with TCS also emits a massive amount of black carbon and GHGs (such as carbon dioxide, methane). LPG, a fossil fuel is the other common cooking fuel used in Nepal, and according to the information contained in the funding proposal reviewed by the independent TAP, has been growing since the household capture of 21 per cent recorded in the census of 2011. Accordingly, recent national household use of LPG for cooking in Nepal has risen to about 40 per cent. The problem however is that not only does the combustion of LPG emit GHG (CO₂), but the fuel is also imported from India. According to statistics included in the funding proposal, in the fiscal year 2019/2020, Nepal imported and consumed 449,063 Mt of LPG. , Due to a widening trade deficit with India, Nepal has become very susceptible to supply risk, with a very negative macroeconomic impact on the country's economy. This presents a double incentive to pursue a policy of phasing out LPG use in cooking to reduce GHG emission footprint of the cooking sector as well as reducing or eliminating the negative impact on the country's economy.

3. Nepal has a very low-carbon electricity grid, with more than 95 per cent of the power generation connected to the grid consisting of low-carbon hydroelectric power plants. Of the 1,278 MW installed power capacity in 2020, slightly more than 1,250 MW (95.7 per cent) is from hydroelectricity.² Furthermore, Nepal has a significant potential for hydroelectricity, estimated at around 83,000 MW. The claim that electricity supplied to the country's grid is clean and from a renewable energy source with no CO₂ emissions was backed up by additional information supplied to the independent TAP by the accredited entity (AE) in the course of this review. A summary of the information is presented as follows:

- (a) The hydroelectric capacity on the grid of Nepal is made of mostly of run-of- river (ROR) and peaking run-of-river (PROR) capacities. ROR facilities do not have storage capacities.
- (b) Clarifications and data presented to the independent TAP showed that apart from the RORs and PRORs, other hydroelectric facilities connected to Nepal's grid include the Khulekhani-I and Khulekhani-II hydropower plants. Khulekhani-I is the only reservoir-type hydro-electric power station in Nepal with installed capacity of 60 MW having two units each with a capacity of 30 MW. Khulekhani-II hydropower station is a cascade of Khulekhani-I with installed capacity of 32 MW.
- (c) PROR projects have on average, four to six hours daily peaking storage, while by design ROR projects do not have any storage. Therefore, the absence of water storage of the ROR coupled with the continuous flow in the ROR and the daily flushing in the PROR are green by design.
- (d) An analysis of the power densities for the Khulekhani-I, Khulekhani-II as well as for the PRORs (due to their peaking storage facility) showed that the power density for all these hydroelectric plants was greater than 10 Watts/square meters (W/m²) According to the guidance provided in the clean development mechanism large-scale consolidated methodology on grid-connected electricity generation from renewable sources (ACM0002), even the storage capacities, the project emission of these installed hydroelectric generating facilities from water reservoir can be considered as zero.
- (e) These clarifications have been added to annex 22 of the funding proposal package to provide clear evidence that emissions for Nepal's grid electricity are zero.

4. As already mentioned, the goal of AEPC for this submission is to introduce an aggressive programme of CCS in Nepal to replace the status quo of cooking fuel end-use that includes, traditional biomass (fuelwood and dung), and LPG. These will be replaced over a period of time

² Department of Electricity Development, 2020. Operating projects available at < <https://doed.gov.np/>>

by CCS, which will include: tier 3+ improved cookstoves (ICS),³ biogas and electric cookstoves (ECS). Electric cooking is included in the cohort of CCS in this programme based on the evidence that electricity supply from the Nepalese grid is low carbon as described in paragraph 3 above. In addition to this 80 per cent of Nepal's population is connected to the national grid.⁴ In the development of this proposed programme, AEPC took into consideration improvements made relating to barriers that constrained the success of an earlier and similar programme. Prior to the announcement of this present intervention, the GoN on its own, and in some cases in collaboration with development partners, introduced CCS in the country. The Government and external development partners (EDPs) invested around 4.5 million USD for CCS in the fiscal year 2018/2019.⁵ Predominantly, AEPC distributed mud ICS which had immediate positive impacts but in the long run mud ICS are not durable and cannot provide a long-lasting impact.

5. The reasons that have been attributed to the very low success rate of these earlier interventions, having the same objectives as the present AEPC proposed project can be summarized as follows:

- (a) The following barriers ensured that these programmes did not yield tangible results that are expected when the status quo-ante cooking culture in the country is phased out:
 - (i) Socioeconomic barriers to transition to CCS, which included consumer's limited awareness about the benefits of CCS, and subsidy policy on renewable energy and reluctance to pay higher prices due to households' low economic income;
 - (ii) Technical barriers were also identified as reasons for the poor performance of earlier CCS interventions in Nepal. The low level of technical knowledge and unavailability of technicians are major constraints to the shift towards CCS. Very few stove designs are available and consumers face a limited choice of models. In addition, there is a dearth of domestic research and development that can overcome the challenge of inefficient performance of cookstove models and products for user requirements in Nepal. Since there are not many manufacturers and vendors of tier 3+ ICS and ECS in Nepal, the cookers had to be imported from India and China. In addition to the low domestic capacity to manufacture CCS stoves in-country, there is also evidence of limited technical workforce availability in many areas of Nepal, coupled with little involvement of women;
 - (iii) Thirdly, financial barriers also constrained the performance of earlier interventions. Most of the TCS users are among the poorest households, who lack the needed capital to switch to CCS technology and they usually do not have access to finance. Although a few cooperatives provide loans for CCS at community level, the interest rate is usually too high for the poorest households;
 - (iv) Weak institutional infrastructure is another barrier that has been identified as a key reason for the failure of earlier CCS interventions to achieve set goals. The new Constitution of Nepal gave the mandate for the development of renewable energy to the local governments. However, their institutional set-up is weak as regards the necessary technologies. They also lack: technical and financial capacities; awareness on the impact of clean energy; dedicated human resources; and expertise and experience required for planning (energy plans),

³ Tier 3+ ICS has an efficiency of ≥ 30 per cent. These stoves emit ≤ 7.2 g/MJ carbon monoxide and ≤ 218 mg/MJ fine particulate matter.

⁴ Ministry of Finance, 2020. Economic Survey 2076/2077. Available at https://mof.gov.np/uploads/document/file/Economic_Survey_2076-77.pdf

⁵ AEPC, 2018. Budget of FY 2075-76. <https://www.aepc.gov.np/uploads/docs/2018-07-09Budget%20of%20FY%202075-76.pdf>; AEPC, 2019. Progress at Glance: A year in Review FY 2075/76 (2018/19). <https://www.aepc.gov.np/uploads/docs/progress-at-glance-a-year-in-review-fy-207576-201819-1572879503.pdf>

- procuring (demand aggregation and bulk tendering approach), implementing and monitoring renewable energy technologies projects and programmes;
- (v) Although e-stoves were not included in the earlier CCS interventions, the barriers that are likely to limit the success of e-stove adoption in the country were also discussed in the funding proposal. Key barriers that are expected to hinder the utilization of e-stoves in the present intervention are expected to include: poor quality of electricity supply in general in many parts of Nepal and voltage fluctuations in many parts of the Nepalese grid. In addition, although the national average electricity access is about 80 per cent, there are still areas where grid access is still low and this will inevitably limit achievement of the objectives for e-stoves. It is also important to ensure that end users can afford the prices of e-stoves;
 - (vi) Another set of barriers that did not allow earlier interventions to deliver tangible results was the bureaucratic approach used to roll out the CCS. The approach had the following essential elements: setting up an annual target for technology deployment; fixing a certain amount of subsidy for each technology type for households and institutions in groups of districts; and shortlisting of companies (who will supply the CCS equipment) based on a set of qualification criteria to implement those targets. These companies are referred to as pre-qualified companies (PQs): they collect demand from households and institutions and submit installation completion forms to the project manager (PM); the PM, who is a GoN agent, then carries out sample-based monitoring to release the final installment of the subsidy amount to PQs. This approach was found to be bureaucratic and lengthy, and since the price of the technology, and hence the subsidy, was pre-determined by the PM, there was no incentive for the PQs to reduce the cost of technology or propose more innovative and better technology;
- (b) Actions and strategies that aim to address the identified barriers to achieving a successful CCS project have been incorporated into the currently proposed intervention and can be summarized as follows:
- (i) To address the bureaucratic process that did not facilitate earlier attempts at implementing a CCS process in Nepal, the following intervention structure has been built into the proposed project:
 - (1) A more cost-effective process that involves bulk tendering via reverse auctioning, which is expected to de-risk the investment by introducing output-based financing;
 - (2) Adopting an implementation process that has a built-in quality assurance procedure with adequate after-sales service. This would include elements of funding to strengthen existing test centres and expand services at the regional and local levels; and
 - (3) Inclusion of local governments in the process to play crucial roles in improving access to CCS through the development of their capacities;
 - (ii) Project design and implementation planning and procedures for the proposed CCS project included the following elements to address the socioeconomic barriers already discussed:
 - (1) Funding of project elements that have been designed to bring about behavioural changes among users. For example, such elements involve programmes to raise the awareness of household users regarding the

- cost and health benefits of ICS through campaigns and promotional activities, stressing gender parity and empowerment of women;
- (2) Implementation of the project activities in 1 million households in the Terai Region of Nepal during a 5-year period with a strong focus on: pre- and post-purchase behaviour and adequate enforcement of implementation rules that will enable the households to switch to the proposed technologies on their own accord given the behavioural changes coupled with intervention on value chain through awareness and promotional campaigns, promotion of emerging CCS technologies, training and support with enhanced repair and maintenance services and technology availability;
- (iii) Actions built into the implementation of this new CCS project are incorporated to overcome key technical barriers evident in earlier interventions represent a lessons learned strategy and include:
- (1) Inclusion of community-level capacity-building, which supports capacity enhancement at community level, human resources development focused at developing local technical expertise on tier 3+ ICS, biogas and ECS technologies;
 - (2) Strengthening of existing test centres to establish the quality of all the stove types throughout the duration of the programme;
 - (3) Addressing of gender issues on technological aspects through focused development of women's involvement in relevant components of the value chain;
 - (4) Promotion of electric cooking in grid connected households through incentives, which will include the introduction of cost-effective tariff structure for household consumption;
 - (5) Ensuring that all households selected to participate in the programme are connected to the electric grid and have a reliable electric supply;
 - (6) Careful and unambiguous selection of the beneficiaries who will include selected clusters in rural settings and selected clusters in urban and peri-urban areas;
 - (7) AEPC will provide Nepal Electricity Authority (NEA) with beneficiary mapping so that the electricity utility can upgrade capacitors and conductors in overloaded areas. In terms of the medium- and long-term measures, AEPC will work with NEA on increasing distribution transformer size and improving distribution and transmission lines to control voltage fluctuations in the project areas;
 - (8) Through the proposed project, the government will play a major role in setting up reliable infrastructure which will assure the wider usage of electric cooking among households;
 - (9) Electric cookstoves with strong reliability have been proposed in the CCS project in order to diminish the usage of LPG in the country;
- (iv) Actions built into the implementation of the new CCS project to overcome the financial barriers that prevented earlier interventions from being effective include:
- (1) Bulk tendering via reverse auctioning that should reduce the price of CCS through competition;

- (2) The concessional GCF debt and the grants, coupled with counterpart funding in the form of equity from the GoN and funds from other development partners would help to mitigate the financial barriers;
- (3) The procurement will take the approach of bulk tendering via reverse auctioning (selection based on least cost) for cost effectiveness and output-based payment to de-risk the investment;
- (v) Some of the actions included in implementation of the programme to overcome institutional barriers identified as lessons learned from previous interventions will include:
 - (1) Empowering of LGs through their engagement in several aspects of the project including their capacity-building throughout the project implementation phase; and
 - (2) The proposed project includes elements that will build the capacity of LGs and provincial governments (PGs) to prioritize CCS promotion in their periodic and annual planning, formulating appropriate policies and mechanisms.

6. It is expected that the inclusion of these various barrier mitigation elements, derived from previous lessons learned, will facilitate the success of the intervention now submitted to the GCF Secretariat for Board approval.

7. The project will be implemented in the Terai Region of Nepal. Terai is bordered by the Indo-Gangetic plain in the south and the Churia physiographic region in the north. As in other parts of Nepal, penetration of clean cooking solution technologies is very low in Terai. According to the 2011 census,⁶ In Terai 56 per cent of households used fuelwood, 22 per cent cow dung and 15 per cent used LPG. Recent statistics indicate that fuel-wood consumption in Terai has remained constant, while LPG consumption is increasing, with about 40 per cent of households using it. It is stated in the funding proposal that access to grid electricity in Nepal is about 80 per cent of the population and that the Terai region has a relatively higher access rate. However, use of electricity for cooking has a low rate of adoption, as merely 0.08 per cent of the target population relies on it.⁷

8. According to the information provided in the funding proposal and related annexes, the project has been developed around the following three components, which are summarized as follows:

- (a) **Component 1:** Scaling up the deployment of clean cooking technologies through accelerated investment and market development

Under Component 1, the proposed project will support clean cooking solutions in Terai. The activities that will be carried out include:

- (i) Activity 1.1: Development of annual procurement and deployment plan;
- (ii) Activity 1.2: Procurement and deployment of annual targeted number of CCS;
- (iii) Activity 1.3: Disbursement of initial 60 per cent payment installment to the vendor upon delivery of CCS to respective municipalities;
- (iv) Activity 1.4: Monitoring and verification of installed CCS units for output-based financing;

⁶Central Bureau of Statistics (CBS), 2011. Nepal Population and Housing Census 2011.

<https://unstats.un.org/unsd/demographic-social/census/documents/Nepal/Nepal-Census-2011-Vol1.pdf>

⁷Baseline survey of Terai clean cooking programme, 2019.

(b) **Component 2:** Strengthening enabling environment through sector-based assessment and quality assurance of the technologies

Activities that will be implemented under Component 2 will include:

- (i) Activity 2.1: Develop partnership agreement between AEPC, provincial governments, local government, and other implementation partners;
- (ii) Activity 2.2: Update existing standards⁸ and set benchmarks for the technologies required for the project implementation;
- (iii) Activity 2.3: Develop a municipal energy plan template by analyzing standards and formats;
- (iv) Activity 2.4: Strengthen and establish testing centres for ensuring quality of the proposed technologies;
- (v) Activity 2.5: Develop a management information system as a national monitoring system with real time data collection from LGs;

(c) **Component 3:** Empowering institutions' capacity for the supply chain and ensuring increased access to clean cooking solutions

Activities that will be implemented under Component 3 will include:

- (i) Activity 3.1: Conduct national stakeholder consultation workshop between AEPC, PGs and LGs;
- (ii) Activity 3.2: Develop LG municipal energy plan;
- (iii) Activity 3.3: Capacity-building for local governments and provincial governments;
- (iv) Activity 3.4: Annual national level review and consultation meeting;
- (v) Activity 3.5: Mobilize clean cooking mobilizers in each LG;
- (vi) Activity 3.6: Development of training materials;
- (vii) Activity 3.7: Identify clean cooking champions;
- (viii) Activity 3.8: Develop materials for awareness on CCS (in national and local languages) with its impact potential on livelihoods and support to vulnerable communities;
- (ix) Activity 3.9: Campaigns and promotional activities in coordination with the PGs, LGs, community champions, non-governmental organizations and volunteers;
- (x) Activity 3.10: Improve awareness materials and guidelines;
- (xi) Activity 3.11: Capacity and training needs assessment;
- (xii) Activity 3.12: Improve project materials and guidelines;
- (xiii) Activity 3.13: Establish and strengthen service centres at provincial and local level;
- (xiv) Activity 3.14: Conduct trainings to operators of service centres and local manufacturers in collaboration with financial and Council for Technical Education and Vocational Training (CTEVT) institutions;

⁸ Standards on biomass cookstoves are as per the National Interim Benchmark for Solid Biomass Cook Stoves, (NIBC) 2016, and national standards for household electrical appliances have recently been developed in coordination with the Nepal Bureau of Standards and Metrology.



- (xv) Activity 3.15: Conduct skills development training to CCS beneficiaries and vulnerable groups in collaboration with CTEVT at province level; and
- (xvi) Activity 3.16: Strengthen existing regional level service centres under AEPC's past CCS projects into fully operational province level service centres.

9. Financing for the three components described above, and sources of the funding (GCF and co-financing) for the activities, are summarized in table 1.

Table 1: Project components and activities with indicative financing

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial instrument	Amount million USD (\$)	Financial instrument	Names of institutions
Component 1: Scaling up the Deployment of Clean Cooking Technologies through Accelerated Investment and Market Development	<u>Output 1:</u> 500,000 electric stoves, 490,000 tier 3+ ICS and 10,000 biogas plants installed	39.8	13.3	Grant	26.5	Grant	AEPC, LGs
Component 2: Strengthening Enabling Environment through Sector-based Assessment and Quality Assurance of the Technologies	<u>Output 2:</u> Enhancing product standards, conducting assessments, surveys and analysis	0.1	0.1	Grant			
	<u>Output 3:</u> Strengthening quality assurance mechanisms	0.4	0.4	Grant			
Component 3: Empowering Institutions' Capacity for Supply Chain and Ensuring Increased Access to Clean Cooking Solutions	<u>Output 4:</u> Capacity development of subnational institutions	0.7	0.7	Grants			
	<u>Output 5:</u> Increased awareness and outreach to enhance demand	2.7	2.7	Grant			
	<u>Output 6:</u> Strengthening service centres, biomass manufacturers to provide quality and	3.0	3.0	Grant			



Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial instrument	Amount million USD (\$)	Financial instrument	Names of institutions
	<i>affordable clean cooking solutions</i>						
Project Management Component	Establish and Operationalize Project Implementation and Management Units	2.5	1.0	Grant	1.5	Grant	AEPC
Indicative total cost		49.2	21.2		28.0		

Source: From section C.2. – Financing by Components on page 51 of the funding proposal,
Abbreviations: AEPC = Alternative Energy Promotion Centre, LG = local government

10. The total financing needed (GCF financing plus co-financing) for this project is USD 49.152 million. This is expected to be delivered as follows:

- (a) **GCF:** A grant funding of USD 21.127 million; and
- (b) **Co-financing:** USD 28.025 million sourced as follows:
 - (i) Grant funding of USD 20.958 million from AEPC; and
 - (ii) Grant funding of USD 7.065 million from local government.

11. According to the AE as presented in the funding proposal, inclusion of the local government (LG) tier in this project is an innovative inclusion that has not been tried in similar initiatives in the past. Mainstreaming and capacitating local governments in the renewable energy sector is one of the three innovative concepts the AEPC has built into the project to increase the use of clean cooking solutions in Nepal. This will ensure grassroots inclusion in the programme. Strong participation by LGs in the project is shown through the provision of grant funding for the activities of Component 1. The LGs will contribute 14.37 per cent (USD 7,065,442) of the total project budget as co-financing. The LG contribution to the budget of Component 1 is 18 per cent, while GCF grant funding for this component constitutes 33 per cent and co-financing from AEPC is 49 per cent. It has been clearly stated in the funding proposal that the LG contribution is not in-kind but actual payment that will be guided by a memorandum of understanding between the LGs and AEPC that will stipulate the disbursement structure. Furthermore, the LG co-financing contribution will be paid out directly by LGs to private technology providers when the installation targets in the respective LGs are verified.

12. The expected potential outcomes that will be delivered by this project as estimated and presented in the funding proposal and in the annexes submitted for this review have been evaluated by the independent TAP as conservative estimates. These are summarized as follows:

- (a) **Output 1:** 500,000 electric stoves, 490,000 tier 3+ ICS and 10,000 biogas plants will be installed;
- (b) **Output 2:** Enhancing product standards, conducting assessments, surveys and analysis. This will consist of:
 - (i) Municipal energy plan template;
 - (ii) Standards on technical specifications of all three CCS technologies; and

- (iii) A robust quality assurance mechanism;
- (iv) Output 3: Strengthening quality assurance mechanisms. This will consist of:
- (v) Establishing and strengthening a supply chain mechanism for CCS;
- (vi) Setting up a management information system database at AEPC; and
- (vii) Strengthening at least one testing centre at central level; and
- (viii) Output 4: Capacity development of subnational institutions. This will consist of:
- (ix) A municipal energy plan developed by at least 150 LGs;
- (x) Four energy units at provincial level with two officers (1 technical and 1 coordinator) and at least 150 CCS mobilizers allocated at local level; and
- (xi) Demand aggregation and cost-effective reverse auction process introduced in municipal energy plan (MEP) of local government with priority given to CCS promotion; and
- (c) Output 5: Increased awareness and outreach to enhance demand:
 - (i) A total of 450 clean cooking champions identified, three in each local government;
 - (ii) 750 awareness campaigns conducted (one in each LG per year for five years); and
 - (iii) Climate resilience of people and communities enhanced with through savings and awareness of climate change adaptation measures; and
- (d) Output 6: Strengthening service centres, biomass manufacturers to provide quality and affordable clean cooking solutions:
 - (i) 24 trainings provided to beneficiaries in collaboration with CTEVT institutions on repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring (training conducted at province level; one training per province per year from year 2 to year 5);
 - (ii) 24 provincial-level trainings provided to service centres or providers in collaboration with CTEVT institutions (trainings conducted at province level; one training per province per year from year 2 to year 5);
 - (iii) Four provincial service centres established and strengthened; and
 - (iv) 900 local service centres established and strengthened with at least 6 in each local government area;

13. The independent TAP reviewed the methodologies⁹ used in the calculation of the GHG emission reductions expected to be delivered by the interventions of the proposed project. The methodological approach in all the calculations presented in the funding proposal were judged to be sound and the tools used are expected to deliver reliable and conservative estimates of GHG emission reductions. As a result, the project will potentially reduce GHG emission by about 6.51 million tonnes of CO₂ eq¹⁰ during the lifetime of the technologies replacing LPG cylinders with ECS, tier 3+ and biogas CCS.

14. The independent TAP concludes that this project has high climate impact potential.

⁹ UNFCCC methodologies Approved Methodologies Small-Scale (AMS) I.C, AMS II.G and AMS I.E were used to estimate the amount of CO₂ that will be reduced by the project.

¹⁰ The estimation of the GHG emission reduction is provided in supporting annex (annex 23).

1.2 Paradigm shift potential

Scale: High

15. The paradigm shift potential of this CCS project in Nepal is discussed considering the extent to which the project can catalyse impacts beyond the proposed intervention. Previous clean cooking solutions in Nepal were not successful as already discussed and to avoid the same errors and barriers certain changes must be made in the process. Potential paradigm shifts are either built into the implementation process or will be delivered by the project; these are discussed below-

1.2.1. Paradigm shift in the procurement process

16. As stated earlier, inherent barriers have prevented the achievement of expected results in earlier projects of this nature. One of these was the lengthy and inefficient process for procurement of clean cookstoves, with bureaucracy hindering cost-effectiveness. In the project proposed for GCF funding for, the concept of bulk tendering via reverse auctioning, along with an output-based financing model was introduced and is a new procurement process in Nepal. The aggregation of the demand for CCS that will be deployed will grow the concept of bulk tendering, while the reverse auctioning method will significantly reduce the cost of CCS technologies compared to the status quo prior to tendering. Moreover, the output-based financing modality will incentivize the vendors, who are private sector entities, by lowering their risk. This process will also engender multiple check and balance mechanisms to ensure quality assurance and de-risk project investment. The success of this proposed procurement approach should incentivize a change in the business-as-usual public procurement process in Nepal in general.

1.2.2. Potential for scaling up

17. The use of biomass (fuelwood and dung) and LPG as cooking fuels is a source of GHG emissions as well as particulates and noxious gases. GHG gases are the focus of climate change mitigation while noxious gases and particulates are well known to negatively impact the health of the users of the TCSs especially in an indoor environment where cooking is mostly done. Women are clearly more severely impacted as the principle users of TCS. Recognition of these problems has catalyzed various CCS interventions by both the GoN and development partners. However, most of the efforts made to date cannot be scaled up. The proposed CCS project was designed with built-in lessons from these previous efforts and new strategies will mitigate the known barriers. Robust results will support replication of this project in other parts of Terai, and other regions currently not involved in the intervention.

18. The second nationally determined contribution (NDC) of Nepal indicated a national goal for CCS for by 2030. Currently, electric cooking accounts for only 5 per cent and the second NDC aims to ensure that 25 per cent of households use electric stoves as their primary mode of cooking by 2030. The project design has a good potential to contribute significantly to the achievement of this target through a strengthened supply chain. Promotion of the use of e-cookstoves, tier 3+ ICS and biogas stoves should lead to their proliferation across the country without external interventions even after the project period.

19. The adopted procurement process is expected to deliver more cost-effective CCS in Nepal. The impact of this will be felt not only in the affordability of the cooking technologies adopted but also in the quality of these solutions. The involvement of LGs, the administrative level closest to the population should also offer potential for the scaling up the project.

1.2.3. Innovation

20. Innovation is partially delivered through changes to the status quo that result in value creation, reforms and renovation of failing systems, and ensuring sustainability of the process. Many of the implementation components of the CCS currently proposed are innovative and will deliver a paradigm shift in the plan of the GoN to expand the use of CCSs in the country. Compared to the business-as-usual scenario, the procurement plan is an example of one such innovation and it is foreseen that it will deliver a better result compared to the bureaucratic and inefficient system which discouraged the involvement of the private sector. Strengthening of the quality assurance in the implementation plan is another innovative feature of the project.

1.2.4. Potential for knowledge-sharing and learning

21. Potential for knowledge-sharing and learning is well built into the project implementation. Empirical knowledge will be generated through project monitoring of the bulk tendering, reverse auctioning and output-based financing. Component 2 of the implementation also covers development of a management information system, which will serve as a national monitoring system with real-time data collection from LGs. This will offer good potential for knowledge-sharing and provide information that will serve as a learning platform during implementation of the CCS intervention and replication in other regions of Nepal.

22. Various activities included in Component 3 of the implementation scheme will provide platforms for knowledge-sharing and learning. These will include: national stakeholders' consultation workshops; annual national-level reviews and consultations; capacity-building activities; and various training and skill development activities. All these will ensure that the potential for knowledge-sharing and learning of the proposal is robust.

23. The proposed project will deliver comprehensive and innovative attributes that the independent TAP believes will contribute strongly to a high paradigm shift potential. It will deliver an enabling environment that will make the tangible outcomes of the intervention sustainable, while also enhancing replicability of the project concept. The knowledge-sharing platform, which is a sub-component of the project delivery, will enhance robust knowledge-sharing and learning attributes that are essential for the scalability and replication of the project. In the light of these observations, the independent TAP concludes that the paradigm shift potential is high.

1.3 Sustainable development potential

Scale: High

1.3.1. Alignment with the Sustainable Development Goals

24. The Sustainable Development Goals (SDGs) that the project intervention is expected to align with are summarized as follows:

- (a) SDG 1: Poverty reduction;
- (b) SDG 3: Promotion of good health and well-being through reduced exposure to indoor air pollution;
- (c) SDG 5: Reducing the chore for women of collecting wood fuel;
- (d) SDG 7: improving access to affordable and clean energy;
- (e) SDG 13: Climate action by reducing GHG emissions; and
- (f) SDG 15: Conserving the forest resources through reduced fuelwood use for cooking.

1.3.2. Economic co-benefits

25. Economic co-benefits that will be delivered by this CCS project will include:

- (a) Job creation – About 2,400 people will be trained as part of this project to provide services to CCS providers and local manufacturers, mobile and other electric appliance repair centres. Additional jobs will be created in the after-sales service, domestic production of CCS technologies, pellets and briquettes, and in the strengthened supply chain. About 1,200 artisans will be trained in making fuel for tier 3+ ICS, repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring to CCS beneficiaries and vulnerable and disadvantaged groups;
- (b) Increased revenue from cost savings – The procurement process that has been adopted for the CCS project will yield a cost-effective acquisition process that will deliver cost savings. In turn, cost savings will yield increased revenue for businesses involved in CCS commerce;
- (c) Time gained for income-generating activities and education – Women who utilize the TCS technology to cook must also spend time gathering fuelwood from sources near and far. As women begin to utilize more efficient and non-fuelwood tier 3+ ICS stoves they will save time from the chore of fuelwood harvesting and become involved in more beneficial economic activities, while young girls will be free to attend school; and
- (d) Savings at the national level from LPG replacement – LPG is one of the baseline cooking fuels that will be displaced by the CCS technologies of this intervention. LPG is imported from India and China. The project is expected to reduce about 30,000 tons of LPG imported from India and China, representing a foreign currency saving of about USD 17.5 million each year, thus reducing the trade deficit.

1.3.3. Social and health co-benefits

26. The social co-benefits, including health impacts, that are expected to be delivered by the CCS project will include increased resilience through improved health. The use of CCS will reduce indoor air pollution caused by emissions of particulates and smoke when fuelwood and other solid biomass fuel is burnt in indoor kitchens. A more than 30 per cent reduction in indoor air pollution is expected to occur when TCS is replaced with tier 3+ ICS.

1.3.4. Environmental co-benefits:

27. The following environmental co-benefits are expected to be delivered by the proposed CCS project:

- (a) Reduction of indoor air pollution;
- (b) Improved air quality; and
- (c) Reduction of forest degradation.

1.3.5. Gender-related co-benefits

28. Key gender-sensitive development impacts of the CCS project can be summarized as follows:

- (a) Reduced workload and reduced exposure to indoor air pollution;
- (b) Time gained for income-generating activities and education; and
- (c) Training and employment to women, with impacts on revising traditional gender norms.

29. In view of the above-mentioned contributions to various elements of sustainable development, the independent TAP concludes that the project will generate significant sustainable development potential and can deliver all the benefits discussed in section 1.3

above. Therefore, the independent TAP concludes that the sustainable development potential of the proposed project is high.

1.4 Needs of the recipient

Scale: High

1.4.1. Needs of the country

30. Nepal is highly vulnerable to climate change. The country is ranked as the ninth most affected by climate change, as per the Global Climate Risk Index 2020. Although it contributes very marginally to global GHG emissions, the country's climate policies and strategies are allied with the global agreement to take more aggressive steps towards reducing the negative impacts of climate change. Nepal is therefore committed to taking active actions to mitigate climate change impacts both through its own efforts and conditional on availability of international efforts. Nepal, therefore, calls on all Parties to increase their ambitions and move collectively onto emission reduction pathways consistent with the 1.5 °C warming limit in the Paris Agreement. Therefore, the proposed CCS intervention is fulfilling a need of the country for climate action, which will deliver tangible GHG emissions reduction with grant funding from GCF and from the GoN.

31. At baseline, access to clean energy cooking solutions is a massive problem for Nepal. A large percentage of households in Nepal depend on traditional emissive energy sources, such as firewood, animal waste and crop residues for cooking purposes. Currently, about 73.5 per cent of Nepalese households depend on firewood for cooking¹¹ using very inefficient traditional cookstoves (TCS).¹² A common fuel used in TCS is animal waste in the form of dung cake. Dung cake is the most polluting fuel for cooking and is therefore at the bottom of the energy ladder. Dung cakes are produced at the household level and burnt in traditional open fires or mud stoves, resulting in the emission of a dangerous cocktail of hundreds of pollutants to which women and young children are exposed daily. Given that the CCS proposal aims to replace the TCS with efficient and less emitting tier 3+ ICS then the proposed project will meet the need to phase out TCS as described in many of the Government's planning documents (e.g. NDCs). The project will therefore contribute to the Government's strategy to reduce GHG emissions as well as improving indoor air quality. The World Health Organization estimates that there are 8,700 deaths a year from indoor air pollution in Nepal. The project will also meet the need of Nepal to phase out the use of dung as a cooking fuel in TCS.

32. Another fuel used in household cooking at baseline is LPG. While fuelwood is the most common household cooking fuel in the rural areas of Nepal, LPG is the most used cooking fuel in the urban areas. LPG is smoke free and does not emit the cocktail of indoor pollution produced when biomass fuel is burnt. It is however a fossil fuel, which when combusted emits GHGs. Given the global plan to appreciably limit GHG emissions from fossil fuels, it is the intention of the GoN to reduce the use of LPG in household cooking. In addition LPG is a costly import for Nepal from China and India. In 2018–19, Nepal imported 429,609 M_t of LPG, which is a 270 per cent increase since 2008/09.¹³ It is the intention of the Government to curb this increasing dependence on LPG to reduce the country's supply risk (as witnessed during the 2015 fuel crisis). Therefore, the inclusion of LPG substitution in the proposed project meets a highly important need for Nepal.

33. A key alternative in the proposed project is the displacement of LPG as a cooking fuel in Nepal by electric cooking. This will be based on the grid electricity supply which is low carbon. A sizeable percentage of grid-connected electricity generation plants in Nepal are hydroelectric

¹¹ World Bank, 2019. NEPAL | Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework

¹² Ibid

¹³ [See Page 11 of the FP under Section B](#)

power plants, produced mainly with run of river (RoR) systems as well as some peaking run of river (PRoR) capacities connected to the grid. PROR projects have an average of four to six hours daily peaking storage, while ROR by design do not have any storage. Both systems are green in that they do not emit GHG after they have been commissioned. The other hydroelectric facilities connected to the grid of Nepal include the Khulekhani-I and Khulekhani-II hydropower plants. Khulekhani-I with a total installed capacity of 60 MW has storage, while Khulekhani-II is a cascade of Khulekhani-I. The AE provided quantitative information on the power density of Khulekhani-I and the PRoR during submission of the independent TAP evaluation. The information on the power density of Khulekhani-I showed that it does not emit GHGs from its water storage dams. Thus there is valid evidence that electricity used for the e-cookstoves will not lead to GHG emissions. Thus, the inclusion of electric cookstoves in household CCS meets the needs of Nepal as noted in some of the national planning documents.

1.4.2. Economic and financial needs of the country

34. It can be concluded that the financial structuring of the project has been designed in a way that meets the country's economic and financial needs. The economy of Nepal in recent years has been suffering from fiscal weakness due to fiscal deficit from imports of LPG from India and China. Nepal being a least developed country and one of the most vulnerable countries to climate change, the project aims to address the vulnerability and exposure to climate risk and considers the economic and social development level of the beneficiaries. Thus, the procurement policy for the CCS will ensure the cost-effectiveness of the process, which will benefit the private sector entities involved in the project. The low-income users who will pay for the project equipment will also benefit; a survey of the Nepalese people carried out in 2018 shows that in the Terai region, the average monthly income of households can be as low as USD 250 per annum. The proposed project design will help to achieve the GoN aim to provide CCS equipment to low-income households.

1.4.3. Absence of alternative sources of financing

35. It can be argued that since previous efforts made to introduce CCS to Nepal attracted funding from both the GoN and international development agencies present in the country, the failure of these programmes was due to inadequate funding. These previous interventions were mostly funded by the GoN and inclusion of additional grant funding from GCF will help to achieve the overall goals of the CCS project.

1.4.4. Needs for strengthening institutions and implementation capacity

36. The need to strengthen institutions in Nepal to ensure the success of the proposed CCS cannot be overemphasized. The intervention aims to include LGs, giving them a strong role in the project, thus extending institutional strengthening.

37. Given the analysis of the project information made available for the review of this project submission, and the discussion of the issues presented in this section, the independent TAP concludes that Nepal's CCS project will deliver the estimated decarbonization of the household cooking sector, while incentivizing the shift to CCS, including using low-carbon grid electricity for cooking. This is expected to deliver a robust economic and financial scenario that will make the project concept successful. Therefore, the independent TAP has scored the needs of the recipient as high.

1.5 Country ownership

Scale: High

1.5.1. Evidence of country ownership through participation of relevant stakeholders

38. Country ownership is strongly related to national stakeholder engagement with the goals and objectives of the proposed project. The funding proposal provides the following information on the level of stakeholder engagement adopted by the accredited entity (AE) and the Government of Nepal (GoN) in the proposed CCS project. These key issues that we believe provide a strong basis for country ownership are summarized as follows:

- (a) The CCS project has been planned within the context of the GoN's historical commitment to international climate change agreements. The country has a long tradition of being at the forefront of combating climate change. The country's institutions have public policy institutional arrangements, which are delivering various strategies that may be considered precursors of the proposed CCS project. The relevant strategies include:
 - (i) An Investment Prospectus and Biomass Energy Strategy was endorsed by the GoN in 2017. This prospectus and strategy aims at attaining clean cooking solutions for all in two stages, namely, by improved solutions for cooking by 2022 (no traditional cooking using open fire burning), and provision of modern cooking solutions with at least tier 3+ level by 2030;
 - (ii) Nepal's NDC,¹⁴ which has as one of its climate change strategies the reduction of by 50 per cent of fossil fuel dependency by 2050 and equipping households in rural areas with smokeless kitchens;
 - (iii) The Fifteenth Plan (2019/20 to 2023/24) that gives high priority to household biogas, ICS, and production of bio-briquettes and pellets; and
 - (iv) The climate change policy of 2019, which aims to contribute to the socioeconomic prosperity of the nation by building a climate resilient society. The policy has been introduced with the objective of providing policy guidance to various levels and thematic areas to develop a resilient society by reducing the risk of climate change impacts. A relevant key strategy of this policy is the requirement that low emission technologies and adaptation programmes will be incorporated in settlement development plans, which would mean CCS technology would be a key priority.
- (b) As stated in the funding proposal and its annexes, the proposed CCS project was developed with the full consultation of all relevant stakeholders, including representatives of indigenous peoples.
- (c) The high degree of participation by the LGs as a project stakeholder with funding contribution responsibilities is an indication of strong grassroots stakeholders' involvement and good evidence of country ownership.

1.5.2. Alignment of the project with relevant national strategies

39. Some of the key national strategies that the CCS project in Nepal is aligned with include: the second NDC of the country submitted to the UNFCCC in December 2020; the most recent fifteenth Periodic Plan of the National Planning Commission; Nepal's SDG Document; the White Paper, 2018 prepared by the Ministry of Energy, Water Resources and Irrigation. Some of the relevant elements in these plans and strategies are summarized as follows:

- (a) The proposed CCS project is well aligned with the NDC. The project targets have been determined to align with the country priorities. For instance, Nepal's second NDC has priorities in the clean cooking sector intended to ensure that 15 per cent of the total energy demand is supplied from renewable energy sources. It is the opinion of the independent TAP that the CCS project as proposed is in alignment with the relevant

¹⁴ Nationally Determined Contributions, October 2016. Ministry of Population and Environment, Government of Nepal.

objectives for climate change mitigation, energy use in the household sector for cooking, and for the energy sector in Nepal. , GHG emissions from the household sector are targeted for reduction as they represent a leading contribution to national emission inventories;

- (b) The second NDC aims to strengthen transmission and distribution links to support scaling of e-cooking, e-heating, e-transport and charging stations. The second NDC also prioritized targets including that 25 per cent of households will use electric stoves by 2025; 500,000 improved cookstoves will be installed specifically in rural areas; and by 2025 an additional 200,000 household biogas plants will be installed;
 - (c) The fifteenth five-year plan of the National Planning Commission envisions the expansion of renewable energy for multipurpose use. It has prioritized the promotion of biogas, ICS, gasifier, bio-briquettes and ECS;
 - (d) Nepal's SDG document envisions CCS for all by 2030; and
 - (e) The White Paper, 2018, prepared by the Ministry of Energy, Water Resources and Irrigation envisions electric stoves in every house and likewise aims to generate 15,000 MW of electricity by 2028,¹⁵ of which 10,000 MW would be consumed internally, justifying the Government's efforts in promoting electric cooking.
40. The project is a priority of the GoN and is linked to the Government-led national initiative, the Terai Clean Cooking Programme. The project will emphasize adequate engagement of relevant stakeholders, which are aligned with national climate change priorities

1.5.3. Engagement with civil society organizations and other relevant stakeholders

- 41. The funding proposal was developed with full consultation of all relevant stakeholders, including representatives of indigenous peoples.
- 42. The independent TAP is of the opinion that some level of stakeholder engagement during the development phase of the CCS project was demonstrated in the funding proposal. The plan to continue these engagement when the CCS project is being implemented and when it becomes operational was also demonstrated robustly in the funding proposal covering all the different stakeholders whose views should be considered in the development of the CCS project.

1.5.4. Capacity of the accredited entities and executing entities

43. As stated in the funding proposal, the accredited entity (AE) and executing entity (EE) for this project will be AEPC and the Government of Nepal acting through the Ministry of Finance will channel GCF proceeds to the project. AEPC is an independent entity established by GoN under the Ministry of Energy, Water Resources, and Irrigation as the national focal agency for promotion of renewable energy technologies and energy efficiency in Nepal. AEPC has implemented projects and programmes in climate change mitigation and adaptation, promotion of renewable energy, and low carbon economic development worth more than USD 328 million up to 2017. In the fiscal year 2018/19, AEPC conducted CCS projects worth USD 4.58 million wherein approximately 98 per cent of the budget was allocated for technology deployment. AEPC have also had a small-scale experience with the bulk tendering process prior to the development of the proposed CCS project. Bulk tendering needs to be mainstreamed and the support from GCF will be crucial in instigating the process. Thus it can be concluded that by mandate and experience, AEPC is experienced and ready to serve in these capacities.

¹⁵ White Paper 2018 (Clause 55); Ministry of Energy, Water Resources and Irrigation, Government of Nepal.

44. In addition to the issues discussed above, the fact that AEPC will contribute about 60 per cent grant funding that will complement grant funding from GCF and some LGs, is also a strong indication of country ownership. Nepal has made a letter of no objection available for this submission to GCF demonstrating strong country ownership.

45. Based on the analysis presented in sections 1.5.1–1.5.4 above, the independent TAP considers that the country ownership of this LRT project submission can be ranked as high.

1.6 Efficiency and effectiveness

Scale: Medium – High

1.6.1. Financial structure

46. The proposed project is requesting grant financing from GCF and co-financing (also as grant) from the GoN to a total of USD 49.2 million. The GCF contribution amounts to USD 21.2 million all in grant funding. The complementary co-financing of USD 28.0 million will be provided by the GoN specifically from the following sources: USD 21.0 million from AEPC sourced as a grant; and USD 7.0 million from local governments of Nepal, which will be sourced as grant funding.

47. The breakdown of the funding contribution from these sources, the types of fund and the total that will be provided for the project, is presented in table 3.

Table 2: Breakdown of project funding

Number	Source	Type of fund	Amount (USD million)
1.	GCF financing		
a.	GCF	Grant	21.2
Subtotal GCF			21.2
2.	Co-financing		
a.	AEPC	Grant	21.0
b.	Local Governments	Grant	7.0
Subtotal co-financing			28.0
Project total funding			49.2

48. The total fund requested from GCF is about 42.95 per cent of the total fund required for implementation of the project.

49. The grant funding from GCF has been earmarked as follows:

- (a) About 63 per cent of the GCF grant funding will be spent on the activities of Component 1, covering the following activities: procurement, disbursement and installation of 500,000 electric stoves, 490,000 tier 3+ ICS and 10,000 biogas plants;
- (b) Of the GCF grant 2 per cent has been earmarked for activities under Component 2, covering: enhancing product standards, conducting assessments, surveys and analysis and strengthening quality assurance mechanisms;
- (c) Of the GCF grant 30 per cent has been earmarked for activities under Component 3, which will cover: capacity development of subnational institutions; increased awareness and outreach to enhance demand; and strengthening service centres and biomass manufacturers to provide quality and affordable clean cooking solutions; and
- (d) The balance of 5 per cent of the GCF grant was earmarked for project management.

1.6.2. Co-financing, leveraging and mobilizing long-term investment

50. The USD 21.2 million requested from GCF will be complemented by USD 28 million from the GoN through its institutions – AEPC (USD 21.0 million) and the LGs (USD 7.0 million). This will yield an indicative leveraging ratio of 1:33. Thus, 1 USD from GCF leveraged 1.33 USD from public sources in Nepal. This can be considered an effective and efficient use of funds for the CCS project.

51. Other key performance indices of the financing scheme can be summarized as follows:
- (a) Approximately 75 per cent co-financing funds will be provided by AEPC, while the balance will be provided by the participating local governments;
 - (b) GCF financing ensures the project viability and enables the effective use of the leveraged fund for the success of the project;
 - (c) A financial analysis showed that the project can deliver a financial internal rate of return of about 29 per cent with GCF funding, a net present value (NPV) of USD 17.3 million and a discounted payback period of 5.24 years;
 - (d) The financial analysis also estimated an economic internal rate of return of about 69 per cent, an NPV of USD 59.5 million, and a discounted payback period of 3.46 years with GCF funding; and
 - (e) Without GCF funding, the financial analysis indicated that the project will not be feasible as the NPV is negative (USD 38.6 million).

52. The independent TAP is of the opinion that the financial performances discussed in paragraphs 48 – 51 above are an indication of the sound cost-effectiveness of the project when compared with benchmarks of similar projects recently funded by GCF.

1.6.3. Cost-efficiency

53. The project is expected to deliver about 6.51 Mt CO₂ eq of GHG emission reductions over its lifetime. The efficiency metrics that will be delivered by this project intervention were estimated as follows:

- (a) Total project financing = USD 49.2 million;
- (b) Requested GCF amount = USD 21.2 million;
- (c) Expected lifetime GHG emissions = 6,513,629 t CO₂ eq;
- (d) Estimated cost per t CO₂ eq ((a)/(c)) = 7.54 USD / t CO₂ eq; and
- (e) Estimated GCF cost per t CO₂ eq ((b)/(c)) = 3.24 USD / t CO₂ eq.

54. The independent TAP concludes that the GCF mitigation cost of USD 3.24 USD/t CO₂ eq as presented above, when compared with other mitigation projects funded by GCF is cost-effective. There is a possibility that there may be inadequate electricity supply from the national grid to some of the consumers who have been connected to the grid, thus negating the benefits expected from CCS. Inadequate supply from the grid can arise if the demand outstrips supply as a result of a boom in demand from the success of electric stove provision and use. This remains a tangible risk to the cost-effectiveness of the project. The existence of this project risk will reduce the otherwise high effectiveness and efficiency of this project and must be mitigated by regular checks for a supply/demand imbalance.

55. The independent TAP therefore concludes that the efficiency and effectiveness of implementation of this proposed project will be moderately high.

II. Overall remarks from the independent Technical Advisory Panel

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56. The independent TAP recommends that the Board approve this CCS project proposed for implementation in the Terai region of Nepal, subject to the following condition:
57. Delivery by the AE to the Fund, in a form and substance satisfactory to the Secretariat:
- (a) An electricity supply/demand balance study on the distribution network to which the disbursed electric stoves of the project are connected, to be prepared in collaboration with Nepal Electricity Corporation (NEC). Such study shall cover the adequacy of the KWh supplies and the quality of the electricity supply.
 - (b) A report of the actions taken to address the findings of the study must be submitted to the Fund prior to the 2nd disbursement.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP172)

Proposal name:	Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)
Accredited entity:	Alternative Energy Promotion Centre (AEPC)
Country/(ies):	Nepal
Project/programme size:	Small

Impact potential
No comment.
Paradigm shift potential
No comment.
Sustainable development potential
No comment.
Needs of the recipient
No comment.
Country ownership
No comment.
Efficiency and effectiveness
No comment.
Overall remarks from the independent Technical Advisory Panel:
No comment.

Mitigating GHG emission through modern, efficient and
climate friendly clean cooking solutions (CCS)

Gender Assessment and Action Plan

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List of Abbreviations

ADB	Asian Development Bank
AEPC	Alternative Energy Promotion Centre
BFI	Bank and Financial Institution
CCS	Clean Cooking Solutions
CEDAW	Convention on the Elimination of All forms of Discrimination against Women
CRT/N	Centre for Rural Technology Nepal
ENPHO	Environment and Public Health Organization
FCHV	Female Community Health Volunteers
FECOFUN	Federation of Community Forest User group Nepal
FHH	Female Headed Households
GDI	Gender Development Index
GESI	Gender Equality and Social Inclusion
GEM	Gender Empowerment Measure
GII	Gender Inequality Index
GS	Gender Specialist
HDI	Human Development Index
ICS	Improved Cook stove
IP	Implementation Phase
MFI	Micro-finance Institution
MoWCSC	Ministry of Women, Children and Senior Citizen
NDHS	Nepal Demographic and Health Survey
NRREP	National Rural Renewable Energy Programme
TCS	Traditional Cook stove
WDP	Women Development Programme
WoNEE	Women's Network of Energy and Environment

1 Introduction

1.1 Gender Background

The International Women's Year (1975) and the United Nations Decade for Women (1976-85) created a wave of activity for bringing women in the development mainstream worldwide. Post this, a series of the status of women in Nepal studies were carried out throughout 1979-81. The policy to promote women's participation in development appeared for the first time in the Sixth National Plan (1980-85) of the country.¹ Nepal has accessed Convention on Political Rights for women on April 26, 1966, followed by ratification of Convention on the Elimination of All forms of Discrimination against Women (CEDAW) on April 22, 1991. Nepal has also ratified Optional Protocol to the Convention on the Elimination of All Forms of Discrimination against Women on June 15, 2007, signed the Beijing Platform for Action, and the International Conference on Population and Development +10. The treaties established a way forward to mainstream Gender Issues in state laws and policies.

The Constitution of Nepal promulgated in 2015, with the commitment to end gender discrimination, has guaranteed the equal right to property, right to safe motherhood and reproductive health rights, right to ensure proportional representation at all concerned levels as well as the right to enjoy fundamental human rights for all women. A separate code of conduct for the prevention of violence against women and girls has been introduced as the Sexual Harassment at Workplace Prevention Act 2015. Also, the Criminal (Penal Code) 2017 protects women and children, particularly minorities.

Women Development Programme (WDP) created in 1981 as a women development office with a vision for an equitable and prosperous society and prime strategies on mainstreaming, empowerment, and inclusion in the domain of Women's development. Since 2000, WDP has been operating with nationwide coverage. For gender issues and women empowerment, the functional ministry is the Ministry of Women, Children and Senior Citizen (MoWCSC). The Department of Women and Children under MoWCSC was created in 2000 and later reformed in 2010 with a mandate to do functional jobs in Women's development. National Women Commission was founded in 2001 as a constitutional body to protect women's rights, women's welfare, and incorporate women in developmental mainstreams and establish women's rights and gender equity.

Institutional Mechanisms for GESI Policy in Ministries, National Commissions for Women, Dalits, Indigenous Nationalities, Madhesis, Muslims, Tharus and a National Inclusion Commission that is mandated to protect the rights of Khas Aryas, Pichardiaka ("backward") class, persons with disabilities, senior citizens, labourers, peasants, minority and marginalized communities, people of the Karnali, and the indigent class, National Planning Commission, MoWCSC, and the Department of Women and Children in Central Level, with similar units in Provincial Level. GESI Mainstreaming Facilitation in Local Level is in process.

Nepal Demographic and Health Survey (NDHS) had conducted a survey in 2016 which provides reliable estimates at the national level, for urban and rural areas, 3 ecological zones, 5 development regions, and 7 provinces. It has summarised the status of women empowerment in three categories.²

- a. Employment: 68% of married women were employed in past 12 months compared to 97% of married women
- b. Ownership of assets: 8% own a home, and 11% own land alone or jointly; among men, 19% own a home and 21% own land alone or jointly

¹ Department of Women and Children, Nepal <https://www.dwc.gov.np/pages/history>

² <https://kathmandupost.com/national/2017/04/24/presidents-women-empowerment-programme-starts-in-26-districts>

- c. Participation in household decisions: 53% to make decisions about major household purchases

Separate governmental programmes such as ‘The President's Women empowerment programme’ started on March 07, 2017 to establish a gender-responsive governance system and a just society by developing and promoting women’s socio-economic empowerment and entrepreneurship.³ There are many parallel women empowerment programmes conducted at national and local levels by several NGOs.

The ministry of Federal affairs and Local development had introduced a Gender Equality and Social Inclusion Policy in 2009 as an uptake in women empowerment in social inclusivity in Local Development entities. The current 15th national plan has allocated the budget of NRs. 14.98 Billion to mainstream Gender equality and inclusion (GESI) principles into the governance system of provincial and local levels as an expected outcome of Provincial and Local government support programmes from 2019/2020 to 2023/24.

The current 15th national plan has the gender equality and women empowerment vision to “Build Nepal as a gender-equal nation.” It has identified the target indicators to fulfill in the year 2023/24 with reference to baseline data from 2018/19 tabulated below.⁴

Indicators	Baseline Year 2018/19	Target year 2023/24
Women representation in Legislation	33.81%	40%
Women representation in Local Governments	40.75%	41%
Women representation in Public Sector	24.7%	34%
Gender Development Index (GDI)	0.897	0.963
Women Empowerment		
Women Owned Enterprise (of total enterprise)	26%	40%
Household with house and land registered in women’s name	26%	33%
Gender Empower Measure (GEM)	0.57	0.64
Ratio of female and male labour participation rate	0.49	0.98
Ratio of female to male wages	0.49	0.98
Violence against Women		
Women who have victimized lifetime of physical, mental or sexual violence	24.4%	13%

³ Nepal Demographic and Health Survey 2016 <https://dhsprogram.com/pubs/pdf/SR243/SR243.pdf>

⁴ 15th national plan, National Planning Commission
https://npc.gov.np/images/category/15th_plan_English_Version.pdf

The World Economic Forum published a recent report on the Global Gender Gap 2020 where Nepal ranks 101th among 153 countries in the Global Gender Gap Index 2020 rankings ahead of China (106) and India (112)⁵. Among South Asia countries, Nepal ranks second after Bangladesh in regional level ranking. The report ranks the countries according to calculated gender gaps between women and men in four key areas: health, education, economy and politics. Key findings from the report are shown in Table 1.

	% Female	% Male	Female to Male Ratio
Economic participation and opportunity			
Labor force participation	84.5%	86.3%	0.98
Estimated earned income (PPP, US\$)	2.1	3.3	0.63
Legislators, senior officials and managers	18.4%	81.6%	0.23
Professional and technical workers	37.5%	62.5%	0.60
Educational attainment			
Literacy rate	59.7%	78.6%	0.76
Enrolment in primary education	74.8%	86.0%	0.87
Enrolment in secondary education	58.6%	56.5%	1.04
Enrolment in tertiary education	12.8%	12.0%	1.07
Health and survival			
Sex ratio at birth	-	-	0.94
Healthy life expectancy, years	62.1	60.5	1.03
Political empowerment			
Women in parliament	32.7%	67.3%	0.49
Women in ministerial positions	14.3%	85.7%	0.17
Years with female/male head of state (last 50)	3.7	46.3	0.08

Table 1: Gender Profile

⁵ World Economic Forum, *Insight Report. The Global Gender Gap Report 2020*. Geneva, Switzerland: World Economic Forum, 2019.

Human Development Report 2019⁶ shows that the Gender Inequality Index (GII) value of Nepal is 0.476, ranking it 115 out of 162 countries in the 2018 index. Whereas, 2018 female Human Development Index (HDI) value for Nepal is 0.549 in contrast with 0.612 for males, resulting in a (Gender Development Index) GDI value of 0.897.

1.2 Gender and Energy Sector

Normally energy policies are approached from a technical and gender neutral perspective and have implications for equality between women and men and for the poor. Improvements in women's health, girls' education opportunities, household nutritional status, and livelihoods can be led by alternative and rural energy technologies like rural electrification, micro hydropower, solar technologies. However, maximizing these benefits requires attention to the barriers that women and excluded groups face in accessing these technologies.

Major policy documents governing the energy sector include the Electricity Act 1992, Hydropower Development Policy, (2001), the National Water Resources Strategy (2002), 15th National Plan (2018-2023), the National Water Plan (2005), and the Rural Energy Policy (2006).

The legal and policy framework governing energy issues is largely silent on recognizing GESI as a factor requiring separate treatment, with some exceptions (e.g., Rural Energy Policy, 2006). A general trend evident in these legislations is that the more recent ones, especially those introduced post-2006, are more sensitive to GESI issues. For example, the Renewable Energy Subsidy Policy, 2016 is more GESI-sensitive in many more parameters compared with something like the Electricity Act, 1992.⁷ For instance, the Rural Energy Policy, 2006, among others, explicitly states "as rural energy is directly linked to activities traditionally carried out by the women, programmes of rural energy technology will be implemented considering it as an integral part of the women's enabling activities" (Article 10.6). The policy also emphasizes issues such community management, enhancing productivity of end users, and representation of excluded groups in user committees.

Policies which are more relevant to the project components would be Renewable Energy Subsidy Policy and Rural Energy Subsidy Delivery Mechanism, 2013 which have specific subsidies targeted to women and the socially excluded groups⁸:

- a. Additional subsidy of Rs. 2,500 per household will be provided to households with single woman, backward, disaster victim, conflict affected, poor and endangered ethnic group as identified by the Government of Nepal.
- b. For solar thermal technologies for rural areas, additional Rs. 20,000 will be provided if at least 50 percent of the target groups are single women, backward, disaster victim, conflict victim, endangered ethnic group.
- c. For biogas and metallic cookstoves, additional subsidies of up to NPR 4000 and 1000 are offered to single women, backward, disaster victims, and endangered poor households, ethnic groups.

The National Rural Renewable Energy Programme (NRREP) managed by AEPC, a 5-year programme that started in 2012, was an integrated version of different smaller programmes or projects. The

⁶ UNDP. (2019). Human Development Report 2019. Beyond income, beyond averages, beyond today: Inequalities in human development in the 21st century. Washington, DC: Author. <http://hdr.undp.org/sites/default/files/hdr2019.pdf>

⁷ ADB, 2018 <https://www.adb.org/sites/default/files/publication/401781/gender-equality-social-inclusion-assessment-energy-nepal.pdf>

⁸ RE Subsidy Policy 2013 [https://www.aepc.gov.np/uploads/docs/2018-06-19_RE%20Subsidy%20Policy,%202073%20\(English\).pdf](https://www.aepc.gov.np/uploads/docs/2018-06-19_RE%20Subsidy%20Policy,%202073%20(English).pdf)

NRREP programme mandated that specific affirmative action will be planned, implemented and monitored across all component activities, aiming to empower women and marginalized groups through enhancement of their technical capabilities and assisting them to take up ownership of technologies. A number of development programmes in Renewable Energy Sector of Nepal, such as the Biogas Support programme (BSP), the Rural Energy Development Programme (REDP), and the ADB-supported Decentralized Rural Infrastructure and Livelihood Project (DRILP) provide good practices in GESI integration that can be learned and integrated into programme planning.⁹

The GESI approach has been considered as a cross-cutting issue in all the programmes of Government institutions and NGOS. For instance, FECOFUN, a federation of Community Forest User Groups has a 50% representation of women with equally divided portfolios in the Executive Committee including people from all regions, indigenous peoples, minority groups, differently abled people, and ultra-poor. They practice Gender Equity and Social Inclusion (GESI) from Federal to local FECOFUN chapters. Similarly, Women's Network of Energy and Environment (WoNEE) is a national network of grassroots women willing to contribute to women's economic empowerment and ensuring energy and environmental justice in Nepal. It is also working to leverage funds from local governments and co-finance the e-cooking sector.

In most of the programs, it is found that women have less access to information through official channels as reviewed by Koojiman A, 2021. Also, the inability of the programmes to involve women in technical trainings and capacity building is also a reason of shortage of women technicians in the market which we have tried to mitigate by setting specific targets of women participation and involvement in technical trainings. Female Community Health volunteers have been the most effective medium of communication and awareness in community, learnt from their experiences in creating awareness in COVID pandemic or post-earthquake scenario.

An instrumental partner in the awareness creation and local monitoring activities of the project could be the Female Community Health Volunteers (FCHVs). It is reported that the community health interventions which are mobilised through FCHV have high population coverage and they can be an important factor for home visits and demonstration projects. They are also mobilised for COVID-19 awareness. Under the Green and Inclusive Energy (GIE) program, FCHVs were trained on prevention and control of household air pollution and further mobilised to train households on the benefits of clean cooking solutions.

Leadership ability of women and girls in the energy sector are undermined by prevalent gender norms and discriminatory gender roles. There is no such consideration of gender issues and inclusive financing as well. There is priority loan provision but it is more focused on youth and poor people, who want to initiate self-employed or micro enterprises. The barriers exist not just in the policy and legal framework, but also the institutions, programmes and implementation of energy and water sectors¹⁰. Moreover, there is no data available on segregated data on GESI in credit financing to market chain actors in RE technology.

1.2.1 Gender and Finances

The supply side of the energy sector has very few involvement of women technicians and experts. In the case of female household heads, they are more vulnerable than male heads to such external changes in terms of acquiring new knowledge and access to new resource bases¹¹. Therefore, special

⁹ Koojiman A., 2021 <https://mecs.org.uk/publications/gender-responsive-electric-cooking-in-nepal/>

¹⁰ Centre for Rural Technology Nepal 2018 <https://www.crtnepal.org/document/Annual%20Report%202018-wxzKeKGutX.pdf>

¹¹ <https://www.adb.org/sites/default/files/institutional-document/32561/women-nepal.pdf>

measures have to be taken to monitor that women family members and female heads of households receive adequate assistance.

Gender stereotypes for women to have less education, early marriage, and total responsibility in household has hindered women to work as technical human resource. Unlike in cottage and small scale industries, interventions to promote women's involvement as technicians are insufficient. In the value chain, representation of women has been the major barrier in policy sectors while their active participation has been the same at local level.

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However, Nepal Rastra Bank directives to the banks and financial institution regulates them to lend up to 5% of their loan portfolio to the deprived sector that covers, low income, socially backward women, janjati, dalit, and differently-abled persons¹³. Loan extended by BFIs to MFIs on corporate guarantee for wholesale lending to cooperatives called deprived sector lending. MFIs and cooperatives being member based organisations usually are successful in targeting GESI objectives which is also usually part of their objective.

Most cooperatives and MFIs have more than 50% female members. Women have loan repayments above 95%. Therefore, these organisations are providing credit targeted beneficiaries for cookstoves. This approach has been a success also with win-win situations for BFIs, MFIs and cooperatives and strongly recommended for future acceleration of clean cooking projects¹⁴.

Credit financing through multipurpose cooperatives have been in practice to support several types of small businesses. Few examples of same to purchase electric cookstoves in Kavre district have been taken as case study by (Koojiman, 2021). Similar models will uplift the beneficiaries' capacity to bear Capital expenses of cleaner cookstoves. This will create a conducive environment to make market sustainable.

1.2.2 Gender and cook stove

More than one half (52%) of the households in the country use mud-stoves for cooking purposes, followed by kerosene/gas stove (21 percent), open fire-place (22 percent), and smokeless stove (3 percent)¹⁵. The gender dimension imposes influence on the household activities specially kitchen,

¹²Centre for Rural Technology Nepal 2018

¹³ <https://www.crtnepal.org/document/Annual%20Report%202018-wxzKeKGutX.pdf>

¹⁴ Landscape Analysis of the cookstove Sector of Nepal, 2021

¹⁵ Central Bureau of Statistics. (2011). Nepal living standards survey 2010–11.

with girls and women bearing the majority of the negative effects of fuel collection and transport, household air pollution, and the use of time-consuming and unsafe cooking technologies. Overharvesting of fuelwood and fodder is also considered one of the major issues of deforestation in Nepal¹⁶.

The present unemployment rate in Nepal is 11.4 percent where the rate of female (13.1%) unemployment is higher in comparison to male (10.3%)¹⁷.

The gender assessment shows Nepal as a country with gender equality issues. Cooking is almost entirely carried out by women, requiring on average about 4 hours a day for cooking using a traditional cooking stoves¹⁸.

Women in Nepal spends about 142 minutes to collect the firewood whereas men spends only about 50 minutes¹⁹. Likewise, the study shows the involvement of women in collecting agricultural residue and cow dung is 70.2% whereas involvement of men is only 29.8%, this shows that in collecting of both firewood and biomass fuel involvement of women is much more than men²⁰.

Despite having plenty of engagement in cooking, women have not been able to make decisions in selection of cookstoves. They have less or no access to information through official channels. For female headed households or single women, if they are illiterate, there are many challenges for them to have privilege of clean cooking technologies. They are so busy in earning their livelihood that they are not even able to utilize the benefits they are eligible to. These are the major barriers in the cookstove from gender's perspective, which would be somewhat addressed by the GAP activities 1.1.2 on selection criteria of beneficiaries, and 3.1.1 and 3.2.1 on selection of CCS mobilizers.

In Nepal, on average, there are 18.3 *bharis* (headloads of about 30 kg on average) of firewood collection per capita per year in Nepal. On average, a household spends 5.01 hours collecting one *bhari* of firewood²¹.

WHO estimates global 3.8 million people yearly die prematurely from illness attributable to the household air pollution caused by the inefficient use of solid fuels and kerosene for cooking. A study on Household Air Pollution by ENPHO found that the average 24-hr mean PM2.5 concentration in Nepali kitchens is 2127 µg/m³ with Traditional Cooking Stove (TCS), and with ICS it is 728 µg/m³.

¹⁶ Chaudhary et al., 2015. Deforestation in Nepal: Causes, consequences and responses.

https://www.researchgate.net/publication/284859236_Deforestation_in_Nepal_Causes_consequences_and_responses

¹⁷ Report on Nepal Labour Force Survey 2017/2018. <https://nepalindata.com/media/resources/items/20/bNLFS-IIIFinal-Report.pdf>

¹⁸ Practical Action, 2014. *Gender and Livelihoods Impacts of Clean Cookstoves in South Asia*.

¹⁹ Venkata Ramana Putti, Michael Tsan, Sumi Mehta, & Kammila, S. (2015). *The State of the Global Clean And Improved Cooking Sector*.

²⁰ Practical Action, 2014. *Gender and Livelihoods Impacts of Clean Cookstoves in South Asia*.

²¹ *ibid*

2 Gender Assessment

2.1 GCF rationale for Gender Assessment

The GCF's commitment to gender equality centers on gender-responsive country programmes and initiatives that benefit everyone, women and men. This requires gender analysis to understand the social, economic and political factors underlying climate change-exacerbated gender inequalities, and the potential contributions of women and men to mitigating and adapting to climate change, and building climate resilience. It further entails adopting methods and tools to promote gender equality and reduce gender disparities in climate funding by mainstreaming gender in the project cycle. Finally, it means measuring the outcomes and impacts of project activities on women's and men's resilience to climate change through gender-responsive monitoring and evaluation (M&E). Gender mainstreaming should shape the entire project cycle, from project identification to M&E²².

2.2 Gender assessment

The project benefits about one million households, comprising 4,800,000 people (average family size of 4.8 people per household) in *Terai* region. 51% of direct beneficiaries are female. The direct beneficiaries of the project comprises about 17% of total population. Also, about 13 million people of *Terai* region are considered as indirect beneficiaries of the project of which 51% of the beneficiaries are female. The indirect beneficiaries of the project accounts for about 50% Of total population.

The TCS users are the most vulnerable community as they are among the poorest and socially excluded groups. 23.44 % of the population in the *Terai* region survive under absolute poverty. Against this, 33 % of the population are disqualified to cross the threshold of human poverty. In addition, 25.9% *Janajati* (Indigenous group) and 38.2% *Dalits* population of *Terai* live under extreme poverty²³.

2.2.1. Status of Gender Equality

i. Social norms and Practices

Social norms adequately contain gender inequalities. Traditionally, it is widely practised for women to be involved more in the household activities like cooking, cleaning, laundry, gardening, and child care. Referring to the statistics, women are behind men in all the indicators in economic participation and opportunity, which is because of the established norm that women should only work inside homes. Women are expected and even forced to have early marriages, which hinders the women to have equal opportunity as men. The women of indigenous communities and those below poverty line will have these hindrances amplified. Working women and women with income have good access to the project activities and benefits.

²² https://www.greenclimate.fund/sites/default/files/document/guidelines-gcf-toolkit-mainstreaming-gender_0.pdf

²³ CBS, 2011. Nepal Living Standard Survey 2010/2011. https://cbs.gov.np/wp-content/uploads/2018/12/Statistical_Report_Vol1.pdf

ii. **Major Indicators and Political Participations**

Female literacy rate, employment in professional sectors, education, political representation and political participation are relatively low, while labour force participation, life expectancy and population is somewhat equivalent or even higher relative to that of male population. Provisions of women reservations in political space and public sectors are there, but despite that, women involvement is less visible due to established boundaries and gender norms set for women. Mandatory nomination of one of chair or vice chairperson of Local government executive body and reserved nomination of 2 out of 6 ward members for female is a progressive sign observed since 2016 local elections. At least 33% of reservation for females in provincial and national assemblies is also supportive.

iii. **Policy environment**

Gradually, the policies have been made to mainstream gender equality and social inclusion in all the sectors. Integration of GESI policies upto Provincial Level has opened up an environment to take them to local levels. In the current fiscal year, several local governments have drafted their policies for GESI integration and many others are in the process. However, a big challenge is to implement those policies at local levels. Parallely, independent efforts have been made at local and national levels by Non-Government Organizations, and other agencies to speed up the process. Several gender assessment reports and reviews submitted to the ministries and authorities have helped to create proper guidelines to amend policies and make them gender friendly. The GESI policy of AEPC envisions to have equal and equitable access, opportunities and benefits of renewable energy to both the female and male based on their social status, ethnicities, income, age, physicality, geographic background and projected affected people. The objectives are to empower women, and less privileged people by creating employment opportunities, to increase participation and capacitate them and to help overcome traditional social norms that hinders the user to have equitable access to Renewable Energy Technologies by providing suitable opportunities.

iv. **Legal Framework**

The legal provisions of action against Gender based violence are secured in the Constitution of Nepal 2015 and Muluki Ain 2017.

v. **Poverty & Income**

The per capita income stood at US \$ 1,004 (Rs 103,335) in the fiscal year 2018/19 and the main portion of income goes for sustaining livelihoods. In 2016, Nepal achieved a score of 0.148 according to the multidimensional poverty index (MPI), with around 22.3% population vulnerable to Multidimensional Poverty (MP), 11.6 in severe MP, and 15 % below income poverty line²⁴.

vi. **Economic Status and Empowerment**

²⁴ Human Development Index Report, 2019 http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/NPL.pdf

Women's labour force participation rate, employment status and overall situation in the labour market show their economic status has improved. Yet the labour market remains marked by many structural barriers and forms of deep-rooted discrimination.

The labour force participation rate in general and women's rate in particular have significantly decreased in recent years. The Labour Force Survey of 2008-2009 showed that the labour force participation rate of women was 80.1 percent, compared to 87.5 percent for men, one of the highest rates in South Asia. Based on the 2017-2018 survey, compared to 48.3 percent of males of working age in employment, only 22.9 percent of females were in employment. Among all working-age women, almost 52.4 percent were either own account workers or contributing family workers compared to 27.7 percent of men²⁵.

vii. Participation in Formal and Informal Economy

Nepali girls and women work for more than boys and men, spending 25 % to 50 % more time on households' tasks, economic and agricultural activities. Yet, due to the nature of women's work, which is often unpaid, on the one hand, and the flawed definition of economic activity, on the other hand, women's economic participation remains statistically invisible. Women in the more orthodox Hindu communities who are largely confined to domestic and subsistence production display a much less significant role in major household economic decisions than those in the Tibeto-Burman communities where women participate actively in the market economy.

The role played by women in the care sector, predominantly their reproductive work, bearing, rearing, nurturing children and household maintenance are activities that fall outside the national accounting systems. While these activities are crucial for household members' well-being and effective participation in different spheres – economic, social and political, they continue to remain non-economic activities. By virtue of women performing these roles, which are statistically not counted as economic and hence not monetarily valued, women's roles and their contribution is assigned low status.

viii. Education, literacy and information

Gender gaps in the education sector continue to persist, where girls and women tend to have lower levels of education, especially in rural areas. The percentage of women participating in primary education has increased substantially in the last two decades, where the number of women completing 12 grades rose from 21% to 42%²⁶. Nonetheless, equality can still be promoted in terms of access to education, especially in rural areas where studies have found that girls' education often remains a lower priority

ix. Land Ownership

Land and house ownership by females is very low compared to that of male. Despite having incentives like Land registration fee waivers for women, it is not able to have instrumental

²⁵ Nepal Human Development Report, 2020. <https://nepal.un.org/sites/default/files/2020-12/UNDP-NP-NHDR-2020.pdf>

²⁶ CBS 2014

change in increasing the percentage of asset ownership by women. The provision in the amendment of Muluki Ain 2017 has made daughters eligible for the inheritance of parental property, which wasn't the same earlier. As ownership relates to the decision-making power inside a family, women are weaker from that perspective too. Involvement of bigger proportions of females in program implementation would empower them and address this barrier.

x. **Health**

Reproductive health has largely been funded by the government. Each local government has a number of health posts to facilitate general and reproductive health issues and health posts with birthing centres to promote hospital based deliveries. Travel and nutrition allowances are provided to all the hospital/ health posts based delivery cases. Further, the Ministry of Health has also implemented family planning programmes in all its subsidiaries.

xi. **Gender Based Violence**

A nationally representative sample of 12,862 women aged 15-49 in 11,040 surveyed households and 4,063 men aged 15-49 in half of the surveyed households were interviewed. Based on that, the 2016 NDHS (Nepal Demographic Health Survey) provides reliable estimates at the national level, for urban and rural areas, 3 ecological zones, 5 development regions, and 7 provinces.

Table: Domestic Violence (among women aged 15-49)

	Women who have ever experienced physical violence since age 15	Ever-married women who have experienced spousal, physical, sexual, or emotional violence
Nepal	22%	26%
Urban Residence	21%	25%
Rural Residence	24%	28%
Focal Areas of the Project		
Province 1	19%	22%
Province 2	34%	37%
Lumbini Province	23%	29%
Sudur paschim Province	17%	22%

A relatively high rate of gender violence has been observed in Province 2, which is also a focal area of the project and requires some additional measures to help reduce the risk of gender violence. However, the project impacts will not instigate any gender based violence.

xii. **Gender and Energy/Environment**

Energy and Environment sectors have been recognized as gender neutral sectors, based on their technicalities. Policies before Rural Energy Policy 2006 have not been gender friendly. As alternative energy is a newer concept and institutionalized with the help of international grants,

this has been more lenient to gender issues. Stakeholder consultation and public hearing programmes under Environmental Protection Acts seek for inclusive participation. Since the sector is of global concern, it is ahead of other sectors in mainstreaming Gender issues.

Conclusion:

Barriers to an equal representation up and down in value chain includes traditional social norms biased against women, poverty as a result of limited access to education, early marriage, limited asset ownership. Furthermore, women's participation is limited to decision making and income generating activities, and technical workforce.

Gender Assessment Team

A core gender assessment team was formed, comprising:

- Key experts on clean cooking solutions
- GESI expert
- Key government authorities and,
- Representatives of civil societies, including individual from the network of women right movement

In addition to the core gender assessment team, stakeholder consultation space was formed to incorporate inputs from energy and clean cooking experts, government officials, GESI experts, women movement representatives and women right activists, women-lead national and local NGOs, private sectors (including the ones headed by women) in both service-supply and service-demand chain.

Recommendations from stakeholder consultations (Annex 1)

- Training and Awareness should target the female (users) and not the household head. Training should be provided to women on electrical safety and basic technical skills on using induction stoves increases the use of the stove and increases their confidence.
- Capacitate females to repair and maintain stoves to ensure sustainability and empower women.
- The poorest and most vulnerable communities cannot purchase the Tier 3+ ICS and even if the stoves are distributed free of cost, fuelwood is another major challenge. Most of them use cow dung as fuel, which is not compatible with the available technologies. Research should be more focused to make technologies compatible with available local fuel sources so the most vulnerable can assess the technologies.
- Women should be involved as CCS promoters to effectively implement the project and the project should be targeted to the female and most vulnerable communities.
- Gender-specific targets should be set to ensure that certain numbers of female and male entrepreneurs are trained as master trainers to reflect the needs of women and men in their products, business models, and livelihood opportunities.

Project Identified Opportunities to Challenge Gender Stereotypes

Some of the actions that are included in the project activities for the positive gender relations in the project as follows:

1. The economically impoverished and socially excluded families, indigenous people, single women, female-headed households, and widows will be priorities while selecting beneficiary households.
2. The project will assist LGs in placing dedicated personnel related to environment safeguard and gender in CCS projects. In addition, they will be supported in conducting gender sensitive monitoring, evaluation and provide training on CCS and its benefits.
3. For the national level consultation workshop between AEPC, LGs, beneficiaries, clean cooking champions, and local private sector service providers, training materials will be developed by drafting mechanisms to effectively address gender equality and supporting vulnerable communities through CCS.
4. The beneficiaries will receive technical assistance and capacity development services that will help to reduce their vulnerability and build resilience to climate change impact through the time saving, health benefits, gender empowerment and increase in agricultural productivity by using cow dung as fertilizer, by replacing its use for cooking fuel.
5. The project enhances women's access and entitlement to benefits. Since women are direct users of the cook stoves, it will benefit women by reducing their exposure to indoor air pollution, thereby improving their health. Besides, reduction of fuelwood after the installation of more efficient stoves will reduce the workload of women to collect fuelwood, resulting in time saving that they can use for other productive activities such as getting an education and care for children or socialize²⁷. More than 50% of the time will be saved via the proposed CCS.
6. The project will address existing barriers in promotion of CCS to poorest households by improving access for vulnerable communities and providing support and training on repair and maintenance of proposed technologies to women and marginalized groups to ensure inclusive sustainability of the project. There are several local NGOs working in the clean cooking sector. The project shall collaborate to mobilize their resources in awareness campaigns and skill development training.
7. A total of 24 Province level trainings will be provided to CCS beneficiaries and vulnerable groups in collaboration with CTEVT institutions. In addition, 1,200 training will be provided on making fuel for Tier 3+ ICS, repair and maintenance, biomass fuel processing, biogas installation, electrical safety and house wiring to CCS beneficiaries and vulnerable and disadvantaged groups (DAG). Training will be provided to women (electric cook stove users) on electrical safety and basic technical skills on using electric stoves to increase their confidence and the use of the stove. Self-employment skilled training related to CCS and others as per the local demand will be provided to the most vulnerable people such as widows, single women, Female headed households, minor ethnic groups, lower caste people (Dalit) and indigenous group (Janjati), ensuring at least 33% female participation.
8. The project provides financial support for the proposed CCS technologies, which will allow vulnerable user communities to transition to better CCS technologies with high returns.

²⁷<http://documents1.worldbank.org/curated/pt/772731563431443718/pdf/Have-Improved-Cookstoves-Benefitted-Rural-Kenyans-Findings-from-the-EnDev-Initiative.pdf>

9. LGs and local communities will learn a lot from the use of CCS as the project will create awareness on and access to CCS, inform the vulnerable people of benefits in terms of health and efficient systems.

3 Gender Action Plan (GAP)

GAP forms the basis for operationalizing the results and recommendation of the gender assessment. It incorporates gender elements during project formulation (and design), project implementation and project monitoring & evaluation. GAP ensures necessary intervention to mainstream gender and integrates gender-perspective in the project to maximize clean cooking benefits. The GAP is compliant with GCF's gender policy and AEPC's Gender Equality & Social Inclusion policy.

The GAP is aligned with the outputs of the log frame and planned output and activities. Additionally, the programme will document the positive and negative effects the program activity will have on gender relations by setting up adequate, gender-sensitive monitoring and collecting sex-disaggregated data. Gender-sensitive language is used in reports, training materials and publications. Different partner entities and stakeholders (AEPC, LG, and vendors) will execute GAP in their program management capacity.

Table 2: Gender Action Plan

	Impact Statement: Reduced GHG emissions and increased resilience and enhanced livelihoods of the most vulnerable people, community, and region by implementing modern, efficient and climate friendly cooking solutions Outcome Statement: Improved access to clean cooking in the southern plains of Nepal including the most vulnerable households (Poor, Female Headed Households (FHH), widowed, minorities, differently abled and elderly) by creating an enabling environment to adopt clean cooking solutions; leading to significant health and economic benefits by reducing exposure to household indoor air pollution and time savings for firewood collection and food preparation.					
	Cross-Cutting					
Activities	Associated GAP activities	Associated activities and Measures	Indicators and Targets	Timelines	Responsibilities	Cost
All Institutional and Managerial		i) Managerial arrangements during project formulation, implementation, and monitoring & evaluation	i) At least one Gender Specialist (GS) throughout the project -Gender balance in stakeholder meet & representations	All years/periodic	AEPC	Use AEPC's existing talent

arrangement during project formulation, implementation, and monitoring & evaluation)		- Gender expert/specialist runs workshop/capacity development trainings to gender focal person in stakeholders			pool (Est. USD 22500 and USD 135000 to hire third party GESI expert consultant)
	Revision of programme documents, training materials, promotional materials to ensure effective gender-mainstreaming, and that the information is gender-sensitive.	All major communication materials and awareness- raising campaigns are reviewed by program's GS and M&E specialist	Periodic (every year)	AEPC (GS) /LG	Included within the activity budget for
	Gender sensitization training for project staff at all levels to maintain desired level of gender awareness	# of trainings (each year), # of project staffs receiving gender awareness training (target-100%)	Periodic	AEPC (GS) /LG	Training, workshop and conference (C1, C5, C7, C11, C12, C12, C16, C17)- Annex

					8
	Encourage the effective participation of women in different points of project formulation, implementation, and monitoring & evaluation. Time and locate training taking into women’s work.	% of female participation (Target: Progressive) (Baseline: varied based on different point of project cycle; location and others)			Include d within the budget of activity
	Include gender specific questionnaires (to include current positioning on socio-economic aspects; effective accessibility- to financial, technical resources; awareness on entitlements and endowments) in baseline survey (sex disaggregated data collection) specific to the local region. All baseline and feasibility surveys to include gendered impact of desired change (access, needs, barriers, potentials, work load benefits) GAP will be revised/updated after the baseline survey	# of baseline survey conducted in year 1 (target: all targeted area)	Year 1 (Prior to PI phase)	AEPC/LG	
	Conduct stakeholder consultation workshop at the local level and revise GAP tailored for them	Inclusion of fieldworkers, female workers, female technicians, potential CC champions, representatives from indigenous communities, LG representative); detailing at PI phase	Year 0 (Prior to PI phase)	AEPC (GS)/LG	
	Output: Installed 490,000 Tier 3+ICS, 10,000 biogas and 500,000 electric cook stove				

	GAP Objective: Ensure women's equal and equitable access, to and benefit from different points of service-supply and service-demand chain in installation of technologies.					
Activity 1.1.1 Development of Annual Procurement and Deployment Plan, 1.1.2 Procurement and Deployment of annual targeted number of CCS	Ensure gender inclusion in both the service-supply (procurement and deployment plan) and service-demand (beneficiaries)	i) Develop beneficiary selection prioritization scoring system (based on socio-economic vulnerability indicators including Female headed households (FHH), single women, widows, and indigenous people. (At PI phase) ii) Increased awareness on tax leverage & incentives, and credit/loan facilities for women entrepreneurs on service-supply and, promotional activities, awareness, and outreach to enhance demand iii) Have gender focal point for all partners and relevant stakeholders (vendors, partners, LGs)	i) Detailed prioritization scoring system to be included at SOP developed at project implementation phase (IP). ii) % of women and men participants in awareness campaign; (target: min 60%) (Baseline: variable in each specific region from 30-60%); region specific targets revisits CCS mobilizers (Target- 50% women Door to door awareness champions (Target- 60% women) (Target revisit after annual survey)	Year: 1-5	AEPC (gender specialist)/ LG (gender focal person)	
Activity 1.1.3 Distribution of initial 60% payment instalment to the vendor upon delivery of CCS to respective municipality			iii) Gender focal point (or gender responsible) in each partner and stakeholder	Year: 1-5	AEPC/LG/P artners	Include d within the budget activity (Annex 8) and uses AEPC's existing talent pool.
Activity 1.1.4		i) Ensure gender parity in training	i) # % of women certified as 'independent verifier'	Year 1-5	AEPC	

Monitoring and Verification of Installed CCS units for Output Based Financing	Gender parity in monitoring and verification	and AEPC-issued certification of 'independent verifier' ii) Sex-disaggregated data collection during monitoring and verification by 'independent verifier'	Target- at least 33% Baseline- 0 ii) survey and verification sheet (to generate gender statistic) - developed at IP	(annual)		
	Output: Enabled environment (enhanced product standards, surveys and analysis and quality assurance mechanism) GAP Objective: Create gender inclusive environment in strengthening quality assessment and assurance mechanism.					
Activity 2.1.1 Develop partnership agreement between AEPC, Province Governments, Local Governments, and other implementing partners 2.1.2 Update existing standards and set benchmarks for the technologies required for the project implementation	Ensure gender inclusiveness in design of normative policies & plans, standardization with strengthened women representative from different stakeholders	i) Create gender diverse/inclusive board and stakeholder's consultation space for the formulation of MEP template and other normative and standardization processes. ii) Provide 'CCS and GESI' related trainings to technical staffs and board members iii) Consider gender (in the context of stakeholder roles, institutional responsibilities, resource access and benefit sharing) in designing municipal energy plan (and guidelines)	i) % of women in board meetings and stakeholder's consultation Target: Min 33% of women in the board and stakeholder's consultation (Revisit target periodically) ii) # of gender session in capacity building and workshops to LG, MEP/standardization formulation board (Target: mandatory gender sessions in all trainings) (Baselines: Technical staff have received no training or in rare cases (limited training) in gender issues pertaining to CCS.	Periodic over 5 years	AEPC(GS)	Included within the budget activity (Annex 8) and uses AEPC's existing talent pool.

2.1.3 Develop Municipal Energy Plan template by analysing standards and format						
2.2.2 Development of Management Information System as a national monitoring system with real time data collection from LGs	Mainstream gender and sensitivity in data collection and MIS design	Collect sex-disaggregated and gender relevant data (To be reviewed with progressive targets annually)	Gender sensitive 'data collection methodology' and questionnaire guide Survey in the 3rd quarter of year 1: (Periodic review, follow up-surveys and progressive targets)		AEPC (GS), Independent Verifier, LG	
	Output: Empowered institutes, capacitate supply chain and ensured increased access to clean cooking solutions GAP Objective: Ensuring fair access to women in service-supply and service-demand chain					
Activity 3.1.1 Conduct national stakeholder consultation workshop between AEPC, PG and LG Activity 3.1.2	Mainstreaming gender in local and provincial institutional setup to increase employment and volunteering opportunities to	i) LG to mainstream gender in their institutional setup for energy units, CCS mobilizers and CC champions in support of GS (AEPC) and GESI responsible at LG. ii) Increase employment and volunteering opportunities to women through preference to women for CCS mobilizers; qualified women	i)50% female (75 nos) for CCS mobilizers of total recruits (150); 60% female (270nos) for CC champions focus on marginalized communities. At least 1 woman as energy officer (out of 2) in energy units, in each province	Periodic (for 5 years)	AEPC	Estimated \$

Develop LG Municipal Energy Plan 3.2.1 Mobilize clean cooking mobilizers in each LG	women	encouraged to apply for energy units in provincial level iii) Ensure effective participation of women, civil society and gender experts, local women led organizations, representations from indigenous group in consultation space for LG municipal energy plan iv) LG municipal energy plan (and guidelines) explicitly need to considers gender related issues(e.g. In the context of stakeholder roles, institutional responsibilities, resource access and benefit-sharing				1,282,500.00 for CCS mobilizers ; Others included within the budget activity
3.1.3 Capacitating Local Government and Provincial Governments	Gender sensitive approach in capacity development approach and plan	i) Place a dedicated person on GESI responsible at LG to conducting gender sensitive monitoring and evaluation ii) Ensure that training materials, promotional to ensure effective gender-mainstreaming, and that the information is gender-sensitive.	# of capacity development trainings (target- 1 in each LG), # of GESI sessions in each workshops (target- 150), # % of female participation in workshops (baseline: varied in each LG (30-50%); target: min 60%)	Periodic (for 5 years)	AEPC (GS), LG (GESI focal point), LG	
3.1.4 Annual national level review and consultation meeting	Gender elements to be reviewed and revised	Qualitative and quantitative survey on gender-based elements (designed at PI phase) Collect and analyse sex-disaggregated and gender relevant	Indicators of GAP will be fully integrated in the M & E framework at SOP and reviewed periodically, # of gender-specific	Periodic (for 5 years)	AEPC (GE)/Independent Verifier/LG	

		data used for correction of measures aiming to hold all partner accountable to gender equality	recommendations, include elements of GBV and other gender related implications of project			
3.3 Strengthen service centre, biomass manufacturers to provide quality and affordable clean cooking solution	Ensure women's participation (from different point of supply-demand chain) and include gender elements	i) Ensure easy access to women in the distribution and after sales service- repair and maintenance by promoting subsidised credit/loan facilities ii) Increase women's participation in the technical trainings (repair and maintenance, biomass fuel processing, biogas installation, electric safety and house wiring) training in provincial level	# of enterprises women-owned and/managed by women participation in trainings (baseline survey to be carried out during PI) (Target- progression) # Of technical trainings provided to beneficiaries (target: total of 24 trainings to beneficiaries and service centres) with minimum 33% participation of women) in collaboration with CTEVT (baseline survey assumes minimum to none) # of women beneficiary participation in skill development trainings (Target: min 33%)	Periodic (for 5 years)	AEPC/Vendors/LG	

Annex 1: Stakeholder consultation Report

Sub-national Stakeholder Consultation Meeting for “Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS)”

AEPC had called for an interaction program on 22nd June 2020 with National Stakeholder and relevant stakeholders on the proposed project Mitigating GHG emission through modern, efficient and climate friendly clean cooking solutions (CCS) to collect their feedback on the project. The online consultation meeting started at 3 Pm sharp (NST) via Zoom. In total, around sixty people attended the meeting. It was facilitated by Mr Mukesh Ghimire, Assistant Director of AEPC, where Mr. Rana Bahadur Thapa, Sr. Officer, AEPC gave a brief overview of the project and solicited necessary information from the participants. Mr Nawa Raj Dhakal, the Deputy Executive Director of AEPC, gave the concluding remarks. The meetings enabled interested and concerned parties to contribute their concerns, which might have been overlooked while preparing the concept note. Majority of the participants appreciated the project and wished for its smooth implementation. Additionally, meaningful consultation will be conducted periodically during the implementation phase with the concerned stakeholders with active participation of women's, Dalit's and vulnerable groups.

The tentative topics for discussion includes but not limited to:

1. Role of LGs and involvement of NGOs at local level

- Awareness on climate change at local level
- Energy planning and Role of local government in CCS promotion including funding (Municipal Energy Plans, Implementation, Any CCS program ongoing or implemented, what %of budget is usually allocated for CCS projects?)
- Any new policy/plans for CCS?
- NGOs currently working on CCS at local level and possibilities to engage them for the project

2. ESS Risks/issues and mitigation measures

- Key social, environmental, gender, economic and culture risks and issues with respect to CCS
- Possible mitigation measures

3. Local level supply Chain

- Baseline fuels for cooking,
- Capacity of local supply chain in meeting local demand,
- Any information of supply and demand of CCS at local level
- Expectations to improve the supply chain

Feedback

- It is very good idea to involve Provincial Energy Coordination Committee (PECC) in province 2, Province 5 and Karnali Province for prioritizing the demand as this committee includes representative from provincial ministries, provincial planning commission, NEA province office, AEPC province team, NARMIN and MuAN.
- Training and Awareness should target the female (users) and not the household head.

- Experience from the past should be analysed to ensure financial, utility, commodity sustainability of the project
- Value chain of ICS (Supply to target group users) should be analyzed to fulfil the gap
- Move forward by catching up bitter experience and golden experienced
- Community should be mobilized in stove distribution (trust fund).
- Local governments should be consulted to reach the poorest of the poor.
- We need to assure the quality of cookstoves and construction of biogas.
- Poverty rate in the Terai region is very high and there are households and communities who don't have NEA meters or the households who cannot keep the meter due to lack of land holding documents. Thus, we need to address these households and communities.
- People were hesitant to adopt the mud ICS may be due to cultural or social barriers. Biogas seems to have much demand compared to ICS so the numbers should increase.
- Community forest spends 25% of their total income in forest conservation and they have been spending the money in biogas and ICS. 37% poor female and vulnerable community can be integrated in community contribution for the project.
- Previous World Bank funded, REED and other projects are more focused in the Terai region so may need to consider other geographical areas.
- Biogas and electric cooking does not benefit the poorest of the poor. Therefore, we need to explore more on how to make these pro-poor.
- Actual demand should be created while collecting demand and exploring the supply side. Options of various technologies should be given to the user to ensure the use of the stoves after deployment.
- Coordination with local government, involvement of females in stove making, distribution, and financial contribution of users creates the enabling environment for stove promotion. At present, each local government has female representatives. They should be enrolled as a facilitator for the long run of the project.
- The poorest and most vulnerable communities cannot purchase the Tier 3+ ICS and even if the stoves are distributed free of cost, fuelwood is another major challenge. Most of them use cow dung as fuel which is not compatible with the available technologies. Research should be more focused to make technologies compatible with available local fuel sources so the most vulnerable can assess the technologies. Research development disseminations and delivery possibilities should be included in the project.
- Cooperation, collaboration, and coordination should be done with the province level agriculture and forestry ministry.
- Activities relating to backward and forward linkage of the project should be carried out parallel along with technology installation. For example, installation of biogas should be linked with promotion of livestock farms and slurry management to promote the CCS.
- Strengthen supply chain by mobilizing the local workforce to provide after sales service.
- Stove design should meet the users' needs such as space heating, agro processing and commercial cooking.
- Access to finance, access to information (how are we giving information to the illiterate), low level of mobility of female should be included
- Awareness campaign should focus on both male and female

- Coordination, collaboration, and coordination should be strong enough. LG has allocated much of its budget in renewable energy and women empowerment. Thus while identifying partners; criteria should be transparent to gain more weightage.
- Free distribution kills the market. Project should not have contradicting policies. In addition, the concept of free distribution should be strongly eliminated.
- Monitoring has to be the part of implementation.
- Coordination with Province level ministries (Agricultural, Tourism, Forest and Environment) to include them in project implementation or monitoring
- Conservation significant area such as buffer zone, vicinity of forest area should be prioritised
- ICS favorable for water heating and space heating to be focused in high altitude area
- More budget should be allocated in technology deployment and should be given to users directly at local level and less in trainings and awareness
- Extended to new geographical area to include Karnali pradesh
- NGOs play an important role in Community mobilization, leadership and awareness campaigns and demand creation. Collaborate with NGO to effectively implement the project.
- Women promoters to effectively implement the project and targeted to the female and most vulnerable communities
- Service centre at each province, repair maintenance centre, development of local manufacturing company makes the project sustainable
- Collaborate with private sector, microfinance, NGOs,
- Projects should be demand driven and should be focused on need.
- Training to women on electrical safety and basic technical skills on using induction stoves increases the use of stove and increase their confidence
- Capacitate female to repair and maintenance to ensure sustainability and empower women
- Research and development to make technologies more efficient rather to replicate the same old technologies
- Strengthen existing Network rather than establishing new ones and allocate more budget to users.
- Sustainability: after sales service and focus on user sensitive programs to increase the use
- Different financial package of support to poorest of the poor people to effectively implement the project
- Shifting to electric cooking may take long time, technical efficiency to user and private sector
- Focused on Socially adaptation of technologies
- Technological capacity development to ensure sustainability
- Provincial/local government should share the cost in technology
- AEPC budget should be focused on enabling environment including the development of
- Need to consider the technological barrier in adopting the electric cookstoves.

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Recording

Government of Nepal
Ministry of Energy, Water Resources and Irrigation
Alternative Energy Promotion Centre
Making Renewable Energy Mainstream Supply in Nepal

Green Climate Fund

- Financial Mechanism within UNFCCC, established (COP 16- Cancun) to help developing countries to invest in low carbon resilient development.
- Acts as operating entity for implementing Paris Agreement
- Can access funding through different three windows.
- Half of the adaptation pool for LDCs, SIDS and African states.
- International Channel or Direct access by country.
- Direct access through Direct Access

Adaptation

GCF funding windows

Mitigation Private sector facility

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